Real Book Exercises, Lectures 1–5: Detailed Solutions

Economics of Organizations and Incentives (MPE Insper)

Solutions to the 50 exercises of `Listas/lista-reais-lectures-01-05.pdf`

November 2026

These are detailed solutions to the real book exercises of the list `lista-reais-lectures-01-05.pdf` in `Listas/` (one block of ten per lecture). Within each lecture the items run from easy to hard, in the list's order, and each heading shows the item's number in the lecture, its number in the combined list, its level and its difficulty from 1 (apply a definition) to 5 (graduate level). Statements are reproduced verbatim in the list and are not repeated here. Every solution has the same six parts: (a) what is asked, in one line; (b) the method, with the book section and page and the lecture slide; (c) the solution, every step shown (for matrices, the full dominance comparisons or best-response marking); (d) the economic interpretation; (e) the verification, by the Python check scripts in this folder and against the book's worked answer or the MWG solutions manual where one exists; (f) common mistakes. Books: Tadelis (2013), Muñoz-Garcia & Toro-Gonzalez (2019, 2nd ed.), Gibbons (1992), Mas-Colell, Whinston & Green (1995) (MWG) and Cabral (2017). Notation follows the slides: a game is $G = (I, \{S_i\}_{i\in I}, \{u_i\}_{i\in I})$, s_{-i} is a profile of the others' strategies, S_i^k is the set of player i's strategies that survive k rounds of IESDS with every strictly dominated strategy removed in each round (Lecture 1, slide 29), and $\sigma_i \in \Delta(S_i)$ is a mixed strategy. In bimatrices player 1 picks rows and the first payoff is player 1's; underlined payoffs are best replies.

Contents

Lecture 1 — Static games I: dominance and iterated elimination **4**

Item 1 (1): Tadelis Exercise 4.5 (p. 77) **4**

Item 2 (2): Muñoz-Garcia Ch. 1, Exercise 13 (p. 26) **6**

Item 3 (3): Gibbons Problem 1.2 (p. 48) **8**

Item 4 (4): Tadelis Exercise 4.1 (p. 76) **10**

Item 5 (5): Muñoz-Garcia Ch. 1, Exercise 12 (p. 23) **12**

Item 6 (6): Tadelis Exercise 4.7 (p. 77–78) **14**

Item 7 (7): Tadelis Exercise 4.3 (p. 76) **16**

Item 8 (8): Tadelis Exercise 4.6 (p. 77) **18**

Item 9 (9): MWG Exercise 8.B.4 (p. 262) **20**

Item 10 (10): MWG Exercise 8.B.5 (p. 262) **23**

Lecture 2 — Static games II: Nash equilibrium and mixed strategies **25**

Item 1 (11): Tadelis Exercise 5.6 (p. 96) **25**

Item 2 (12): Muñoz-Garcia Ch. 2, Exercise 2 (p. 41) **27**

Item 3 (13): Muñoz-Garcia Ch. 3, Exercise 1 (p. 86–87) **29**

Item 4 (14): Gibbons Problem 1.12 (p. 51) **31**

Item 5 (15): Tadelis Exercise 5.8 (p. 96–97) **33**

Item 6 (16): Gibbons Problem 1.6 (p. 49) **35**

Item 7 (17): Gibbons Problem 1.7 (p. 49–50) **37**

Item 8 (18): Tadelis Exercise 6.8 (p. 125) **39**

Item 9 (19): MWG Exercise 8.D.2 (p. 263) **41**

Item 10 (20): MWG Exercise 8.D.6 (p. 264) **43**

Lecture 3 — Dynamic games: extensive form, backward induction, SPE **45**

Item 1 (21): Tadelis Exercise 7.1 (p. 147) **45**

Item 2 (22): Muñoz-Garcia Ch. 4, Exercise 3 (p. 160) **47**

Item 3 (23): Tadelis Exercise 7.2 (p. 147) **50**

Item 4 (24): Tadelis Exercise 8.5 (p. 170) **53**

Item 5 (25): Tadelis Exercise 8.10 (p. 172) **56**

Item 6 (26): Tadelis Exercise 7.4 (p. 148) **59**

Item 7 (27): Tadelis Exercise 7.9 (p. 149–150) **62**

Item 8 (28): Tadelis Exercise 7.10 (p. 150) **64**

Item 9 (29): MWG Exercise 9.B.4 (p. 302) **67**

Item 10 (30): Gibbons Problem 2.1 (p. 130) **69**

Lecture 4 — Credible threats and subgame perfection **71**

Item 1 (31): Gibbons Problem 2.12 (p. 135) **71**

Item 2 (32): Tadelis Exercise 7.6 (p. 148–149) **73**

Item 3 (33): Tadelis Exercise 8.8 (p. 171–172) **75**

Item 4 (34): Tadelis Exercise 8.6 (p. 170–171) **77**

Item 5 (35): Muñoz-Garcia Ch. 4, Exercise 11 (p. 180) **79**

Item 6 (36): Tadelis Exercise 8.11 (p. 172–173) **82**

Item 7 (37): Tadelis Exercise 8.4 (p. 170) **85**

Item 8 (38): Tadelis Exercise 8.13 (p. 173–174) **87**

Item 9 (39): Tadelis Exercise 8.7 (p. 171) **90**

Item 10 (40): MWG Exercise 9.B.5 (p. 302) **94**

Lecture 5 — Repeated games and bargaining **97**

Item 1 (41): Cabral Exercise 7.13 (p. 182) **97**

Item 2 (42): Tadelis Exercise 10.2 (p. 215) **99**

Item 3 (43): Muñoz-Garcia Ch. 6 Exercise 1 (p. 274) **102**

Item 4 (44): Muñoz-Garcia Ch. 6 Exercise 9 (p. 305) **104**

Item 5 (45): Tadelis Exercise 10.3 (p. 215–216) **107**

Item 6 (46): Tadelis Exercise 9.1 (p. 186–187) **109**

Item 7 (47): Gibbons Problem 2.11 (p. 135) **112**

Item 8 (48): Tadelis Exercise 11.5 (p. 236–237) **114**

Item 9 (49): Tadelis Exercise 10.7 (p. 217–218) **117**

Item 10 (50): Gibbons Problem 2.20 (p. 137–138) **120**

Lecture 1 — Static games I: dominance and iterated elimination

Item 1 (item 1 of the combined list)**: Tadelis Exercise 4.5** Foundations · difficulty 1/5

Source: Tadelis Exercise 4.5, page 77. The statement is reproduced verbatim in the list (item 1); it is not repeated here.

(a) What is asked. Which strategy profiles of the 3×3 game survive IESDS?

(b) Method. Strict dominance (Lecture 1, slide 23; Tadelis §4.1, pp. 59–61) and IESDS round by round (Lecture 1, slides 28–29; Tadelis §4.2, pp. 63–65). In each round k a strategy of player i is deleted if some other surviving strategy of i gives a strictly higher payoff against *every* surviving strategy of the opponent; all such strategies are deleted at once. The procedure stops when a round deletes nothing. A best-response marking is added at the end to show how the survivors relate to Nash equilibrium (Lecture 2, slides 14–17).

(c) Solution. Matrix as printed in the book (p. 77):

$1 \setminus 2$	L	C	R
U	6, 8	2, 6	8, 2
M	8, 2	4, 4	9, 5
D	8, 10	4, 6	6, 7

Write each strategy as its payoff vector against the opponent's strategies: player 1's rows over columns (L, C, R) are $U = (6, 2, 8)$, $M = (8, 4, 9)$, $D = (8, 4, 6)$; player 2's columns over rows (U, M, D) are $L = (8, 2, 10)$, $C = (6, 4, 6)$, $R = (2, 5, 7)$.

Round 1 (S^0 = the whole game). Every pair of strategies is compared.

Player	Pair	Componentwise comparison	Verdict
1	U vs M	$6 < 8$, $2 < 4$, $8 < 9$	U strictly dominated by M
1	U vs D	$6 < 8$, $2 < 4$, $8 > 6$	not ranked
1	M vs D	$8 = 8$, $4 = 4$, $9 > 6$	D only weakly dominated by M
2	L vs C	$8 > 6$, $2 < 4$, $10 > 6$	not ranked
2	L vs R	$8 > 2$, $2 < 5$, $10 > 7$	not ranked
2	C vs R	$6 > 2$, $4 < 5$, $6 < 7$	not ranked

Only U is deleted: $S_1^1 = \{M, D\}$, $S_2^1 = \{L, C, R\}$.

Round 2. Player 2 now compares columns on the surviving rows (M, D) only: $L = (2, 10)$, $C = (4, 6)$, $R = (5, 7)$.

- C vs R: $4 < 5$ and $6 < 7$, so C is strictly dominated by R.
- L vs C: $2 < 4$ but $10 > 6$; L vs R: $2 < 5$ but $10 > 7$. Not ranked.

Player 1 still faces all three columns: $M = (8, 4, 9)$ and $D = (8, 4, 6)$ tie against L and C, so neither strictly dominates the other. $S_1^2 = \{M, D\}$, $S_2^2 = \{L, R\}$.

Round 3. Player 1 on (L, R): $M = (8, 9)$, $D = (8, 6)$, a tie against L, so no strict dominance. Player 2 on (M, D): $L = (2, 10)$, $R = (5, 7)$, not ranked. Nothing is deleted and the process stops:

$$S^\infty = \{M, D\} \times \{L, R\} = \{(M, L), (M, R), (D, L), (D, R)\}.$$

Why no mixed strategy deletes more. In the final game each surviving strategy is a best reply to some surviving strategy of the opponent: L is the unique best reply to D ($10 > 7$), R the unique best reply to M ($5 > 2$), M the unique best reply to R ($9 > 6$), and D a best reply to L (tie at 8). A strategy that is a best reply to some belief cannot be strictly dominated by any pure or mixed strategy, so the answer does not depend on whether mixtures are allowed as dominators.

Best-response marking (underlined: the payoff is a best reply of its player to the other's strategy).

$1 \setminus 2$	L	C	R
U	6, $\underline{8}$	2, 6	8, 2
M	$\underline{8}$, 2	$\underline{4}$, 4	$\underline{9}$, $\underline{5}$
D	$\underline{8}$, $\underline{10}$	$\underline{4}$, 6	6, 7

Player 1: $BR_1(L) = \{M, D\}$, $BR_1(C) = \{M, D\}$, $BR_1(R) = \{M\}$. Player 2: $BR_2(U) = \{L\}$, $BR_2(M) = \{R\}$, $BR_2(D) = \{L\}$. Both payoffs are underlined in exactly two cells, (M, R) and (D, L): these are the pure Nash equilibria, and both lie in S^∞, as Lecture 2's Proposition 1 (slide 16) requires.

(d) Interpretation. Round 1 uses only player 1's rationality: U is worse than M whatever player 2 does. Round 2 uses player 2's knowledge that player 1 is rational: C is a sensible reply only to beliefs that put weight on U, and once U is gone R beats it everywhere (Lecture 2, slide 7). After that, no further layer of knowledge helps. The iterated-elimination equilibrium is a set of four profiles, not a point; Nash equilibrium, which adds correct beliefs, cuts it to (M, R) with payoffs $(9, 5)$ and (D, L) with $(8, 10)$. Deleting the weakly dominated D would single out (M, R), but weak dominance is not part of IESDS and its outcome can depend on the order of deletion.

(e) Verification. `check_A1.py` runs IESDS with the lecture's definition and finds exactly these rounds (round 1: U; round 2: C; then stop), that U is dominated only by M and C only by R, that D is weakly dominated by M in the final set, and, by brute force over the nine cells, that the pure Nash equilibria are (M, R) and (D, L). Tadelis gives no answer.

(f) Common mistakes.

- Deleting D because "M is at least as good everywhere". The ties against L and C make that weak dominance, which IESDS does not use.
- Deleting C in round 1. Against U it beats R ($6 > 2$); it becomes dominated only after U is gone, which is why the order of rounds matters for the explanation.
- Reporting the answer as one profile. The question asks for the surviving profiles: all four in $\{M, D\} \times \{L, R\}$.

Answer. Round 1: delete U (dominated by M). Round 2: delete C (dominated by R once U is gone). Nothing else is strictly dominated: $S^\infty = \{M, D\} \times \{L, R\}$, four profiles. (Of these, (M, R) and (D, L) are the pure Nash equilibria.)

Item 2 (item 2 of the combined list)**: Muñoz-Garcia Ch. 1, Exercise 13** Foundations · difficulty 1/5

Source: Muñoz-Garcia Ch. 1, Exercise 13, page 26. The statement is reproduced verbatim in the list (item 2); it is not repeated here.

(a) What is asked. For the 3×2 game of Fig. 1.33, find strictly dominant and strictly dominated strategies, then run IDSDS and explain each step.

(b) Method. Strictly dominant and strictly dominated strategies (Tadelis Defs. 4.1–4.2, pp. 60–61; Lecture 1, slide 23) and iterated deletion (Muñoz-Garcia's "IDSDS" is the slides' IESDS: Lecture 1, slides 28–29; Tadelis §4.2, pp. 63–65). The point of the exercise is the difference between strict and weak dominance: only strict dominance is used.

(c) Solution.

$1 \setminus 2$	L	R
U	$-10,\ 10$	$0,\ 12$
M	$-12,\ 0$	$2,\ 2$
D	$-9,\ 0$	$1,\ 0$

(U, M, D, L, R abbreviate Up, Middle, Down, Left, Right.) Payoff vectors: player 1's rows over (L, R) are $U = (-10, 0)$, $M = (-12, 2)$, $D = (-9, 1)$; player 2's columns over (U, M, D) are $L = (10, 0, 0)$, $R = (12, 2, 0)$.

(a) Strictly dominant strategies. A strictly dominant strategy must beat every other strategy of the player against every opponent strategy.

- Player 1: against L the best row is D ($-9 > -10 > -12$); against R it is M ($2 > 1 > 0$). No row is best against both columns, so player 1 has no strictly dominant strategy.
- Player 2: R beats L in row U ($12 > 10$) and row M ($2 > 0$) but ties in row D ($0 = 0$). So R is weakly, not strictly, dominant. Player 2 has no strictly dominant strategy.

(b) Strictly dominated strategies. All pairs:

Player	Pair	Componentwise comparison	Verdict
1	U vs M	$-10 > -12$, $0 < 2$	not ranked
1	U vs D	$-10 < -9$, $0 < 1$	U strictly dominated by D
1	M vs D	$-12 < -9$, $2 > 1$	not ranked
2	L vs R	$10 < 12$, $0 < 2$, $0 = 0$	L only weakly dominated by R

The only strictly dominated strategy is player 1's U (by D). Player 2 has none.

(c) IDSDS.

- *Step 1.* Delete U: player 1 is rational and D pays more than U whatever player 2 does.
- *Step 2.* In the reduced game (rows M, D), player 2 knows player 1 is rational, so she compares columns on rows (M, D) only: $L = (0, 0)$, $R = (2, 0)$. R is better in row M and ties in row D: weak dominance only, so nothing is deleted. Player 1 on columns (L, R): $M = (-12, 2)$, $D = (-9, 1)$, not ranked.

Nothing else can be deleted, and the surviving pairs are

$$\text{IDSDS} = \{M, D\} \times \{L, R\} = \{(M, L), (M, R), (D, L), (D, R)\}.$$

Beyond what is asked: best-response marking and Nash equilibria.

$1 \setminus 2$	L	R
U	$-10,\ 10$	$0,\ \underline{12}$
M	$-12,\ 0$	$\underline{2},\ \underline{2}$
D	$\underline{-9},\ \underline{0}$	$1,\ \underline{0}$

$BR_1(L) = \{D\}$, $BR_1(R) = \{M\}$; $BR_2(U) = \{R\}$, $BR_2(M) = \{R\}$, $BR_2(D) = \{L, R\}$. The pure Nash equilibria are (M, R), payoffs $(2, 2)$, and (D, L), payoffs $(-9, 0)$. The second one uses player 2's weakly dominated L, which is why it survives IDSDS but would be removed by weak dominance.

(d) Interpretation. Rationality alone removes U. A second layer of reasoning does not help, because player 2 is indifferent between L and R exactly when player 1 plays D, and D survives. The prediction is the four profiles $\{M, D\} \times \{L, R\}$. If player 2 broke ties toward R (weak dominance), player 1 would play M and the outcome would be (M, R); that is a stronger assumption than IDSDS makes. The book's own remark makes the same point: if player 2's payoff at (D, R) were 1 instead of 0, R would be strictly dominant.

(e) Verification. `check_A1.py` checks every pairwise comparison in the table, that IESDS deletes only U (round 1) and then stops at $\{M, D\} \times \{L, R\}$, that R weakly dominates L, and that the pure Nash equilibria are (M, R) and (D, L). *Comparison with the book's answer* (pp. 26–28): it agrees on all three parts (no strictly dominant strategy; only U strictly dominated, by D; IDSDS $= \{(M, L), (M, R), (D, L), (D, R)\}$). One slip: part (c) opens "As shown in part (a) of the exercise, strategy U is strictly dominated", but that was shown in part (b).

(f) Common mistakes.

- Deleting L because R is "always at least as good". The tie in row D makes it weak dominance, which IDSDS does not use.
- Calling D strictly dominant for player 1 because it dominates U. A dominant strategy must beat *every* other strategy, and M beats D against R.
- Giving the answer as payoffs $((2, 2), \dots)$ instead of strategy pairs.

Answer. (a) Neither player has a strictly dominant strategy (R is only weakly dominant for player 2). (b) Only player 1's U is strictly dominated (by D); player 2 has no strictly dominated strategy. (c) Delete U; then nothing else is strictly dominated: $\{(M, L), (M, R), (D, L), (D, R)\}$ survive.

Item 3 (item 3 of the combined list)**: Gibbons Problem 1.2** Foundations · difficulty 2/5

Source: Gibbons Problem 1.2, page 48. The statement is reproduced verbatim in the list (item 3); it is not repeated here.

(a) What is asked. Which strategies of the 3×3 game survive IESDS, and which profiles are pure-strategy Nash equilibria?

(b) Method. IESDS (Gibbons §1.1.B, pp. 4–7; Lecture 1, slides 28–29) and pure-strategy Nash equilibrium (Gibbons §1.1.C, pp. 7–12; Lecture 2, slide 14), found by marking best responses (Lecture 2, slide 10). Gibbons's Proposition B (p. 12; Lecture 2, Proposition 1, slide 16): every Nash equilibrium survives IESDS, so after the elimination only the surviving cells need to be searched. The full marking below confirms it independently.

(c) Solution. The matrix as printed (the top-left cell reads "2.0" in the scan; it is $(2, 0)$):

$1 \setminus 2$	L	C	R
T	2, 0	1, 1	4, 2
M	3, 4	1, 2	2, 3
B	1, 3	0, 2	3, 0

Payoff vectors: player 1's rows over (L, C, R) are $T = (2, 1, 4)$, $M = (3, 1, 2)$, $B = (1, 0, 3)$; player 2's columns over (T, M, B) are $L = (0, 4, 3)$, $C = (1, 2, 2)$, $R = (2, 3, 0)$.

Round 1. All pairs:

Player	Pair	Componentwise comparison	Verdict
1	T vs M	$2 < 3$, $1 = 1$, $4 > 2$	not ranked
1	T vs B	$2 > 1$, $1 > 0$, $4 > 3$	B strictly dominated by T
1	M vs B	$3 > 1$, $1 > 0$, $2 < 3$	not ranked
2	L vs C	$0 < 1$, $4 > 2$, $3 > 2$	not ranked
2	L vs R	$0 < 2$, $4 > 3$, $3 > 0$	not ranked
2	C vs R	$1 < 2$, $2 < 3$, $2 > 0$	not ranked

C is *not* dominated yet: row B is what keeps it alive ($2 > 0$ against R). $S_1^1 = \{T, M\}$, $S_2^1 = \{L, C, R\}$.

Round 2. Player 2 on rows (T, M): $L = (0, 4)$, $C = (1, 2)$, $R = (2, 3)$. C vs R: $1 < 2$ and $2 < 3$, so C is strictly dominated by R. L vs C ($0 < 1$, $4 > 2$) and L vs R ($0 < 2$, $4 > 3$) are not ranked. Player 1 on all three columns: T vs M unchanged, not ranked. $S_1^2 = \{T, M\}$, $S_2^2 = \{L, R\}$.

Round 3. Player 1 on (L, R): $T = (2, 4)$, $M = (3, 2)$, not ranked. Player 2 on (T, M): $L = (0, 4)$, $R = (2, 3)$, not ranked. Stop:

$$S^\infty = \{T, M\} \times \{L, R\}.$$

Pure Nash equilibria: best-response marking of the whole matrix.

$1 \setminus 2$	L	C	R
T	2, 0	$\underline{1}$, 1	$\underline{4}$, $\underline{2}$
M	$\underline{3}$, $\underline{4}$	$\underline{1}$, 2	2, 3
B	1, $\underline{3}$	0, 2	3, 0

Player 1: $BR_1(L) = \{M\}$ $(3 > 2 > 1)$, $BR_1(C) = \{T, M\}$ (tie at 1), $BR_1(R) = \{T\}$ $(4 > 3 > 2)$. Player 2: $BR_2(T) = \{R\}$ $(2 > 1 > 0)$, $BR_2(M) = \{L\}$ $(4 > 3 > 2)$, $BR_2(B) = \{L\}$ $(3 > 2 > 0)$. Both payoffs are underlined only in (T, R) and (M, L). Cell by cell inside S^∞:

- (T, L): player 1 would switch to M $(3 > 2)$. Not NE.
- (T, R): $T \in BR_1(R)$ and $R \in BR_2(T)$. NE, payoffs $(4, 2)$.
- (M, L): $M \in BR_1(L)$ and $L \in BR_2(M)$. NE, payoffs $(3, 4)$.
- (M, R): player 1 would switch to T $(4 > 2)$. Not NE.

(d) Interpretation. C is a sensible reply only to beliefs that put weight on B; once player 2 knows that player 1 is rational, C goes (Lecture 2, slide 7). IESDS stops at four profiles because each surviving strategy is a best reply to one of the opponent's surviving strategies. Nash equilibrium adds the requirement that beliefs be correct and cuts the four to two. The two equilibria are ranked differently by the players ((T, R) favours player 1, (M, L) player 2), so the game has a coordination problem that neither IESDS nor Nash equilibrium resolves.

(e) Verification. `check_A1.py` reproduces the rounds (round 1: B; round 2: C), checks that C is not dominated in the full game, and finds by brute force over all nine cells that the pure Nash equilibria are exactly (T, R) and (M, L). Gibbons gives no answer. Kumar's unofficial Gibbons manual (Answer 1.2, not in `Livros/`) gives the same deletions and equilibria; it writes the survivors as "(T,M) and (L,R)", meaning $\{T, M\}$ for player 1 and $\{L, R\}$ for player 2.

(f) Common mistakes.

- Deleting C in round 1 because it looks poor. In the full game C beats R in row B and L in row T; it is dominated only once B is gone.
- Reading "2.0" as the number two and leaving player 2's payoff blank. It is the pair $(2, 0)$; with any other reading the column comparisons change.
- Calling (T, L) or (M, R) an equilibrium because they survive IESDS. Survival is necessary for Nash equilibrium, not sufficient.

Answer. IESDS: delete B (by T), then C (by R); survivors $\{T, M\} \times \{L, R\}$. Pure Nash equilibria: (T, R) with payoffs $(4, 2)$ and (M, L) with payoffs $(3, 4)$.

Item 4 (item 4 of the combined list)**: Tadelis Exercise 4.1** Exam · difficulty 2/5

Source: Tadelis Exercise 4.1, page 76. The statement is reproduced verbatim in the list (item 4); it is not repeated here.

(a) What is asked. Prove that a strictly dominant strategy equilibrium, when it exists, is the only dominant strategy equilibrium.

(b) Method. Definitions only: strict dominance (Tadelis Def. 4.1, p. 60), strictly dominant strategy (Def. 4.2, p. 61), strict dominant strategy equilibrium (Def. 4.3, p. 61) and Proposition 4.1 itself (p. 61); in the slides, Lecture 1, slides 23 and 25. The proof is by contradiction, one player at a time. The statement says "the unique dominant strategy equilibrium" without the word "strict", so the proof also rules out a second profile of *weakly* dominant strategies (Lecture 1, slide 23), which is the stronger reading.

(c) Solution. Notation. $\Gamma = \langle N, \{S_i\}_{i=1}^n, \{v_i\}_{i=1}^n \rangle$; for a profile s, s_{-i} is the profile of the others. Every S_j is non-empty, so S_{-i} is non-empty. s_i is *strictly dominant* if $v_i(s_i, s_{-i}) > v_i(s_i', s_{-i})$ for all $s_i' \neq s_i$ and all s_{-i}; it is *weakly dominant* if $v_i(s_i, s_{-i}) \geq v_i(s_i', s_{-i})$ for all s_i' and all s_{-i}, with strict inequality somewhere for each $s_i' \neq s_i$. A (strict, weak) dominant strategy equilibrium is a profile in which every player uses a (strictly, weakly) dominant strategy.

Step 1: a player has at most one dominant strategy when one of them is strictly dominant. Let s_i^D be strictly dominant for player i and let $t_i \neq s_i^D$. By Def. 4.2, for every $s_{-i} \in S_{-i}$,

$$v_i(s_i^D, s_{-i}) > v_i(t_i, s_{-i}). \tag{1}$$

Suppose t_i were also dominant, strictly or weakly. Then, in particular,

$$v_i(t_i, s_{-i}) \geq v_i(s_i^D, s_{-i}) \quad \text{for every } s_{-i} \in S_{-i}. \tag{2}$$

Pick any $\hat{s}_{-i} \in S_{-i}$ (possible because $S_{-i} \neq \emptyset$). Then (1) and (2) give $v_i(s_i^D, \hat{s}_{-i}) > v_i(t_i, \hat{s}_{-i}) \geq v_i(s_i^D, \hat{s}_{-i})$, a contradiction. So no $t_i \neq s_i^D$ is dominant in either sense.

Step 2: the proposition. Let s^D be a strict dominant strategy equilibrium, and suppose some profile $s' \neq s^D$ is also a dominant strategy equilibrium. Since $s' \neq s^D$, there is a player i with $s_i' \neq s_i^D$. In s', player i's strategy s_i' is dominant; but s_i^D is strictly dominant for i (because s^D is a strict dominant strategy equilibrium), so by Step 1 no strategy other than s_i^D can be dominant for i. This contradicts $s_i' \neq s_i^D$. Hence s^D is the unique dominant strategy equilibrium. ■

Two remarks that complete the picture.

- The proposition does not claim existence: Battle of the Sexes has no dominant strategy equilibrium (Tadelis p. 62).
- Uniqueness extends to the other solution concepts of the course: s^D is the unique profile that survives IESDS (Lecture 1, slide 29: in round 1 every $s_i' \neq s_i^D$ is strictly dominated by s_i^D) and hence the unique Nash equilibrium (Lecture 2, slide 17). Without strictness uniqueness can fail: if dominance only required "$\geq$ against every s_{-i}", a player whose payoffs are all equal would have every strategy dominant, and every profile of such players would be a dominant strategy equilibrium.

(d) Interpretation. A strictly dominant strategy is a best reply that does not depend on beliefs. Two such strategies for the same player would each have to do strictly better than the other against the same opponents, which is impossible. That is why the dominant-strategy prediction, when it exists, is a single profile, and why it needs only rationality, not common knowledge of it.

(e) Verification. `check_A1.py` draws 3,000 random finite games (two or three players, two to four strategies each, integer payoffs), plants a strictly dominant strategy for every player in half of them, and checks in each game that, whenever a strict dominant strategy equilibrium exists, no other profile consists of weakly or strictly dominant strategies, and that it is the only IESDS survivor. Tadelis gives no answer.

(f) Common mistakes.

- Proving only that there is no second *strict* dominant strategy equilibrium. The proposition says "dominant", and the same two-line argument rules out weakly dominant ones.
- Comparing the two profiles s^D and s' as a whole. Dominance is a property of one player's strategy against all of the others' profiles; the contradiction comes from a single player whose strategies differ.
- Forgetting that S_{-i} must be non-empty for the contradiction (one $\hat{s}_{-i}$ is needed). It always is in a game, but the step should be stated.

Answer. If s^D is a strict dominant strategy equilibrium and $s' \neq s^D$, some player has $s'_i \neq s^D_i$. Strict dominance of s^D_i gives $v_i(s^D_i, s_{-i}) > v_i(s'_i, s_{-i})$ for every s_{-i}, so s'_i cannot be dominant, strictly or weakly (that would need $v_i(s'_i, s_{-i}) \geq v_i(s^D_i, s_{-i})$ for every s_{-i}). Hence s' is not a dominant strategy equilibrium, and s^D is the unique one.

Item 5 (item 5 of the combined list)**: Muñoz-Garcia Ch. 1, Exercise 12**Exam · difficulty 2/5

Source: Muñoz-Garcia Ch. 1, Exercise 12, page 23. The statement is reproduced verbatim in the list (item 5); it is not repeated here.

(a) What is asked. For the 3×3 game of Fig. 1.30, find dominant and dominated strategies, then what is deleted under rationality alone and under common knowledge of rationality.

(b) Method. Strict dominance and dominant strategies (Lecture 1, slides 23 and 25; Tadelis §4.1, pp. 59–61) and the epistemic reading of IESDS (Lecture 2, slide 7; Tadelis §4.2.1, pp. 63–65). Round 1 of IESDS uses only each player's own rationality; round 2 adds "each knows the other is rational"; round k adds $k-1$ layers of that knowledge. Part (c) stops after round 1, part (d) runs IESDS to the end (Lecture 1, slide 29).

(c) Solution. Matrix from the book (p. 23):

$1 \setminus 2$	F	C	B
F	0, 5	2, 4	2, 3
C	2, 3	0, 5	3, 2
B	5, 0	2, 4	2, 3

Payoff vectors: player 1's rows over columns (F, C, B) are $F = (0, 2, 2)$, $C = (2, 0, 3)$, $B = (5, 2, 2)$; player 2's columns over rows (F, C, B) are $F = (5, 3, 0)$, $C = (4, 5, 4)$, $B = (3, 2, 3)$.

Best-response marking (used in (a) and at the end).

$1 \setminus 2$	F	C	B
F	0, $\underline{5}$	$\underline{2}$, 4	2, 3
C	2, 3	0, $\underline{5}$	$\underline{3}$, 2
B	$\underline{5}$, 0	$\underline{2}$, $\underline{4}$	2, 3

$BR_1(F) = \{B\}$, $BR_1(C) = \{F, B\}$, $BR_1(B) = \{C\}$; $BR_2(F) = \{F\}$, $BR_2(C) = \{C\}$, $BR_2(B) = \{C\}$.

(a) Strictly dominant strategies. A strictly dominant strategy would be the unique best reply to every strategy of the opponent. Player 1's best reply changes with the column (B, then $\{F, B\}$, then C), and player 2's changes with the row (F, then C). Neither player has a strictly dominant strategy.

(b) Strictly dominated strategies. All pairs:

Player	Pair	Componentwise comparison	Verdict
1	F vs C	$0 < 2$, $2 > 0$, $2 < 3$	not ranked
1	F vs B	$0 < 5$, $2 = 2$, $2 = 2$	F only weakly dominated by B
1	C vs B	$2 < 5$, $0 < 2$, $3 > 2$	not ranked
2	F vs C	$5 > 4$, $3 < 5$, $0 < 4$	not ranked
2	F vs B	$5 > 3$, $3 > 2$, $0 < 3$	not ranked
2	C vs B	$4 > 3$, $5 > 2$, $4 > 3$	B strictly dominated by C

No mixed strategy dominates anything else either: F, C, B of player 1 and F, C of player 2 are each a best reply to some opponent strategy (marking above). The only strictly dominated strategy is player 2's B.

(c) Rationality, no common knowledge. Each player deletes his own strictly dominated strategies but does not use the fact that the other does the same. Player 2 drops B; player 1 has nothing to drop and still evaluates his rows against all three columns. The surviving profiles are

$$\{F, C, B\} \times \{F, C\} = \{(F, F), (F, C), (C, F), (C, C), (B, F), (B, C)\}.$$

(d) Common knowledge of rationality.

- Round 2: player 1 knows player 2 is rational, so he compares rows on columns $\{F, C\}$ only: $F = (0, 2)$, $C = (2, 0)$, $B = (5, 2)$. C vs B: $2 < 5$, $0 < 2$, so C is strictly dominated by B. F vs B: $0 < 5$, $2 = 2$, only weak. F vs C: not ranked. Player 2 still faces all three rows; F vs C is not ranked. $S_1^2 = \{F, B\}$, $S_2^2 = \{F, C\}$.
- Round 3: player 2 on rows (F, B): $F = (5, 0)$, $C = (4, 4)$, not ranked. Player 1 on columns (F, C): $F = (0, 2)$, $B = (5, 2)$, only weak. Stop.

$$S^\infty = \{F, B\} \times \{F, C\} = \{(F, F), (F, C), (B, F), (B, C)\}.$$

Beyond what is asked. In the marking, both payoffs are underlined only in (B, C): it is the unique pure-strategy Nash equilibrium, with payoffs $(2, 4)$, and it lies in S^∞.

(d) Interpretation. Rationality alone removes only strategies that are bad against *everything*. Player 1's C is the best reply to player 2's B, so it can be discarded only by someone who knows that player 2 will not play B, that is, by someone who uses player 2's rationality. Each extra layer of "I know that you know" can remove more strategies; here the second layer removes C and the third removes nothing. The difference between (c) and (d) is the difference between S^1 and S^∞ in the slides.

(e) Verification. `check_A1.py` checks the dominance statements in (a)–(b), that IESDS removes B for player 2 in round 1 and C for player 1 in round 2, that F is only weakly dominated by B, and that (B, C) is the only pure Nash equilibrium. The book's answer (pp. 23–26) agrees with (a)–(d). Its text for (b) and (d) describes the comparisons loosely (for example, it says player 1 deletes C because his payoff "is higher from B ... when player 2 plays F"), but the deletions and the final sets are the same as ours. The figure caption "three-players game" is a slip in the book: the game has two players.

(f) Common mistakes.

- Deleting player 1's F in (b) because B is never worse. The ties against C and B make it weak dominance only; F survives IESDS.
- In (c), letting player 1 compare rows only on $\{F, C\}$. That already uses player 2's rationality, which is the step (c) forbids.
- Thinking that "no dominant strategy" implies "no dominated strategy": (a) and (b) are separate questions, and player 2's B is dominated although no strategy is dominant.

Answer. (a) None. (b) Only player 2's B (strictly dominated by C). (c) Rationality alone: $\{F, C, B\} \times \{F, C\}$, six profiles. (d) With common knowledge, player 1 then drops C (dominated by B on $\{F, C\}$): $S^\infty = \{F, B\} \times \{F, C\}$, four profiles.

Item 6 (item 6 of the combined list)**: Tadelis Exercise 4.7** Exam · difficulty 3/5

Source: Tadelis Exercise 4.7, page 77–78. The statement is reproduced verbatim in the list (item 6); it is not repeated here.

(a) What is asked. Turn the campaigning story into a normal-form game and matrix, then run IESDS and say whether it gives a clear prediction.

(b) Method. Normal-form game $G = (I, \{S_i\}, \{u_i\})$ (Lecture 1, slides 13–14; Tadelis §3.1, pp. 46–50); IESDS (Lecture 1, slides 28–29; Tadelis §4.2, pp. 63–65). The only modelling step is to fill the payoffs the story leaves implicit, using the fact that exactly one candidate wins.

(c) Solution. **(a) The game.** Players $I = \{1, 2\}$; strategies $S_1 = S_2 = \{P, B, N\}$. Let $\pi(s_i, s_j)$ be the probability that the candidate playing s_i wins against one playing s_j. The book gives four rules:

$$\pi(s, s) = 0.5, \qquad \pi(P, B) = 0, \qquad \pi(P, N) = 0.3, \qquad \pi(N, B) = 0.6.$$

Exactly one candidate wins, so $\pi(s_i, s_j) + \pi(s_j, s_i) = 1$. This fills the other three entries:

$$\pi(B, P) = 1 - \pi(P, B) = 1, \qquad \pi(N, P) = 1 - 0.3 = 0.7, \qquad \pi(B, N) = 1 - 0.6 = 0.4.$$

Payoffs: $u_1(s_1, s_2) = \pi(s_1, s_2)$ and $u_2(s_1, s_2) = \pi(s_2, s_1) = 1 - u_1(s_1, s_2)$. Player 2's payoff function is the same rule π with the roles swapped (the game is symmetric), and it is the complement of player 1's because win probabilities add up to one (the game is constant-sum).

(b) The matrix, with best responses underlined:

$1 \setminus 2$	P	B	N
P	0.5, 0.5	0, $\underline{1}$	0.3, $\underline{0.7}$
B	$\underline{1}$, 0	0.5, 0.5	0.4, $\underline{0.6}$
N	0.7, 0.3	$\underline{0.6}$, 0.4	$\underline{0.5}$, $\underline{0.5}$

$BR_1(P) = \{B\}$, $BR_1(B) = \{N\}$, $BR_1(N) = \{N\}$, and by symmetry $BR_2(P) = \{B\}$, $BR_2(B) = \{N\}$, $BR_2(N) = \{N\}$.

(c) IESDS. Player 1's rows over columns (P, B, N): $P = (0.5, 0, 0.3)$, $B = (1, 0.5, 0.4)$, $N = (0.7, 0.6, 0.5)$.

- *Round 1.* P vs B: $0.5 < 1$, $0 < 0.5$, $0.3 < 0.4$, so P is strictly dominated by B. P vs N: $0.5 < 0.7$, $0 < 0.6$, $0.3 < 0.5$, so P is also dominated by N. B vs N: $1 > 0.7$ but $0.5 < 0.6$, not ranked. By symmetry player 2's columns satisfy the same comparisons. $S_i^1 = \{B, N\}$ for $i = 1, 2$.
- *Round 2.* On the columns $\{B, N\}$: $B = (0.5, 0.4)$, $N = (0.6, 0.5)$. $0.5 < 0.6$ and $0.4 < 0.5$, so B is strictly dominated by N. The same for player 2. $S_i^2 = \{N\}$.
- *Round 3.* One strategy each: nothing to compare.

$S^\infty = \{(N, N)\}$: both run negative campaigns and each wins with probability 0.5. The procedure gives a clear prediction, a unique profile, after two rounds. In the marking (N, N) is the only cell with both payoffs underlined, so it is also the unique Nash equilibrium (as it must be: Lecture 2, Proposition 2, slide 17).

(d) Interpretation. B survives round 1 only because it crushes a positive campaign. Once each candidate knows the other will not run a purely positive campaign, going fully negative wins. The outcome has the same win probabilities as (P, P), and in a constant-sum game every profile is Pareto optimal *for the candidates*. If voters dislike negative campaigning, the equilibrium is a prisoner's-dilemma outcome for them: the candidates end up exactly where they would have been with positive campaigns, but with a nastier campaign.

(e) Verification. `check_A1.py` builds the matrix from the four rules plus complementarity, checks symmetry, that P is dominated by both B and N, that IESDS removes P (round 1) and B (round 2) for both players, leaving (N, N), and that (N, N) is the only pure Nash equilibrium. Tadelis gives no answer.

(f) Common mistakes.

- Filling the missing cells by symmetry of *payoffs* ($\pi(B, P) = \pi(P, B) = 0$) instead of by complementarity. A candidate who beats P for sure must win with probability 1.
- Deleting B in round 1. Against P it is the unique best reply ($1 > 0.7$); it becomes dominated only after P is gone.
- In (a), writing only the matrix. The question asks for players, strategy sets and payoff functions, and for why player 2's function differs ($u_2 = 1 - u_1$).

Answer. (a) $I = \{1, 2\}$, $S_i = \{P, B, N\}$, $u_1 = \pi(s_1, s_2)$ from the rules plus $\pi(s_i, s_j) + \pi(s_j, s_i) = 1$, and $u_2 = 1 - u_1$ (the same rule from candidate 2's side). (b) Matrix above. (c) Round 1 deletes P (dominated by B and by N); round 2 deletes B (dominated by N); unique prediction (N, N), each wins with probability 0.5.

Item 7 (item 7 of the combined list)**: Tadelis Exercise 4.3** Exam · difficulty 3/5

Source: Tadelis Exercise 4.3, page 76. The statement is reproduced verbatim in the list (item 7); it is not repeated here.

(a) What is asked. Write the two-bidder first-price auction with bids $\{0, 1, 2\}$ as a matrix, find the strictly dominated strategies, and run IESDS.

(b) Method. Building the normal form from the rules (Lecture 1, slides 13–19; Tadelis §3.1, pp. 46–50); strict dominance (Tadelis Def. 4.1, p. 60; Lecture 1, slide 23); IESDS with every dominated strategy removed in each round (Lecture 1, slide 29; Tadelis §4.2, pp. 63–65). Ties are broken by a fair coin, so a tie gives each bidder half of (value − bid): payoffs are expected payoffs over the coin toss.

(c) Solution. (a) The matrix. Player 1 values the item at 3, player 2 at 5; $b_i \in \{0, 1, 2\}$. Player i's payoff is $\theta_i - b_i$ if $b_i > b_j$, 0 if $b_i < b_j$, and $\frac{1}{2}(\theta_i - b_i)$ if $b_i = b_j$, with $\theta_1 = 3$, $\theta_2 = 5$. Cell by cell:

- ties: $(0,0)$: $(\frac{3}{2}, \frac{5}{2})$; $(1,1)$: $(\frac{3-1}{2}, \frac{5-1}{2}) = (1, 2)$; $(2,2)$: $(\frac{1}{2}, \frac{3}{2})$;
- player 1 wins: $(1,0)$: $(3-1, 0) = (2, 0)$; $(2,0)$ and $(2,1)$: $(3-2, 0) = (1, 0)$;
- player 2 wins: $(0,1)$: $(0, 5-1) = (0, 4)$; $(0,2)$ and $(1,2)$: $(0, 5-2) = (0, 3)$.

1 \ 2	0	1	2
0	1.5, 2.5	0, 4	0, 3
1	2, 0	1, 2	0, 3
2	1, 0	1, 0	0.5, 1.5

Payoff vectors: player 1's rows over $b_2 = (0, 1, 2)$ are $0 \mapsto (1.5, 0, 0)$, $1 \mapsto (2, 1, 0)$, $2 \mapsto (1, 1, 0.5)$; player 2's columns over $b_1 = (0, 1, 2)$ are $0 \mapsto (2.5, 0, 0)$, $1 \mapsto (4, 2, 0)$, $2 \mapsto (3, 3, 1.5)$.

(b) Strictly dominated strategies (by pure strategies). All pairs:

Player	Pair	Componentwise comparison	Verdict
1	0 vs 1	$1.5 < 2$, $0 < 1$, $0 = 0$	0 only weakly dominated by 1
1	0 vs 2	$1.5 > 1$, $0 < 1$, $0 < 0.5$	not ranked
1	1 vs 2	$2 > 1$, $1 = 1$, $0 < 0.5$	not ranked
2	0 vs 1	$2.5 < 4$, $0 < 2$, $0 = 0$	0 only weakly dominated by 1
2	0 vs 2	$2.5 < 3$, $0 < 3$, $0 < 1.5$	0 strictly dominated by 2
2	1 vs 2	$4 > 3$, $2 < 3$, $0 < 1.5$	not ranked

Yes: player 2's bid 0 is strictly dominated by bid 2. Player 1 has no strategy strictly dominated by another pure strategy. (*Remark, mixed dominators*, Tadelis §6.1 and Lecture 2, slide 42: player 1's bid 0 *is* strictly dominated by the mixed strategy $\frac{3}{4}\cdot 1 + \frac{1}{4}\cdot 2$, which pays $(1.75,\ 1,\ 0.125) > (1.5,\ 0,\ 0)$. With Chapter 4's pure-strategy definition this does not count, and it does not change the answer to (c).)

(c) IESDS.

- *Round 1.* Delete player 2's 0 (by 2). Nothing for player 1. $S_1^1 = \{0, 1, 2\}$, $S_2^1 = \{1, 2\}$.

- *Round 2.* Player 1 over $b_2 \in \{1,2\}$: $0 \mapsto (0,0)$, $1 \mapsto (1,0)$, $2 \mapsto (1,0.5)$. 0 vs 2: $0 < 1$, $0 < 0.5$, so 0 is strictly dominated by 2. 1 vs 2: $1 = 1$, only weak. Player 2 over $b_1 \in \{0,1,2\}$: $1 \mapsto (4,2,0)$, $2 \mapsto (3,3,1.5)$, not ranked. $S_1^2 = \{1,2\}$, $S_2^2 = \{1,2\}$.
- *Round 3.* Player 2 over $b_1 \in \{1,2\}$: $1 \mapsto (2,0)$, $2 \mapsto (3,1.5)$): 1 is strictly dominated by 2. Player 1 over $b_2 \in \{1,2\}$: $1 \mapsto (1,0)$, $2 \mapsto (1,0.5)$, only weak. $S_1^3 = \{1,2\}$, $S_2^3 = \{2\}$.
- *Round 4.* Player 1 against $b_2 = 2$ only: bid 1 pays 0, bid 2 pays 0.5, so 1 is strictly dominated by 2. $S_1^4 = \{2\}$, $S_2^4 = \{2\}$.

$S^\infty = \{(2,2)\}$: both bid 2, each wins with probability $\frac{1}{2}$, and the expected payoffs are $(0.5, 1.5)$. Since the survivor is unique, it is the unique Nash equilibrium (Lecture 2, slide 17); the best-response marking confirms it: $BR_1(2) = \{2\}$ and $BR_2(2) = \{2\}$, while every other cell has a profitable deviation.

(d) Interpretation. Each round adds one layer of "I know that you know": player 2 never bids 0 (her value is high enough that bidding 2 is always better); knowing that, player 1 never bids 0 either; then player 2 bids 2 to avoid losing ties at 1; then player 1 matches. The coarse bid grid matters: the high-value bidder wins only half the time, so the item goes to the bidder who values it less with probability $\frac{1}{2}$, an inefficiency created by the tie rule and the bid cap.

(e) Verification. `check_A1.py` builds the matrix from the auction rules in exact arithmetic, checks the dominance table, runs IESDS and finds the four rounds above (player 2's 0; player 1's 0; player 2's 1; player 1's 1), confirms that $(2,2)$ is the only pure Nash equilibrium, and checks that $\frac{3}{4} \cdot 1 + \frac{1}{4} \cdot 2$ strictly dominates player 1's bid 0. Tadelis gives no answer.

(f) Common mistakes.

- Giving the tie winner the whole surplus or splitting the item rather than the expected payoff. With a fair coin each tied bidder gets $\frac{1}{2}(\theta_i - b_i)$.
- Charging the loser his bid. In a first-price auction only the winner pays.
- Stopping after round 1 or 2. The deletions alternate between the players, and only the fourth round leaves a single profile.

Answer. (a) Matrix above (rows: player 1's bid; columns: player 2's bid). (b) Player 2's bid 0 is strictly dominated by bid 2; player 1 has no strictly dominated pure strategy (his bid 0 is only weakly dominated by 1). (c) IESDS deletes, in order, player 2's 0, player 1's 0, player 2's 1, player 1's 1; only $(2,2)$ survives, with expected payoffs $(0.5, 1.5)$.

Item 8 (item 8 of the combined list)**: Tadelis Exercise 4.6** Exam · difficulty 4/5

Source: Tadelis Exercise 4.6, page 77. The statement is reproduced verbatim in the list (item 8); it is not repeated here.

(a) What is asked. For cleaning times $t_i \geq 0$ with payoff $(10 - t_j)t_i - t_i^2$, find the best response, the survivors of one round of IESDS, and the survivors of IESDS.

(b) Method. Best response (Lecture 2, slide 10; Tadelis §4.3.1, p. 69) and IESDS with a continuum of strategies (Lecture 1, slide 29), done exactly as in Tadelis's Cournot example (§4.2.2, pp. 65–67, Fig. 4.1) and in the slides' Cournot remark (Lecture 2, slide 24): bound the set of possible best replies and show that every action outside it is strictly dominated by the nearest end of the range.

(c) Solution. **(a) Best response.** Fix $t_j \geq 0$ and maximise $u_i(t_i, t_j) = (10 - t_j)t_i - t_i^2$ over $t_i \geq 0$.

- $\partial u_i/\partial t_i = 10 - t_j - 2t_i$ and $\partial^2 u_i/\partial t_i^2 = -2 < 0$: u_i is strictly concave, so the first-order condition with the constraint $t_i \geq 0$ gives the unique maximiser.
- If $t_j < 10$, the FOC $10 - t_j - 2t_i = 0$ gives $t_i = (10 - t_j)/2 > 0$.
- If $t_j \geq 10$, $\partial u_i/\partial t_i = 10 - t_j - 2t_i < 0$ for every $t_i > 0$, so the maximum is at the corner $t_i = 0$.

$$BR_i(t_j) = \max\Big\{0,\ 5 - \frac{t_j}{2}\Big\}.$$

It is single-valued and decreasing in t_j: cleaning is a strategic substitute.

Dominance lemma. Write $x = (10 - t_j)/2$ for the unconstrained peak, so $u_i(t, t_j) = 2xt - t^2 = t(2x - t)$. For any two times a and t,

$$u_i(a, t_j) - u_i(t, t_j) = 2x(a - t) - (a^2 - t^2) = (a - t)(2x - a - t) = (t - a)(t + a - 2x). \quad (1)$$

If $t > a \geq x$, both factors on the right are positive; if $t < a \leq x$, both are negative. So: if every peak x that the opponent's surviving strategies can generate lies in $[\underline{a}, \bar{a}]$, then every $t > \bar{a}$ is strictly dominated by $\bar{a}$ and every $t < \underline{a}$ by $\underline{a}$. Conversely, a t that is the unique best reply to some surviving t_j is never strictly dominated.

(b) One round. With $t_j \in [0, \infty)$, the best replies range over $[0, 5]$ ($BR_i(0) = 5$, $BR_i(t_j) = 0$ for $t_j \geq 10$), and the peaks $x = 5 - t_j/2$ are all ≤ 5.

- Every $t_i > 5$ is strictly dominated by 5: by (1) with $a = 5 \geq x$, for every t_j (if $x < 0$ the inequality still holds because $t + a - 2x > 0$).
- Every $t_i \in [0, 5]$ is the unique best reply to $t_j = 10 - 2t_i \in [0, 10]$, so it is not dominated.

Hence $S_i^1 = [0, 5]$ for both roommates.

(c) Iterating. Suppose $S_j^k = [L_k, U_k]$. Because BR_i is continuous and decreasing, the best replies to $t_j \in [L_k, U_k]$ fill exactly $[BR_i(U_k), BR_i(L_k)]$; by (1) everything outside that interval is strictly dominated by its nearer endpoint, and everything inside is a best reply. So

$$L_{k+1} = 5 - \frac{U_k}{2}, \qquad U_{k+1} = 5 - \frac{L_k}{2}, \qquad (L_1, U_1) = (0, 5).$$

(The $\max\{0, \cdot\}$ never binds from round 2 on because $U_k \leq 5 < 10$.)

k	1	2	3	4	5	∞
L_k	0	$5-5/2=2.5$	2.5	$5-3.75/2=3.125$	3.125	10/3
U_k	5	$5-0/2=5$	$5-2.5/2=3.75$	3.75	$5-3.125/2=3.4375$	10/3

Convergence. Subtracting the two recursions, $U_{k+1}-L_{k+1}=\frac{1}{2}(U_k-L_k)$, so $U_k-L_k=5/2^{k-1}\to 0$. Adding them, the midpoint $M_k=(L_k+U_k)/2$ satisfies $M_{k+1}=5-M_k/2$. Its fixed point solves $M=5-M/2$, i.e. $M=10/3$, and

$$M_k-\tfrac{10}{3}=(-\tfrac{1}{2})^{k-1}(M_1-\tfrac{10}{3})=(-\tfrac{1}{2})^{k-1}(\tfrac{5}{2}-\tfrac{10}{3})=(-\tfrac{1}{2})^{k-1}(-\tfrac{5}{6})\to 0.$$

The intervals are nested, their widths go to zero and their midpoints go to 10/3, so $S_i^\infty=\bigcap_k S_i^k=\{10/3\}$ for both roommates. IESDS predicts the unique profile $t_1=t_2=10/3$. It is also the Nash equilibrium: $t=5-t/2$ gives $t=10/3$.

(d) Interpretation. Each round uses one more layer of knowledge: round 1, "I am rational" (never clean more than 5 hours); round 2, "I know you are rational" (you clean at most 5, so I clean at least 2.5); and so on (Lecture 2, slide 7). Each roommate gets $u_i=(10-10/3)(10/3)-(10/3)^2=200/9-100/9=100/9\approx 11.1$. Joint welfare $2[(10-t)t-t^2]=2(10t-2t^2)$ is maximised at $10-4t=0$, i.e. $t=2.5$, where each gets $(7.5)(2.5)-6.25=12.5>100/9$. The IESDS outcome over-cleans relative to the joint optimum: each roommate ignores that his cleaning lowers the value of the other's (a negative externality, $\partial u_j/\partial t_i=-t_j<0$).

(e) Verification. `check_A1.py` checks the best reply and the factorization (1) with sympy, runs the interval recursion for 40 rounds in exact arithmetic (the table, width $5/2^{k-1}$ and the midpoint formula at every round), checks the payoffs 100/9 and 12.5 and the joint optimum 2.5, and, as a second route, runs brute-force IESDS on the grid $\{0,\frac{1}{6},\dots,10\}$, which leaves exactly $t_1=t_2=10/3$. Tadelis gives no answer.

(f) Common mistakes.

- Answering (b) with $[0,\infty)$ minus nothing, or with $[0,10]$. The bound comes from the best replies, which never exceed 5, not from where the payoff is positive.
- Writing $L_{k+1}=5-L_k/2$: the *upper* bound of the opponent gives the *lower* bound of the best reply, because BR_i is decreasing.
- Claiming that a t outside the best-reply range is "dominated" without naming the dominating strategy. Lemma (1) supplies it: the nearer endpoint.

Answer. (a) $BR_i(t_j)=\max\{0,5-t_j/2\}$. (b) $S_i^1=[0,5]$: any $t_i>5$ is strictly dominated by 5. (c) Rounds give $[2.5,5]$, $[2.5,3.75]$, $[3.125,3.75]$, $[3.125,3.4375]$, …, with widths $5/2^{k-1}$; only $t_1=t_2=10/3$ survives IESDS.

Item 9 (item 9 of the combined list)**: MWG Exercise 8.B.4** Advanced · difficulty 4/5

Source: MWG Exercise 8.B.4, page 262. The statement is reproduced verbatim in the list (item 9); it is not repeated here.

(a) What is asked. Prove that, in iterated deletion of strictly dominated strategies, the set of surviving strategies does not depend on the order in which dominated strategies are deleted.

(b) Method. Strict dominance and its iterated deletion (MWG §8.B, Defs. 8.B.2–8.B.4 and the iterated procedure, pp. 236–241; Lecture 1, slides 23 and 29; Tadelis §4.2, pp. 63–65). Two facts carry the proof: (i) dominance is *inherited* when the opponents' strategy sets shrink; (ii) in a finite game, a dominated strategy has a dominator that is itself undominated, and such a dominator is never deleted. The game is finite, as in MWG §8.B (finite S_i); a final remark shows why finiteness is needed.

(c) Solution. Setting. $\Gamma = [I, \{S_i\}, \{u_i\}]$ with every S_i finite. For a product set $T = \prod_i T_i \subseteq S$, say that $s_i \in T_i$ is *strictly dominated in T by $t_i \in S_i$* if $u_i(t_i, s_{-i}) > u_i(s_i, s_{-i})$ for every $s_{-i} \in T_{-i}$; it is *dominated in T* if this holds for some $t_i \in T_i$.

A *deletion sequence* is a chain $S = T^0 \supseteq T^1 \supseteq \cdots \supseteq T^K$ of product sets in which each T^{k+1} is obtained from T^k by deleting a non-empty set of strategies (of one or several players) each of which is dominated in T^k, and which stops only when T^K contains no dominated strategy. "Order of deletion" means the choice of which dominated strategies to delete at each step; the lecture's procedure (Lecture 1, slide 29) is the sequence that deletes all of them at once. Since S is finite and each step deletes something, every deletion sequence ends after finitely many steps. Claim: all deletion sequences end at the same T^K.

Lemma 1 (inheritance). If $X \subseteq Y$ (that is, $X_j \subseteq Y_j$ for all j), $s_i \in X_i$, and s_i is strictly dominated by t_i in Y, then s_i is strictly dominated by t_i in X.

Proof. The inequality $u_i(t_i, s_{-i}) > u_i(s_i, s_{-i})$ holds for every $s_{-i} \in Y_{-i}$, hence for every $s_{-i} \in X_{-i} \subseteq Y_{-i}$. □

Lemma 2 (an undominated dominator). Fix a product set X and a player i. If $s_i \in X_i$ is strictly dominated in X by some strategy in S_i, then there is a $t_i^* \in S_i$ that strictly dominates s_i in X and is itself not strictly dominated in X by any strategy in S_i.

Proof. Let $D = \{t \in S_i : t \text{ strictly dominates } s_i \text{ in } X\}$, non-empty and finite. "r strictly dominates t in X" is irreflexive and transitive (if $u_i(r, \cdot) > u_i(t, \cdot)$ and $u_i(t, \cdot) > u_i(s_i, \cdot)$ on X_{-i}, then $u_i(r, \cdot) > u_i(s_i, \cdot)$ there), so it is a strict partial order, and the finite set D has an element t_i^* not dominated by any other element of D. If some $r \in S_i$ strictly dominated t_i^* in X, then by transitivity r would dominate s_i, so $r \in D$, a contradiction. □

Lemma 3 (the end point of any sequence is contained in every set of any other sequence). Let $(T^k)_{k=0}^K$ and $(R^l)_{l=0}^L$ be two deletion sequences, with end points $T = T^K$ and $R = R^L$. Then $T \subseteq R^l$ for every l; in particular $T \subseteq R$.

Proof, by induction on l. For $l = 0$, $T \subseteq S = R^0$. Suppose $T \subseteq R^l$, and suppose, for a contradiction, that some $s_i \in T_i$ is deleted at step $l+1$ of the second sequence. Then s_i is strictly dominated in R^l by some $t_i \in R_i^l$. Since $T \subseteq R^l$, Lemma 1 shows that s_i is strictly dominated in T by t_i. By Lemma 2 (with $X = T$) there is a $t_i^* \in S_i$ that strictly dominates s_i in T and is not strictly dominated in T by any strategy in S_i.

Now $t_i^* \in T_i$. Indeed, if t_i^* had been deleted at some step $k+1$ of the first sequence, it would be strictly dominated in T^k by some $r \in T_i^k$; since $T \subseteq T^k$, Lemma 1 makes r strictly dominate t_i^* in

T, contradicting the choice of t_i^*.

So $s_i \in T_i$ is strictly dominated in T by $t_i^* \in T_i$: s_i is dominated in T. But T is the end point of a deletion sequence and contains no dominated strategy. This contradiction shows that no strategy of T is deleted at step $l+1$, so $T \subseteq R^{l+1}$. □

The theorem. Apply Lemma 3 twice: $T \subseteq R$, and, exchanging the roles of the two sequences, $R \subseteq T$. Hence $T = R$: every order of deletion ends at the same set, which is the set S^∞ produced by deleting everything dominated in each round. ■

(No T_i^k ever becomes empty: dominance in T^k is a strict partial order on the finite set T_i^k, so T_i^k has an element that no other element of T_i^k dominates, and that element is not deleted at step $k+1$.)

Why finiteness matters. Take one player with $S = [0,1]$ and $u(s) = s$ for $s < 1$, $u(1) = 0$. Every $s < 1$ is strictly dominated by $(1+s)/2$, and $s = 1$ by $s = \frac{1}{2}$. Deleting everything dominated at once leaves the empty set. Deleting only the interval $(0,1)$ first leaves $\{0,1\}$, where $u(0) = u(1) = 0$ and nothing is dominated, so the process stops at $\{0,1\}$. Lemma 2 is what fails: the set of dominators of a strategy has no undominated element.

Why "strictly" matters (the manual's note, made concrete).

$1 \setminus 2$	L	R
T	1, 1	0, 0
M	1, 1	2, 1
B	0, 0	2, 1

With weak dominance: deleting T (weakly dominated by M) first makes L weakly dominated by R on rows $\{M, B\}$, and the process ends at $\{M, B\} \times \{R\}$ with payoffs $(2,1)$. Deleting B (weakly dominated by M) first makes R weakly dominated by L on rows $\{T, M\}$, and it ends at $\{T, M\} \times \{L\}$ with payoffs $(1,1)$. Lemma 1 fails for weak dominance: removing opponent strategies can remove the only profile where the inequality was strict.

(d) Interpretation. IESDS is a well-defined solution concept: what common knowledge of rationality rules out does not depend on the order in which an analyst (or the players) notices dominated strategies. That is why the slides can delete everything at once and why textbook solutions that delete one strategy at a time reach the same answer. Weak dominance does not have this property, which is one reason the course does not iterate it.

(e) Verification. `check_A1.py` draws 2,000 random finite games (two or three players, up to four strategies each, small integer payoffs so that ties are common), runs 20 random deletion orders in each (at each step a random non-empty subset of the currently dominated strategies is deleted) and checks that all of them end at the maximal-deletion set; it also checks the weak-dominance example (two orders, two different end points) and the one-player continuum example on a grid of exact rationals. *Comparison with the MWG solutions manual* (pp. 8-2–8-3): it gives the same key idea in two lines (a dominated s_i has a dominator that is not itself dominated; if s_i is not deleted now it is still dominated, and deleted, later) and the same remark that order matters for weak dominance. It does not show that the undominated dominator survives every later round, nor where finiteness enters; Lemmas 2–3 supply both. The conclusion agrees.

(f) Common mistakes.

- Arguing "a dominated strategy stays dominated, so it is deleted eventually" without checking

that its dominator is still there. The dominator may be deleted; Lemma 2 picks one that cannot be.

- Proving only that the maximal procedure deletes more at each round. The claim is about the end points of *every* order, including orders that delete one strategy at a time.
- Extending the result to weak dominance, or to infinite games without further assumptions; both counterexamples above show it fails.

Answer. In a finite game, strict dominance is inherited by smaller games (Lemma 1), and every dominated strategy has a dominator that is itself undominated (Lemma 2), which no deletion sequence ever removes. So the end point T of any deletion sequence is contained in every stage of any other sequence (Lemma 3), hence in its end point R; by symmetry $R \subseteq T$. All orders give the same surviving set S^∞. (False for weak dominance and, in general, for infinite games: examples above.)

Item 10 (item 10 of the combined list)**: MWG Exercise 8.B.5** Advanced · difficulty 5/5

Source: MWG Exercise 8.B.5, page 262. The statement is reproduced verbatim in the list (item 10); it is not repeated here.

(a) What is asked. Show that IESDS gives a unique prediction in linear Cournot with two firms, and say whether it still does with three.

(b) Method. Iterated deletion of strictly dominated strategies (MWG §8.B, Defs. 8.B.2–8.B.4 and the iterated procedure, pp. 236–241; Lecture 1, slide 29), applied to a continuum of quantities by bounding best replies, as in Tadelis §4.2.2, pp. 65–67, and the slides' Cournot remark (Lecture 2, slide 24: "eliminating outputs that are never optimal yields the sequence $\frac{1}{2}q^m, \frac{3}{4}q^m, \frac{5}{8}q^m, \cdots \to \frac{2}{3}q^m$"). The tool is the dominance lemma of the roommates exercise (Tadelis 4.6), here with a slope b. Assumptions: $a > c \geq 0$, $b > 0$, $q_i \in [0, \infty)$; write $m = (a-c)/(2b)$ for the monopoly output.

(c) Solution. Payoff and best reply. With Q_{-i} the rivals' total output,

$$u_i = \big(a - b(q_i + Q_{-i}) - c\big)\, q_i = b\, q_i\Big(\frac{a-c}{b} - Q_{-i} - q_i\Big) = b\, q_i\,(2x - q_i), \qquad x = m - \tfrac{Q_{-i}}{2},$$

since $(a-c)/b - Q_{-i} = 2m - Q_{-i} = 2x$. This is a concave parabola in q_i with peak at x: $\partial u_i/\partial q_i = b(2x - 2q_i)$, $\partial^2 u_i/\partial q_i^2 = -2b < 0$. With $q_i \geq 0$,

$$BR_i(Q_{-i}) = \max\{0,\ x\} = \max\Big\{0,\ m - \frac{Q_{-i}}{2}\Big\}.$$

Expanding, for any two outputs a' and t,

$$u_i(a', Q_{-i}) - u_i(t, Q_{-i}) = b\big[2x(a'-t) - (a'^2 - t^2)\big] = b\,(t - a')(t + a' - 2x). \tag{1}$$

So if every possible peak x lies in $[\underline{a}, \bar{a}]$, outputs above $\bar{a}$ are strictly dominated by $\bar{a}$ (both factors positive) and outputs below $\underline{a}$ by $\underline{a}$ (both negative). An output that is the unique best reply to some surviving Q_{-i} is not dominated.

(a) Two firms. Here $Q_{-i} = q_j$. If $S_j^k = [L_k, U_k]$, the best replies to $q_j \in [L_k, U_k]$ fill $[BR(U_k), BR(L_k)]$, so

$$L_{k+1} = \max\{0,\ m - \tfrac{U_k}{2}\}, \qquad U_{k+1} = m - \tfrac{L_k}{2}.$$

- Round 1: $q_j \in [0, \infty)$, so the peaks $x = m - q_j/2$ are all $\leq m$ and the best replies fill $[0, m]$ ($BR = m$ at $q_j = 0$, $BR = 0$ for $q_j \geq 2m$). By (1) every $q_i > m$ is strictly dominated by m. $S_i^1 = [0, m]$.
- Round 2: $q_j \in [0, m]$ gives peaks in $[m - m/2, m] = [m/2, m]$; every $q_i < m/2$ is dominated by $m/2$. $S_i^2 = [m/2, m]$.
- Round 3: $q_j \in [m/2, m]$ gives $[m - m/2, m - m/4] = [m/2, 3m/4]$. Round 4: $[m - 3m/8, m - m/4] = [5m/8, 3m/4]$. Round 5: $[5m/8, 11m/16]$.

From round 2 on the bounds are interior ($U_k \leq m < 2m$), so subtracting the recursions gives $U_{k+1} - L_{k+1} = \frac{1}{2}(U_k - L_k)$: the width is $m/2^{k-1} \to 0$. Adding them, the midpoint satisfies $M_{k+1} = m - M_k/2$, whose fixed point solves $q = m - q/2$, i.e. $q = 2m/3$, and $M_k - 2m/3 = (-\frac{1}{2})^{k-1}(M_1 - 2m/3) \to 0$. The nested intervals shrink to one point:

$$S_i^\infty = \Big\{\tfrac{2m}{3}\Big\} = \Big\{\frac{a-c}{3b}\Big\}, \qquad i = 1, 2.$$

IESDS predicts the Cournot outcome $q_1 = q_2 = (a-c)/(3b)$, the slides' $\frac{2}{3}q^m$ (Lecture 2, slide 24; Tadelis's example has $m = 45$ and gives 30).

(b) Three firms. Now $Q_{-i} = q_j + q_h$.

- Round 1: $Q_{-i} \in [0, \infty)$, so the best replies fill $[0, m]$ and every $q_i > m$ is strictly dominated by m. $S_i^1 = [0, m]$.
- Round 2: now $Q_{-i} \in [0, 2m]$, so the peaks are $x = m - Q_{-i}/2 \in [0, m]$ and $BR(2m) = \max\{0, m - m\} = 0$. The best replies still fill all of $[0, m]$: each $q \in [0, m]$ is the unique best reply to $Q_{-i} = 2(m - q)$, which is reached by $q_j = q_h = m - q \in [0, m]$, two strategies that survived round 1. Nothing is deleted: $S_i^2 = S_i^1$.

Since round 2 deletes nothing, the process stops: $S_i^\infty = [0, (a-c)/(2b)]$ for each firm. *No*, IESDS does not give a unique prediction with three firms. The three-firm Cournot equilibrium $q_i = (a-c)/(4b) = m/2$ (the n-firm exercise, Tadelis 5.8, with $n = 3$ and $b = 1$) lies inside, but so does every other output in $[0, m]$.

(d) Interpretation. With two firms the best-reply map is a contraction (slope $-\frac{1}{2}$): each round of "I know that you know ..." halves the range of plausible outputs, so common knowledge of rationality alone pins down the Cournot outcome. With three firms the rivals' *combined* output can be as large as $2m$, enough to make zero output a best reply, so the first-round bounds reproduce themselves and higher-order reasoning adds nothing. IESDS is weak in games with several players whose actions are strategic substitutes; Nash equilibrium (Lecture 2) still pins down the outcome, but only by assuming correct beliefs.

(e) Verification. `check_A1.py` verifies with sympy that the Cournot payoff equals $bq_i(2x - q_i)$ with $x = m - q_j/2$, the factorization (1), and that the fixed point of $q = m - q/2$ is $(a-c)/(3b)$. It runs the duopoly interval recursion for 60 rounds in exact arithmetic ($m = 1$: $[0, 1]$, $[\frac{1}{2}, 1]$, $[\frac{1}{2}, \frac{3}{4}]$, $[\frac{5}{8}, \frac{3}{4}]$, $[\frac{5}{8}, \frac{11}{16}]$, ..., $\to \frac{2}{3}$), checks that every $q \in [0, m]$ is the best reply to $Q = 2(m - q)$ with both rivals in $[0, m]$, and runs brute-force IESDS for three firms on a grid ($a - c = 12$, $b = 1$, steps of 0.5), which leaves exactly $[0, 6] = [0, m]$. *Comparison with the MWG solutions manual* (pp. 8-3–8-4): it agrees on both parts, $(a-c)/(3b)$ for two firms and $q_i \in [0, (a-c)/(2b)]$ with no unique prediction for three.

(f) Common mistakes.

- Stopping (a) at "the best replies cross at $(a-c)/(3b)$". That is the Nash argument; the exercise asks for dominance, so each round must name what is deleted and by what.
- In (b), bounding Q_{-i} by m instead of $2m$. Each rival can produce up to m, so the sum reaches $2m$, and that is what makes $q_i = 0$ a best reply.
- Concluding from (b) that the three-firm game has many Nash equilibria. It has one; only the dominance argument fails to find it.

Answer. (a) $BR_i(q_j) = \max\{0, (a-c)/(2b) - q_j/2\}$; IESDS gives $[0, m]$, $[m/2, m]$, $[m/2, 3m/4]$, $[5m/8, 3m/4]$, ..., with widths halving, so the unique survivor is $q_1 = q_2 = (a-c)/(3b)$. (b) No: with three firms the rivals' total can reach $2m$, every $q_i \in [0, (a-c)/(2b)]$ is a best reply to some surviving profile, and IESDS stops at $[0, (a-c)/(2b)]$ for each firm.

Lecture 2 — Static games II: Nash equilibrium and mixed strategies

Item 1 (item 11 of the combined list)**: Tadelis Exercise 5.6** Foundations · difficulty 1/5

Source: Tadelis Exercise 5.6, page 96. The statement is reproduced verbatim in the list (item 11); it is not repeated here.

(a) What is asked. Write Hawk-Dove as a matrix in (v, k, d), then find its pure-strategy Nash equilibria at $v = 10$, $k = 6$, $d = 4$.

(b) Method. Building a matrix from a story (Lecture 1, slides 17–19); pure-strategy Nash equilibrium (Tadelis §5.1, p. 80; Lecture 2, slide 14), found by marking best responses cell by cell (Lecture 2, slide 10; Tadelis §5.1.1, p. 81). The mixed equilibrium mentioned at the end uses the principle of indifference (Lecture 2, slide 44).

(c) Solution. Notation. $v > 0$: value of the resource; k: injury cost when both fight, $k > v/2$; d: display cost when both display, $d < v/2$ (and $d \geq 0$, a cost). $S_1 = S_2 = \{H, D\}$.

(a) The matrix. Each cell from the story:

- both H: they split v and each loses k: $v/2 - k$ each;
- both D: they split v and each pays d: $v/2 - d$ each;
- one H, one D: the hawk gets all of v, the dove leaves with no benefit and no cost, 0.

$1 \setminus 2$	H	D
H	$\frac{v}{2} - k,\ \frac{v}{2} - k$	$v,\ 0$
D	$0,\ v$	$\frac{v}{2} - d,\ \frac{v}{2} - d$

(b) Pure equilibria at $v = 10$, $k = 6$, $d = 4$. Substituting, $v/2 - k = 5 - 6 = -1$ and $v/2 - d = 5 - 4 = 1$. Best responses underlined:

$1 \setminus 2$	H	D
H	$-1,\ -1$	$\underline{10},\ \underline{0}$
D	$\underline{0},\ \underline{10}$	$1,\ 1$

- Player 1 against H: $u_1(D, H) = 0 > -1 = u_1(H, H)$, so $BR_1(H) = \{D\}$.
- Player 1 against D: $u_1(H, D) = 10 > 1 = u_1(D, D)$, so $BR_1(D) = \{H\}$.
- Player 2, by symmetry: $BR_2(H) = \{D\}$, $BR_2(D) = \{H\}$.

Cell by cell:

- (H, H): each would switch to D $(0 > -1)$. Not NE.
- (D, D): each would switch to H $(10 > 1)$. Not NE.
- (H, D): $H \in BR_1(D)$ and $D \in BR_2(H)$. NE, payoffs $(10, 0)$.
- (D, H): by symmetry, NE, payoffs $(0, 10)$.

The same holds for every admissible (v, k, d). Against H, D is strictly better because $0 > v/2 - k \iff k > v/2$. Against D, H is strictly better because $v > v/2 - d \iff d > -v/2$, true for any $d \geq 0$. So the two asymmetric profiles are the only pure equilibria under the book's assumptions; the numbers only make it concrete.

Beyond what is asked: the symmetric mixed equilibrium. Let each play H with probability p. A player is indifferent when

$$\underbrace{p\left(\tfrac{v}{2} - k\right) + (1-p)v}_{EU(H)} = \underbrace{(1-p)\left(\tfrac{v}{2} - d\right)}_{EU(D)} \iff v - p\left(\tfrac{v}{2} + k\right) = \tfrac{v}{2} - d - p\left(\tfrac{v}{2} - d\right) \iff p^* = \frac{v/2 + d}{k + d}.$$

At $(10, 6, 4)$: $p^* = 9/10$, and each expects $EU(D) = (1 - 0.9)(1) = 0.1$.

(d) Interpretation. Hawk-Dove is an anti-coordination game (the same structure as Chicken, Muñoz-Garcia Ch. 3 Ex. 1): each player wants to do the opposite of the other. The two pure equilibria are asymmetric: one animal takes everything, the other backs off. Nothing in the symmetric game says *which* animal is the hawk, which is why the biology literature looks at the symmetric mixed (population) answer: a share $p^* = 0.9$ of hawks, where hawks and doves do equally well. Fighting is frequent because the prize ($v = 10$) is large relative to the injury loss ($k - v/2 = 1$).

(e) Verification. `check_A2.py` builds the matrix from the story with symbols, checks that $BR(H) = D$ and $BR(D) = H$ for 2,000 random admissible (v, k, d), finds by brute force that the pure NE at $(10, 6, 4)$ are exactly (H, D) and (D, H), and checks the general formula for p^*, $p^* = 9/10$ and the payoff $1/10$. Tadelis gives no answer.

(f) Common mistakes.

- Giving the dove a negative payoff when it meets a hawk. The story says it leaves "with no benefits and no costs": 0.
- Forgetting that both H and both D split v: the cells are $v/2 - k$ and $v/2 - d$, not $v - k$ and $-d$.
- Answering "(H, D)" only. The game is symmetric, so (D, H) is an equilibrium too.

Answer. (a) (H, H): $(\frac{v}{2} - k, \frac{v}{2} - k)$; (H, D): $(v, 0)$; (D, H): $(0, v)$; (D, D): $(\frac{v}{2} - d, \frac{v}{2} - d)$. (b) With $v = 10$, $k = 6$, $d = 4$ the matrix is $\begin{pmatrix} -1,-1 & 10,0 \\ 0,10 & 1,1 \end{pmatrix}$; the pure-strategy Nash equilibria are (H, D) and (D, H), with payoffs $(10, 0)$ and $(0, 10)$.

Item 2 (item 12 of the combined list)**: Muñoz-Garcia Ch. 2, Exercise 2** Foundations · difficulty 1/5

Source: Muñoz-Garcia Ch. 2, Exercise 2, page 41. The statement is reproduced verbatim in the list (item 12); it is not repeated here.

(a) What is asked. Find the set of Nash equilibria of the Battle of the Sexes in Fig. 2.4.

(b) Method. Pure-strategy Nash equilibrium by best-response marking (Tadelis §5.1, pp. 80–81; Lecture 2, slides 10 and 14; the slides' own Battle of the Sexes, slide 9). Muñoz-Garcia's Chapter 2 covers pure strategies only, so the book's "set of Nash Equilibria" is the pure set; for completeness the mixed equilibrium is added by the principle of indifference (Tadelis §6.2 and Prop. 6.1, pp. 107–108; Lecture 2, slides 44–50, where the same game is solved in mixed strategies).

(c) Solution. Notation. Husband H (rows), wife W (columns); F = Football, O = Opera.

Best responses, underlined:

$H \backslash W$	F	O
F	$\underline{3},\ \underline{1}$	$0,\ 0$
O	$0,\ 0$	$\underline{1},\ \underline{3}$

- Husband: against F, $3 > 0$, so $BR_H(F) = \{F\}$; against O, $1 > 0$, so $BR_H(O) = \{O\}$.
- Wife: against F, $1 > 0$, so $BR_W(F) = \{F\}$; against O, $3 > 0$, so $BR_W(O) = \{O\}$.

Each player's best reply is to go where the other goes.

Pure equilibria, cell by cell:

- (F, F): $F \in BR_H(F)$ and $F \in BR_W(F)$. NE, payoffs $(3, 1)$.
- (O, O): NE by the same check, payoffs $(1, 3)$.
- (F, O): the husband would switch to O $(1 > 0)$. Not NE.
- (O, F): the husband would switch to F $(3 > 0)$. Not NE.

psNE $= \{(F, F), (O, O)\}$.

The mixed equilibrium. Let $p = \Pr(H \text{ plays } F)$ and $q = \Pr(W \text{ plays } F)$.

- Husband indifferent: $EU_H(F) = 3q$, $EU_H(O) = 1 \cdot (1 - q)$, so $3q = 1 - q$ and $q^* = \frac{1}{4}$.
- Wife indifferent: $EU_W(F) = 1 \cdot p$, $EU_W(O) = 3(1 - p)$, so $p = 3 - 3p$ and $p^* = \frac{3}{4}$.

At $(p^*, q^*) = (\frac{3}{4}, \frac{1}{4})$ each player is indifferent between both pure strategies, so each mix is a best reply to the other (Lecture 2, slide 44). Expected payoffs: $EU_H = 3q^* = \frac{3}{4}$ and $EU_W = p^* = \frac{3}{4}$. The full set of Nash equilibria is

$$\Big\{(F, F),\ (O, O),\ ((\tfrac{3}{4}F, \tfrac{1}{4}O),\ (\tfrac{1}{4}F, \tfrac{3}{4}O))\Big\}.$$

There are no others: $BR_H(q)$ is $p = 1$ for $q > \frac{1}{4}$, $p = 0$ for $q < \frac{1}{4}$ and all of $[0, 1]$ at $q = \frac{1}{4}$; $BR_W(p)$ is $q = 1$ for $p > \frac{3}{4}$, $q = 0$ for $p < \frac{3}{4}$ and $[0, 1]$ at $p = \frac{3}{4}$; the two graphs cross only at $(0, 0)$, $(1, 1)$ and $(\frac{3}{4}, \frac{1}{4})$.

(d) Interpretation. A coordination game with conflict: both want to be together, but they disagree on where. Nash equilibrium identifies the two conventions but cannot say which one the couple will use; that needs something outside the game (a focal point, a habit, who argued last night). In the mixed equilibrium each goes to his or her *own* favourite event with probability $\frac{3}{4}$, they end up apart with probability $\frac{3}{4} \cdot \frac{3}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{5}{8}$, and each expects $\frac{3}{4}$, less than the 1 either gets in the worse pure equilibrium.

(e) Verification. `check_A2.py` finds by brute force that the pure equilibria are exactly (F, F) and (O, O), solves both indifference conditions ($q^* = \frac{1}{4}$, $p^* = \frac{3}{4}$), checks the payoffs $\frac{3}{4}$ and the miscoordination probability $\frac{5}{8}$, and finds on a grid of mixed profiles that the mutual best responses are exactly $(0, 0)$, $(1, 1)$ and $(\frac{3}{4}, \frac{1}{4})$. *Comparison with the book's answer* (pp. 41–42): it finds the same best responses and psNE $= \{(F, F), (O, O)\}$; it does not report the mixed equilibrium, which belongs to its Chapter 3. The row label "Housband" in Fig. 2.4 is the book's typo.

(f) Common mistakes.

- Swapping the mixes: the husband's indifference pins down the *wife's* probability ($q^* = \frac{1}{4}$), not his own.
- Reading the answer as "the husband goes to football": an equilibrium is a profile, here (F, F) or (O, O), not one player's choice.
- Claiming the mixed equilibrium is "fairer" and therefore better. Both players get $\frac{3}{4}$, less than in either pure equilibrium.

Answer. Best replies: each goes where the other goes. Pure Nash equilibria: (F, F) with payoffs $(3, 1)$ and (O, O) with payoffs $(1, 3)$ (the book's answer). Including mixed strategies, there is one more: the husband plays F with probability $\frac{3}{4}$, the wife plays F with probability $\frac{1}{4}$, each expecting $\frac{3}{4}$.

Item 3 (item 13 of the combined list)**: Muñoz-Garcia Ch. 3, Exercise 1** Foundations · difficulty 2/5

Source: Muñoz-Garcia Ch. 3, Exercise 1, page 86–87. The statement is reproduced verbatim in the list (item 13); it is not repeated here.

(a) What is asked. For Chicken, find dominated strategies, best responses, the pure and the mixed Nash equilibria, and draw the best-response functions.

(b) Method. Dominance (Lecture 1, slide 23; Tadelis §4.1, pp. 59–61); best responses and pure NE (Tadelis §4.3.1, p. 69, and §5.1, p. 80; Lecture 2, slides 10 and 14); mixed strategies and expected payoffs (Tadelis §6.1, p. 105; Lecture 2, slides 42–43); mixed NE through the principle of indifference (Tadelis §6.2 and Prop. 6.1, pp. 107–108; Lecture 2, slides 44–46): a player mixes only if the other's mix makes him indifferent. The graph in (e) follows the battle-of-the-sexes example (Lecture 2, slides 47–50).

(c) Solution. Notation. St = Straight, Sw = Swerve. Matrix (Fig. 3.1): $(St, St) = (0, 0)$, $(St, Sw) = (3, 1)$, $(Sw, St) = (1, 3)$, $(Sw, Sw) = (2, 2)$. The game is symmetric.

(a) Player 1's rows over columns (St, Sw): $St = (0, 3)$, $Sw = (1, 2)$. Sw is better against St ($1 > 0$), St is better against Sw ($3 > 2$): neither row dominates the other. By symmetry the same holds for player 2. No strictly dominated strategy.

(b) Best responses underlined:

$1 \setminus 2$	St	Sw
St	$0,\ 0$	$\underline{3},\ \underline{1}$
Sw	$\underline{1},\ \underline{3}$	$2,\ 2$

$BR_1(St) = Sw$ (payoff $1 > 0$) and $BR_1(Sw) = St$ (payoff $3 > 2$). By symmetry $BR_2(St) = Sw$ and $BR_2(Sw) = St$. Each player wants the opposite of the other: an anti-coordination game.

(c) A cell is a pure NE when both strategies are best responses to each other. (St, Sw): $St = BR_1(Sw)$ and $Sw = BR_2(St)$, NE. (Sw, St): NE by symmetry. (St, St) and (Sw, Sw) are not, since in each both players would switch. The psNE are (St, Sw) with payoffs $(3, 1)$ and (Sw, St) with payoffs $(1, 3)$.

(d) Mixed equilibrium. Player 1's expected payoffs when player 2 plays St with probability q:

$$EU_1(St) = 0 \cdot q + 3(1 - q) = 3 - 3q, \tag{1}$$
$$EU_1(Sw) = 1 \cdot q + 2(1 - q) = 2 - q. \tag{2}$$

Player 1 is willing to mix only if (1) = (2):

$$3 - 3q = 2 - q \implies 1 = 2q \implies q^* = \tfrac{1}{2}.$$

Player 2's expected payoffs when player 1 plays St with probability p (her payoffs are 0 at (St, St), 3 at (Sw, St), 1 at (St, Sw), 2 at (Sw, Sw)):

$$EU_2(St) = 0 \cdot p + 3(1 - p) = 3 - 3p, \tag{3}$$
$$EU_2(Sw) = 1 \cdot p + 2(1 - p) = 2 - p. \tag{4}$$

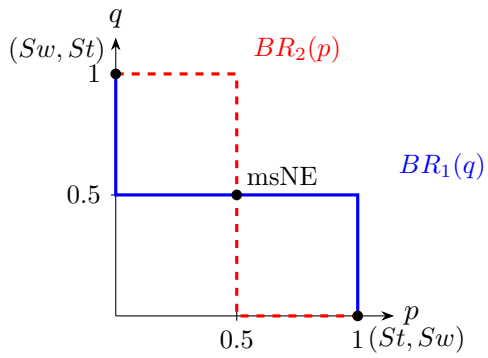
Setting (3) = (4): $3 - 3p = 2 - p$, so $p^* = \frac{1}{2}$. The msNE is

$$\sigma^* = \Big((\tfrac{1}{2}\, St, \tfrac{1}{2}\, Sw),\ (\tfrac{1}{2}\, St, \tfrac{1}{2}\, Sw) \Big),$$

and each player's expected payoff is $EU_1 = 3 - 3 \cdot \frac{1}{2} = 1.5$ (by (1) at $q^* = \frac{1}{2}$).

(e) Best-response functions. From (1)–(2), $EU_1(St) - EU_1(Sw) = 1 - 2q$, so

$$BR_1(q) = \begin{cases} p = 1 & q < \frac{1}{2}, \\ p \in [0,1] & q = \frac{1}{2}, \\ p = 0 & q > \frac{1}{2}, \end{cases} \qquad BR_2(p) = \begin{cases} q = 1 & p < \frac{1}{2}, \\ q \in [0,1] & p = \frac{1}{2}, \\ q = 0 & p > \frac{1}{2}. \end{cases}$$



The two graphs cross three times: at $(p, q) = (1, 0)$, i.e. (St, Sw); at $(0, 1)$, i.e. (Sw, St); and at $(\frac{1}{2}, \frac{1}{2})$, the msNE.

(d) Interpretation. Chicken has the shape of Hawk-Dove (Tadelis 5.6). In the mixed equilibrium each driver goes straight half of the time, so they crash with probability $\frac{1}{4}$, and each expects 1.5, less than the 2 of mutual swerving and less than the 3 of the favourable pure equilibrium. The mix is pinned by the *other* player's indifference, not by one's own payoffs (Lecture 2, slide 50). Raising one's own payoff from swerving would not change one's own mix; it would change the *opponent's*.

(e) Verification. `check_A2.py` checks (a)–(c) by brute force, solves both indifference conditions with sympy ($q^* = p^* = \frac{1}{2}$, payoff $\frac{3}{2}$ for both players, crash probability $\frac{1}{4}$), and finds on a grid of $(p, q) \in \{0, \frac{1}{100}, \ldots, 1\}^2$ that the mutual best responses are exactly $(1, 0)$, $(0, 1)$ and $(\frac{1}{2}, \frac{1}{2})$. *Comparison with the book's answer* (pp. 87–91): it agrees on all five parts (no dominated strategies, the same best responses, the two psNE, msNE with $p = q = \frac{1}{2}$, the same best-response graph, Figs. 3.4–3.6). One typo in the book: on p. 89 the rearranged indifference condition is printed "$3+3q = q+2-2q \Rightarrow q = 1/2$"; from $0q+3(1-q) = 1q+2(1-q)$ it should read $3-3q = q+2-2q$ (as printed, $3+3q = 2-q$ gives $q = -\frac{1}{4}$). The book's result $q = \frac{1}{2}$ is correct.

(f) Common mistakes.

- Solving player 1's indifference for p. Player 1's indifference condition contains only q: it pins down the *opponent's* mix.
- Copying the book's typo line $3 + 3q = q + 2 - 2q$; it gives $q = -\frac{1}{4}$, which is not a probability and should be a warning sign.
- Drawing BR_1 as a function of p on the horizontal axis. BR_1 maps q to p; with p on the horizontal axis its graph has a horizontal segment at $q = \frac{1}{2}$.

Answer. (a) No strictly dominated strategy for either player. (b) $BR_i(St) = Sw$, $BR_i(Sw) = St$. (c) psNE: (St, Sw) and (Sw, St). (d) msNE: each plays Straight with probability $\frac{1}{2}$ ($p^* = q^* = \frac{1}{2}$), expected payoff 1.5 each. (e) $BR_1(q)$: $p = 1$ for $q < \frac{1}{2}$, $p = 0$ for $q > \frac{1}{2}$, any p at $q = \frac{1}{2}$ (symmetric for player 2); the graphs cross at $(1, 0)$, $(0, 1)$ and $(\frac{1}{2}, \frac{1}{2})$.

Item 4 (item 14 of the combined list)**: Gibbons Problem 1.12** Exam · difficulty 2/5

Source: Gibbons Problem 1.12, page 51. The statement is reproduced verbatim in the list (item 14); it is not repeated here.

(a) What is asked. Find the mixed-strategy Nash equilibrium of the 2×2 game with T, B against L, R.

(b) Method. Mixed strategies and expected payoffs (Gibbons §1.3.A, pp. 29–33; Lecture 2, slides 42–43); mixed Nash equilibrium through the principle of indifference (Gibbons §1.3.B, pp. 33–44; Tadelis Prop. 6.1, p. 108; Lecture 2, slides 44–46). First check that there is no pure equilibrium, so any equilibrium must have both players mixing.

(c) Solution. Best responses, underlined:

$1 \setminus 2$	L	R
T	$\underline{2}$, 1	0, $\underline{2}$
B	1, $\underline{2}$	$\underline{3}$, 0

$BR_1(L) = \{T\}$ $(2 > 1)$, $BR_1(R) = \{B\}$ $(3 > 0)$; $BR_2(T) = \{R\}$ $(2 > 1)$, $BR_2(B) = \{L\}$ $(2 > 0)$. No cell has both payoffs underlined: the best responses cycle $(T, L) \to (T, R) \to (B, R) \to (B, L) \to (T, L)$, so there is no pure-strategy Nash equilibrium.

No equilibrium with only one player mixing. If player 2 played a pure strategy, player 1's best reply would be unique and pure, and then player 2's best reply to it would be the other column; so in equilibrium player 2 mixes, and by the same argument so does player 1. Both must be indifferent between their two pure strategies (Lecture 2, slide 44).

Player 2's mix, from player 1's indifference. Let $q = \Pr(L)$.

$$EU_1(T) = 2q + 0(1-q) = 2q, \qquad EU_1(B) = 1 \cdot q + 3(1-q) = 3 - 2q.$$

$$2q = 3 - 2q \implies 4q = 3 \implies q^* = \tfrac{3}{4}.$$

Player 1's mix, from player 2's indifference. Let $p = \Pr(T)$.

$$EU_2(L) = 1 \cdot p + 2(1-p) = 2 - p, \qquad EU_2(R) = 2p + 0(1-p) = 2p.$$

$$2 - p = 2p \implies 3p = 2 \implies p^* = \tfrac{2}{3}.$$

Equilibrium and payoffs.

$$\sigma^* = \Big((\tfrac{2}{3}\,T,\ \tfrac{1}{3}\,B),\ (\tfrac{3}{4}\,L,\ \tfrac{1}{4}\,R) \Big), \qquad u_1^* = 2q^* = \tfrac{3}{2}, \qquad u_2^* = 2p^* = \tfrac{4}{3}.$$

Uniqueness. $EU_1(T) - EU_1(B) = 4q - 3$, so $BR_1(q)$ is $p = 1$ for $q > \frac{3}{4}$, $p = 0$ for $q < \frac{3}{4}$, and $[0, 1]$ at $q = \frac{3}{4}$. $EU_2(L) - EU_2(R) = 2 - 3p$, so $BR_2(p)$ is $q = 1$ for $p < \frac{2}{3}$, $q = 0$ for $p > \frac{2}{3}$, and $[0, 1]$ at $p = \frac{2}{3}$. The graphs cross only at $(p, q) = (\frac{2}{3}, \frac{3}{4})$: it is the unique Nash equilibrium.

(d) Interpretation. This is a Matching-Pennies-like game (Gibbons §1.3.A): player 1 wants to match (T with L, B with R), player 2 wants to mismatch. Each player randomises so as to leave the *other* indifferent, which is why player 1's probabilities come from player 2's payoffs. For instance, raising player 1's payoff at (B, R) from 3 to 4 would leave p^* unchanged and raise q^* to $\frac{4}{5}$ (from $2q = 4 - 3q$).

(e) Verification. `check_A2.py` confirms by brute force that there is no pure equilibrium, solves both indifference conditions with sympy ($q^* = \frac{3}{4}$, $p^* = \frac{2}{3}$), checks the payoffs $\frac{3}{2}$ and $\frac{4}{3}$ and the comparative-statics remark, and finds on a grid of mixed profiles that $(\frac{2}{3}, \frac{3}{4})$ is the only mutual best response. Gibbons gives no answer.

(f) Common mistakes.

- Solving player 1's indifference for p. The condition $EU_1(T) = EU_1(B)$ contains only q, player 2's probability.
- Reporting $(\frac{3}{4}, \frac{2}{3})$ in the wrong order. State which probability belongs to which player and which strategy.
- Stopping at "no pure NE". Nash's theorem (Lecture 2, slide 51) guarantees a mixed one, and the question asks for it.

Answer. No pure-strategy equilibrium. The unique (mixed) Nash equilibrium: player 1 plays T with probability $\frac{2}{3}$ and B with $\frac{1}{3}$; player 2 plays L with probability $\frac{3}{4}$ and R with $\frac{1}{4}$. Expected payoffs $\frac{3}{2}$ for player 1 and $\frac{4}{3}$ for player 2.

Item 5 (item 15 of the combined list)**: Tadelis Exercise 5.8** Exam · difficulty 3/5

Source: Tadelis Exercise 5.8, page 96–97. The statement is reproduced verbatim in the list (item 15); it is not repeated here.

(a) What is asked. Write the n-firm Cournot game in normal form, find its Nash equilibrium, and find the limit of the price as $n \to \infty$.

(b) Method. Normal-form games with continuous strategies (Tadelis §3.1, pp. 46–50; Lecture 1, slide 13) and Nash equilibrium as the crossing of best responses (Tadelis §5.2.3, pp. 87–88; Lecture 2, slides 20–27, with the n-firm case on slides 26–27).

(c) Solution. Book typo. The statement prints "$Q = q_i + \cdots + q_n$". Aggregate production is the sum over all firms, $Q = q_1 + \cdots + q_n$; the subscript i is a typo for 1, and the solution uses $Q = \sum_{j=1}^{n} q_j$.

Notation. $N = \{1, \ldots, n\}$; $q_i \geq 0$ firm i's output; $Q = \sum_j q_j$; $Q_{-i} = Q - q_i$ the rivals' total; $a > c \geq 0$. The book defines $P(Q) = a - Q$ for $Q \leq a$; for $Q > a$ we take $P = 0$ (Gibbons's convention, p. 15). This only matters off equilibrium: there a firm earns $-c\,q_i \leq 0$.

(a) Normal form. Players N; strategies $S_i = [0, \infty)$; payoffs

$$u_i(q_i, q_{-i}) = \max\{0,\ a - q_i - Q_{-i}\}\, q_i - c\, q_i. \tag{1}$$

(b) Best response. Fix Q_{-i}. Raising q_i beyond $a - Q_{-i}$ only adds cost, so consider $q_i \leq a - Q_{-i}$, where $u_i = (a - c - Q_{-i} - q_i)\, q_i$ is strictly concave ($\partial^2 u_i / \partial q_i^2 = -2$). The first-order condition is

$$a - c - Q_{-i} - 2q_i = 0, \tag{2}$$

and with the constraint $q_i \geq 0$,

$$BR_i(Q_{-i}) = \max\left\{0,\ \frac{a - c - Q_{-i}}{2}\right\}. \tag{3}$$

Every firm produces in equilibrium. Suppose some firm has $q_i^* = 0$. By (3) that requires $Q_{-i}^* \geq a - c$, so some rival j produces $q_j^* > 0$ and, by (2) for j, $q_j^* = a - c - Q^*$ (because $Q_{-j}^* + 2q_j^* = Q^* + q_j^*$). But $Q^* \geq Q_{-i}^* \geq a - c$ makes $q_j^* \leq 0$, a contradiction. So every firm is interior and (2) holds for all i.

Solving. Rewrite (2) with $Q_{-i} = Q - q_i$:

$$a - c - (Q - q_i) - 2q_i = 0 \implies q_i = a - c - Q. \tag{4}$$

The right side of (4) is the same for every firm, so all firms produce the same amount, $q_i = q$. Summing (4) over the n firms:

$$Q = n(a - c - Q) \implies (n+1)Q = n(a - c) \implies Q^* = \frac{n(a-c)}{n+1}. \tag{5}$$

Substituting (5) into (4):

$$q_i^* = a - c - \frac{n(a-c)}{n+1} = \frac{a-c}{n+1}, \qquad i = 1, \ldots, n. \tag{6}$$

Since $Q^* < a - c < a$, the price is on the linear part of demand:

$$P^* = a - Q^* = a - \frac{n(a-c)}{n+1} = \frac{a + nc}{n+1}, \tag{7}$$

and each firm earns

$$u_i^* = (P^* - c)\, q_i^* = \Big(\frac{a+nc}{n+1} - c\Big)\frac{a-c}{n+1} = \Big(\frac{a-c}{n+1}\Big)^2.$$

The equilibrium is unique: the argument above showed every NE is interior, and (4)–(6) have one solution. In the slides' notation, $q^m = (a-c)/2$ and (6) reads $q_i^* = 2q^m/(n+1)$; $n=1$ gives the monopoly output q^m and $n=2$ gives $\frac{2}{3}q^m$ (slide 24).

(c) Limit. From (7),

$$P^* = \frac{a+nc}{n+1} = c + \frac{a-c}{n+1} \xrightarrow[n\to\infty]{} c,$$

and $Q^* \to a - c = q^{cp}$, the competitive output, with each firm's output and profit going to zero. This is the familiar perfectly competitive outcome: price equals marginal cost.

(d) Interpretation. The markup $P^* - c = (a-c)/(n+1)$ falls as $1/(n+1)$. Each firm's output lowers the price for everyone, but a firm counts only the loss on its own units; with more firms its own share is smaller, so it expands more aggressively and the outcome approaches competition (slide 27). Contrast with Bertrand (Lecture 2, slides 29–40), where two firms already price at cost.

(e) Verification. `check_A2.py` solves the n first-order conditions with sympy for $n = 1, \ldots, 6$ and checks (5)–(7) and the profit formula, checks that $P^* - c = (a-c)/(n+1)$ and that the limit is c, and confirms numerically ($a = 100$, $c = 10$, $n = 3$) that no unilateral deviation on a fine grid beats $q_i^* = 22.5$, including the kinked part of demand. No answer in the book.

(f) Common mistakes.

- Treating Q_{-i} as $(n-1)q$ *before* differentiating. Symmetry may be imposed only after the first-order condition is written for a general profile; otherwise firm i seems to move its rivals' output too.
- Assuming the equilibrium is symmetric without proof. The step "$q_i = a - c - Q$ for every i" shows that every equilibrium is symmetric, and the corner argument rules out firms with $q_i = 0$.
- Reading (c) as "each firm's output goes to zero, so the price goes to a". Total output rises to $a - c$, and the price falls to c.

Answer. (a) $N = \{1, \ldots, n\}$, $S_i = [0, \infty)$, $u_i = (a - \sum_j q_j)\, q_i - c\, q_i$ (with price 0 if $\sum_j q_j > a$); the printed "$Q = q_i + \cdots + q_n$" should be $q_1 + \cdots + q_n$. (b) Unique NE $q_i^* = \frac{a-c}{n+1}$, $Q^* = \frac{n(a-c)}{n+1}$, $P^* = \frac{a+nc}{n+1}$, profit $\big(\frac{a-c}{n+1}\big)^2$. (c) $P^* \to c$: the perfectly competitive price (price = marginal cost).

Item 6 (item 16 of the combined list)**: Gibbons Problem 1.6** Exam · difficulty 3/5

Source: Gibbons Problem 1.6, page 49. The statement is reproduced verbatim in the list (item 16); it is not repeated here.

(a) What is asked. Find the Cournot equilibrium with marginal costs c_1, c_2 when both costs are below $a/2$, and when $2c_2 > a + c_1$.

(b) Method. Cournot equilibrium as the crossing of best responses (Gibbons §1.2.A, pp. 14–21; Tadelis §5.2.3, eq. (5.2), p. 87; Lecture 2, slides 20–24), now with a corner: a best response can be $q_i = 0$.

(c) Solution. Notation. $q_i \geq 0$; $P(Q) = a - Q$ for $Q < a$ and 0 for $Q \geq a$ (the referenced model, Gibbons p. 15); firm i's cost $c_i q_i$, $0 \leq c_i < a$; $j \neq i$.

Best responses. For $q_i < a - q_j$ the payoff is

$$u_i(q_i, q_j) = (a - q_i - q_j - c_i)\, q_i, \tag{1}$$

strictly concave in q_i; for $q_i \geq a - q_j$ the price is zero and $u_i = -c_i q_i \leq 0$. The first-order condition of (1) is $a - c_i - q_j - 2q_i = 0$, so

$$BR_i(q_j) = \max\Big\{0,\ \frac{a - c_i - q_j}{2}\Big\}. \tag{2}$$

The corner $BR_i = 0$ occurs exactly when $q_j \geq a - c_i$: the rival alone already pushes the price down to c_i or below.

Interior candidate. If both are interior, (2) gives $2q_1 = a - c_1 - q_2$ and $2q_2 = a - c_2 - q_1$. Substituting $q_2 = (a - c_2 - q_1)/2$ into the first:

$$2q_1 = a - c_1 - \Big(\frac{a - c_2 - q_1}{2}\Big) \implies 4q_1 = 2a - 2c_1 - a + c_2 + q_1 \implies q_1^* = \frac{a - 2c_1 + c_2}{3}, \tag{3}$$

and symmetrically

$$q_2^* = \frac{a - 2c_2 + c_1}{3}. \tag{4}$$

This is a NE exactly when both (3) and (4) are positive.

Case 1: $0 < c_i < a/2$ **for** $i = 1, 2$**.** Then $a - 2c_i > 0$, so $a - 2c_i + c_j > c_j > 0$: (3) and (4) are positive and form a NE. No other NE exists. A corner NE with $q_2 = 0$ would need $q_1 = BR_1(0) = (a - c_1)/2$ and, by (2), $q_1 \geq a - c_2$, that is, $(a - c_1)/2 \geq a - c_2 \iff 2c_2 \geq a + c_1$, impossible because $2c_2 < a < a + c_1$. The same holds for $q_1 = 0$, and $(0, 0)$ fails because $BR_i(0) = (a - c_i)/2 > 0$. The unique NE is

$$q_1^* = \frac{a - 2c_1 + c_2}{3}, \quad q_2^* = \frac{a - 2c_2 + c_1}{3}, \quad Q^* = \frac{2a - c_1 - c_2}{3}, \quad P^* = \frac{a + c_1 + c_2}{3},$$

with profits $u_i^* = (P^* - c_i)\, q_i^* = (q_i^*)^2$, because $P^* - c_i = (a + c_1 + c_2 - 3c_i)/3 = (a - 2c_i + c_j)/3 = q_i^*$.

Case 2: $c_1 < c_2 < a$ **and** $2c_2 > a + c_1$**.** Now (4) is negative, $a - 2c_2 + c_1 < 0$, so there is no interior NE. Try firm 1 as a monopolist:

$$q_1 = BR_1(0) = \frac{a - c_1}{2}, \qquad q_2 = 0.$$

Firm 2's best response to it, by (2):

$$BR_2\Big(\frac{a - c_1}{2}\Big) = \max\Big\{0,\ \frac{a - c_2 - \frac{a - c_1}{2}}{2}\Big\} = \max\Big\{0,\ \frac{a + c_1 - 2c_2}{4}\Big\} = 0,$$

since $a + c_1 - 2c_2 < 0$. So $(\frac{a-c_1}{2}, 0)$ is a NE. It is the only one: the other corner, $q_1 = 0$ and $q_2 = (a - c_2)/2$, would need $(a - c_2)/2 \geq a - c_1$, that is, $2c_1 \geq a + c_2$, but $2c_1 < c_1 + c_2 < a + c_2$; and $(0, 0)$ fails as before. At the NE the price is the monopoly price of firm 1,

$$P^* = a - \frac{a - c_1}{2} = \frac{a + c_1}{2} < c_2,$$

so firm 2 cannot sell any unit profitably; firm 1 earns $\left(\frac{a-c_1}{2}\right)^2$.

(d) Interpretation. With similar costs both firms produce, and the cheaper one produces more: $q_1^* - q_2^* = c_2 - c_1$. When firm 2's cost exceeds firm 1's monopoly price, $c_2 > (a + c_1)/2$, firm 2 is priced out even though it is not shut out by any rule: firm 1 simply behaves as a monopolist and firm 2's best reply is to stay out. The condition $2c_2 > a + c_1$ is exactly "c_2 above firm 1's monopoly price". (Case 1 is sufficient, not necessary, for the interior NE: (3)–(4) are positive whenever $2c_i < a + c_j$ for both firms, even if some $c_i \geq a/2$.)

(e) Verification. `check_A2.py` derives (2)–(4) with sympy, checks $u_i^* = (q_i^*)^2$, and for random cost pairs in each case finds the NE by iterating the exact best responses (2) and confirms it equals the formulas above; it also checks on a fine grid that no unilateral deviation pays, including outputs that push Q above a. No answer in the book. (Kumar's unofficial Gibbons manual, not in `Livros/`, was not consulted.)

(f) Common mistakes.

- Reporting $q_2^* = (a - 2c_2 + c_1)/3$ in the second case. It is negative there; the formula is a Nash equilibrium only when it is non-negative, and the corner of the best response takes over.
- Dropping the $\max\{0, \cdot\}$ from the best response. Without it the second case has no solution at all.
- Checking only that firm 2 does not want to produce. Uniqueness also needs the other corner and $(0, 0)$ to be ruled out, as done above.

Answer. If $0 < c_i < a/2$: unique NE $q_1^* = \frac{a-2c_1+c_2}{3}$, $q_2^* = \frac{a-2c_2+c_1}{3}$, price $\frac{a+c_1+c_2}{3}$, profits $(q_i^*)^2$. If $c_1 < c_2 < a$ and $2c_2 > a + c_1$: unique NE $q_1^* = \frac{a-c_1}{2}$, $q_2^* = 0$; firm 1 is a monopolist at price $\frac{a+c_1}{2} < c_2$.

Item 7 (item 17 of the combined list)**: Gibbons Problem 1.7** Exam · difficulty 3/5

Source: Gibbons Problem 1.7, page 49–50. The statement is reproduced verbatim in the list (item 17); it is not repeated here.

(a) What is asked. Show that in homogeneous-good Bertrand duopoly with unit cost $c < a$ the unique Nash equilibrium is $p_1 = p_2 = c$.

(b) Method. Nash equilibrium (Gibbons §1.1.C, pp. 7–12; Lecture 2, slide 14) in a game with a discontinuous payoff, so there is no first-order condition: existence by checking every deviation, uniqueness by showing that every other profile has a profitable deviation. Gibbons §1.2.B (pp. 21–22) treats the differentiated case; the slides prove this exact result (Lecture 2, slides 29–39, "(c, c) is the unique equilibrium"), and the case split below follows them.

(c) Solution. The game. Players $1, 2$; strategies $p_i \in [0, \infty)$. Demand for firm i is

$$D_i(p_i, p_j) = \begin{cases} a - p_i & p_i < p_j, \\ \frac{1}{2}(a - p_i) & p_i = p_j, \\ 0 & p_i > p_j, \end{cases}$$

read as $\max\{0, \cdot\}$ when $p_i > a$ (no negative sales), and profit is $\pi_i = (p_i - c)\, D_i(p_i, p_j)$, with $0 \le c < a$.

Step 1: (c, c) is a Nash equilibrium. At (c, c) each firm sells $(a - c)/2$ at margin 0 and earns 0. Fix $p_j = c$ and let firm i deviate.

- $p_i > c$: $D_i = 0$, so $\pi_i = 0$. Not better.
- $p_i < c$: firm i takes the whole market, $D_i = a - p_i > 0$ (because $p_i < c < a$), at a negative margin: $\pi_i = (p_i - c)(a - p_i) < 0$. Worse.

No deviation is profitable, so (c, c) is a Nash equilibrium.

Step 2: no other profile is a Nash equilibrium. Take $(p_1, p_2) \neq (c, c)$ and label the firms so that $p_1 \le p_2$ (the game is symmetric). Every such profile falls in exactly one case.

(i) $p_1 < c$. Firm 1 sells a positive amount ($a - p_1$ or $(a - p_1)/2$, both > 0 because $p_1 < c < a$) at a negative margin, so $\pi_1 < 0$. Deviating to any $p_1' > p_2$ gives $D_1 = 0$ and $\pi_1 = 0 > \pi_1$. Not NE.

(ii) $p_1 = c < p_2$. Firm 1 earns 0. Let $\bar{p} = \min\{p_2, a\} > c$ and deviate to $p_1' = (c + \bar{p})/2 \in (c, \bar{p})$: still below p_2, so firm 1 keeps the whole market, $D_1 = a - p_1' > 0$, and $\pi_1 = (p_1' - c)(a - p_1') > 0$. Not NE.

(iii) $c < p_1 < p_2$. Firm 2 sells nothing and earns 0. Let $\bar{p} = \min\{p_1, a\} > c$ and let firm 2 deviate to $p_2' = (c + \bar{p})/2 \in (c, \bar{p})$: it undercuts firm 1, sells $a - p_2' > 0$ at a positive margin, $\pi_2 > 0$. Not NE.

(iv) $c < p_1 = p_2 = p$. If $p \ge a$, demand is 0 and both earn 0; firm 1 deviating to $(c + a)/2$ sells $(a - c)/2 > 0$ at margin $(a - c)/2 > 0$. Not NE. If $c < p < a$, each earns $\pi_1 = \frac{1}{2}(p - c)(a - p) > 0$. Firm 1 undercuts to the midpoint $p' = (p + c)/2$, which is in (c, p), takes the whole market and earns

$$(p' - c)(a - p') = \frac{p - c}{2}\Big(a - p + \frac{p - c}{2}\Big) = \frac{1}{2}(p - c)(a - p) + \frac{(p - c)^2}{4} > \frac{1}{2}(p - c)(a - p).$$

Not NE.

The cases cover every profile with $p_1 \leq p_2$ other than (c, c): $p_1 < c$ is (i); $p_1 = c$ forces $p_2 > c$, case (ii); $p_1 > c$ is (iii) or (iv). With Step 1, (c, c) is the unique Nash equilibrium. ■

(d) Interpretation. With identical products and constant marginal cost, whoever is cheaper takes the whole market, so any price above cost invites a slight undercut that roughly doubles the undercutter's sales. Competition drives price to marginal cost and profits to zero with only two firms: the *Bertrand paradox* (Lecture 2, slide 39), in sharp contrast with Cournot, where two firms keep a markup of $(a - c)/3$ (the n-firm exercise, Tadelis 5.8). The result rests on the continuous price grid, unlimited capacity and homogeneous goods; relaxing any of them (a smallest currency unit, capacity constraints, differentiation as in Gibbons §1.2.B) restores positive margins.

(e) Verification. `check_A2.py` works in exact rational arithmetic with $a = 10$, $c = 2$: it checks that no deviation on a fine price grid (including prices above a) improves on (c, c), and, for 3,000 random profiles $(p_1, p_2) \neq (c, c)$, that the deviation named in the case that applies is strictly profitable. Gibbons gives no answer.

(f) Common mistakes.

- Using a first-order condition. Profit jumps at $p_i = p_j$, so the best response does not exist when $p_j > c$ (one always wants to be slightly below p_j); the argument must be by deviations.
- Forgetting case (ii): $p_1 = c < p_2$ looks "competitive", but firm 1 can raise its price toward p_2 and earn a positive profit.
- Undercutting by "ε" without checking the gain. The midpoint deviation in (iv) shows explicitly that the undercutter earns more than its half of the market.

Answer. At (c, c) both earn 0; raising the price loses all sales and cutting it sells at a loss, so no deviation pays. Any other profile has a profitable deviation: a firm below cost raises its price above the rival's; a firm at cost facing a higher rival price raises its price toward the rival's; a firm priced out undercuts; and at a common price above cost a firm undercuts slightly and nearly doubles its profit. Hence $p_1 = p_2 = c$ is the unique Nash equilibrium.

Item 8 (item 18 of the combined list)**: Tadelis Exercise 6.8** Exam · difficulty 4/5

Source: Tadelis Exercise 6.8, page 125. The statement is reproduced verbatim in the list (item 18); it is not repeated here.

(a) What is asked. Find all pure Nash equilibria of the three-firm entry game, and the symmetric mixed equilibrium.

(b) Method. Pure NE by checking unilateral deviations (Tadelis Def. 5.1, p. 80); mixed NE by the principle of indifference (Tadelis Prop. 6.1, p. 108; Lecture 2, slides 44–46), here with three players.

(c) Solution. Notation. $I = \{1, 2, 3\}$, $S_i = \{E, O\}$ (enter, stay out). With n entrants an entrant's payoff is $\pi(n) = 150/n - 62$; staying out pays 0:

$$\pi(1) = 150 - 62 = 88, \qquad \pi(2) = 75 - 62 = 13, \qquad \pi(3) = 50 - 62 = -12.$$

(a) Pure equilibria. Classify profiles by the number n of entrants.

- $n = 0$: any firm that enters gets $\pi(1) = 88 > 0$. Not NE.
- $n = 1$: an outsider that enters gets $\pi(2) = 13 > 0$. Not NE.
- $n = 2$: each entrant gets $13 > 0$, so it does not want to leave; the outsider would get $\pi(3) = -12 < 0$ by entering, so it stays out. NE.
- $n = 3$: each gets $-12 < 0$ and would rather stay out. Not NE.

The pure NE are the three profiles in which exactly two firms enter: (E, E, O), (E, O, E), (O, E, E), each with payoffs 13 to the entrants and 0 to the outsider.

(b) Symmetric mixed equilibrium. Let each firm enter with probability p. The pure symmetric profiles are not NE (by (a), $p = 0$ and $p = 1$ fail), so $p \in (0, 1)$ and every firm must be indifferent between E and O. Fix firm i. The number of other entrants is 0, 1 or 2 with probabilities $(1-p)^2$, $2p(1-p)$, p^2, so

$$u_i(E, p) = (1-p)^2 \cdot 88 + 2p(1-p) \cdot 13 + p^2 \cdot (-12), \qquad u_i(O, p) = 0. \tag{1}$$

Expanding the first expression in (1):

$$88(1 - 2p + p^2) + 26(p - p^2) - 12p^2 = 88 - 176p + 88p^2 + 26p - 26p^2 - 12p^2 = 88 - 150p + 50p^2.$$

Indifference, $u_i(E, p) = 0$, gives

$$50p^2 - 150p + 88 = 0 \iff 25p^2 - 75p + 44 = 0,$$

$$p = \frac{75 \pm \sqrt{75^2 - 4 \cdot 25 \cdot 44}}{2 \cdot 25} = \frac{75 \pm \sqrt{5625 - 4400}}{50} = \frac{75 \pm 35}{50} \in \left\{\frac{4}{5}, \frac{11}{5}\right\}.$$

Only $p = 4/5$ is a probability. The symmetric mixed NE has each firm entering with probability $p^* = \frac{4}{5}$, and each firm's expected payoff is 0 (the payoff of staying out).

Check: at $p = 0.8$, $(0.2)^2 \cdot 88 + 2(0.8)(0.2) \cdot 13 + (0.8)^2 \cdot (-12) = 3.52 + 4.16 - 7.68 = 0$. Why only interior p can work: $u_i(E, p) = 88 - 150p + 50p^2$ is positive for $p < \frac{4}{5}$ (every firm would enter for sure) and negative for $p \in (\frac{4}{5}, 1]$ (every firm would stay out), so $p^* = \frac{4}{5}$ is the only symmetric equilibrium.

The number of entrants at $p^* = \frac{4}{5}$. It is binomial with $n = 3$: $\Pr(0) = 0.2^3 = 0.008$, $\Pr(1) = 3(0.8)(0.2)^2 = 0.096$, $\Pr(2) = 3(0.8)^2(0.2) = 0.384$, $\Pr(3) = 0.8^3 = 0.512$; the expected number of entrants is $3p^* = 2.4$.

Beyond what is asked: an asymmetric mixed equilibrium. Let firm 1 enter for sure and firms 2 and 3 enter with probability r each. Firm 2 faces one sure entrant and firm 3 mixing, so it is indifferent when $(1-r)\cdot 13 + r\cdot(-12) = 0$, i.e. $r = \frac{13}{25}$; the same holds for firm 3. Firm 1 must prefer entering: $(1-r)^2\cdot 88 + 2r(1-r)\cdot 13 + r^2\cdot(-12) = 88 - 150r + 50r^2 = 88 - 78 + 13.52 = 23.52 > 0$ at $r = 0.52$. So firm 1 entering and firms 2, 3 entering with probability $\frac{13}{25}$ (and the two permutations) are also Nash equilibria.

(d) Interpretation. Entry is a coordination problem over *who* enters: the market supports two firms, not three. The pure equilibria solve it but need the firms to know who stays out. Without such a convention, the symmetric equilibrium has each firm entering with probability 0.8. Then all three enter with probability $0.8^3 = 0.512$ and lose money, and the expected profit of the industry is zero: all the rents of the two-firm market are dissipated by miscoordination.

(e) Verification. `check_A2.py` brute-forces the 2^3 profiles (pure NE are exactly the three two-entrant profiles), solves the indifference condition with sympy (roots $\frac{4}{5}$ and $\frac{11}{5}$), and checks that at $p = \frac{4}{5}$ entering and staying out both pay 0 while $u_i(E, p)$ is positive for $p < \frac{4}{5}$ and negative for $p \in (\frac{4}{5}, 1]$. No answer in the book.

(f) Common mistakes.

- Using a binomial over *three* firms in the indifference condition. Firm i's payoff depends on how many of the *other two* enter; the firm's own choice is what is being compared.
- Keeping the root $p = \frac{11}{5}$, or discarding $\frac{4}{5}$ because the expected payoff is zero. A zero expected payoff is exactly what indifference with staying out requires.
- Concluding that the symmetric equilibrium is the only mixed one. The question restricts attention to it; asymmetric mixed equilibria exist (see above).

Answer. (a) The three profiles with exactly two entrants, (E, E, O), (E, O, E), (O, E, E): each entrant earns $75 - 62 = 13$, and the outsider would earn $50 - 62 = -12$ by entering. (b) Each firm enters with probability $p^* = \frac{4}{5}$ (from $50p^2 - 150p + 88 = 0$); expected payoff 0.

Item 9 (item 19 of the combined list)**: MWG Exercise 8.D.2** Advanced · difficulty 4/5

Source: MWG Exercise 8.D.2, page 263. The statement is reproduced verbatim in the list (item 19); it is not repeated here.

(a) What is asked. Prove that if exactly one strategy profile survives iterated removal of strictly dominated strategies, that profile is a Nash equilibrium.

(b) Method. Iterated deletion of strictly dominated strategies (MWG §8.B, pp. 236–241; Lecture 1, slide 29) and Nash equilibrium (MWG Def. 8.D.1, pp. 246–247; Lecture 2, slide 14). The proof is the slides' Proposition 2 (Lecture 2, slide 17): follow a chain of dominating strategies, along which the payoff against the survivor's profile strictly rises, until it reaches the surviving strategy. The game is finite (MWG §8.B). The same chain works whether the dominators are pure strategies (MWG Def. 8.B.2) or mixed (Def. 8.B.4); both versions are proved.

(c) Solution. Setting. $\Gamma = [I, \{S_i\}, \{u_i\}]$, every S_i finite. $S^0 = S$, and S^{k+1} removes from S^k every strategy strictly dominated given S^k (by a pure strategy in S_i^k, or by a mixed strategy $\sigma_i \in \Delta(S_i^k)$). The process stops after finitely many rounds at S^∞. Hypothesis: $S^\infty = \{s^*\}$.

Claim. For every player i and every $s_i \in S_i$, $u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*)$.

Step 1: one link of the chain. Let $s_i \neq s_i^*$. Then $s_i \notin S_i^\infty$, so s_i was deleted in some round, say round $k+1$: there is a dominator, pure $t_i \in S_i^k$ or mixed $\sigma_i \in \Delta(S_i^k)$, with

$$u_i(\sigma_i, s_{-i}) > u_i(s_i, s_{-i}) \quad \text{for every } s_{-i} \in S_{-i}^k. \tag{1}$$

Since s^* is never deleted, $s_{-i}^* \in S_{-i}^k$, and (1) holds at $s_{-i} = s_{-i}^*$:

$$\sum_{t_i \in S_i^k} \sigma_i(t_i)\, u_i(t_i, s_{-i}^*) = u_i(\sigma_i, s_{-i}^*) > u_i(s_i, s_{-i}^*).$$

A weighted average exceeds $u_i(s_i, s_{-i}^*)$ only if at least one of its terms does, so there is a pure t_i with $\sigma_i(t_i) > 0$ (in the pure case, t_i itself) such that

$$t_i \in S_i^k \quad \text{and} \quad u_i(t_i, s_{-i}^*) > u_i(s_i, s_{-i}^*). \tag{2}$$

Call t_i a *successor* of s_i. Since $t_i \in S_i^k$, it survived at least until the round in which s_i was deleted.

Step 2: the chain. Start at any $s_i^{(0)} = s_i \neq s_i^*$. While the current strategy is not s_i^*, it was deleted at some round, so by Step 1 it has a successor; let $s_i^{(m+1)}$ be a successor of $s_i^{(m)}$. By (2),

$$u_i(s_i^{(0)}, s_{-i}^*) < u_i(s_i^{(1)}, s_{-i}^*) < u_i(s_i^{(2)}, s_{-i}^*) < \cdots$$

The payoffs strictly increase, so no strategy appears twice; S_i is finite, so the chain cannot go on forever. It stops only at a strategy with no successor, that is, at a strategy that is never deleted: s_i^*, the only one in S_i^∞. Hence

$$u_i(s_i, s_{-i}^*) < u_i(s_i^*, s_{-i}^*) \quad \text{for every } s_i \neq s_i^*.$$

Step 3: conclusion. No player gains by a unilateral pure deviation from s^*, so s^* is a Nash equilibrium (in fact a *strict* one). Mixed deviations do not help either, since $u_i(\sigma_i, s_{-i}^*)$ is a weighted average of the pure payoffs, each at most $u_i(s^*)$. ■

Remarks.

- *Uniqueness.* Every Nash equilibrium survives IESDS (Lecture 2, Proposition 1, slide 16), so s^* is also the only pure Nash equilibrium; and in any mixed equilibrium every strategy in the support survives (MWG manual's argument below), so s^* is the only Nash equilibrium of any kind.
- *Finiteness* is used in Step 2: in an infinite game the chain of successors need not end at s^*, and a unique survivor need not be an equilibrium. (Outside the course books: Dufwenberg and Stegeman, *Econometrica* 2002, study this.) Infinite games in the course where IESDS has a unique survivor, such as the Cournot duopoly of MWG 8.B.5, satisfy the conclusion by direct computation.

(d) Interpretation. If common knowledge of rationality alone pins down a single profile, that profile is self-enforcing: no player can do better against what the others are predicted to do. Nash equilibrium is weaker than IESDS in what it assumes about beliefs, and this result says the two agree whenever IESDS is decisive, as in the campaigning game (Tadelis 4.7) or the discrete auction (Tadelis 4.3).

(e) Verification. `check_A2.py` generates random finite games (two players, two to four strategies, integer payoffs) until it has 1,000 with a unique IESDS survivor under pure dominance, and checks that each survivor is a strict Nash equilibrium and the only pure one; it also builds the chain of Step 2 explicitly in each game and checks that payoffs rise along it and that it ends at s_i^*. *Comparison with the MWG solutions manual* (p. 8-9): the manual proves the converse inclusion, that every (mixed) Nash equilibrium survives iterated strict dominance, and concludes that the equilibrium must be the unique survivor. That route needs an equilibrium to exist in the first place, which in a finite game is Nash's theorem (MWG Prop. 8.D.2) but is not stated in the manual's answer. The direct chain argument above needs no existence theorem. The conclusions agree.

(f) Common mistakes.

- Proving the converse (every Nash equilibrium survives IESDS) and stopping. That shows the survivor is the only *candidate*; existence of an equilibrium is a separate step.
- Assuming the strategy that dominated s_i survives to the end. It may itself be deleted later; the chain handles this.
- Comparing payoffs against all s_{-i} instead of against s^*_{-i}. Only the inequality at s^*_{-i} is needed, and it holds because s^*_{-i} is never deleted.

Answer. Let s^* be the unique survivor and $s_i \neq s_i^*$. When s_i was deleted, its dominator (or a pure strategy in its support) t_i was still present and, since s^*_{-i} was also present, $u_i(t_i, s^*_{-i}) > u_i(s_i, s^*_{-i})$. Repeating from t_i gives a chain with strictly rising payoffs against s^*_{-i}; in a finite game it ends at a never-deleted strategy, s_i^*. So $u_i(s_i^*, s^*_{-i}) > u_i(s_i, s^*_{-i})$ for all $s_i \neq s_i^*$ and all i: s^* is a (strict) Nash equilibrium.

Item 10 (item 20 of the combined list)**: MWG Exercise 8.D.6** Advanced · difficulty 5/5

Source: MWG Exercise 8.D.6, page 264. The statement is reproduced verbatim in the list (item 20); it is not repeated here.

(a) What is asked. Prove that every 2×2 game, with arbitrary payoffs, has a mixed-strategy Nash equilibrium.

(b) Method. The hint's route: describe each player's strategy by one probability, write the best-response correspondences, and show they always intersect (MWG §8.D, Def. 8.D.2 and Prop. 8.D.1, pp. 250–251, which is the slides' principle of indifference, Lecture 2, slides 44–46; Tadelis §6.4, pp. 121–122; Gibbons §1.3.B, pp. 41–44). Pure equilibria count: they are degenerate mixed strategies. Nash's theorem (slide 51) gives the result in general; the point here is a direct proof.

(c) Solution. Notation. $p = \sigma_1(a_1) \in [0,1]$, $q = \sigma_2(b_1) \in [0,1]$. Payoffs: $u_1(a_1,b_1) = u$, $u_1(a_1,b_2) = \ell$, $u_1(a_2,b_1) = w$, $u_1(a_2,b_2) = y$; $u_2(a_1,b_1) = v$, $u_2(a_1,b_2) = m$, $u_2(a_2,b_1) = x$, $u_2(a_2,b_2) = z$.

Step 1: the payoff gains are affine. Player 1's gain from a_1 over a_2 when player 2 plays q:

$$D_1(q) = [qu + (1-q)\ell] - [qw + (1-q)y] = q(u-w) + (1-q)(\ell - y). \tag{1}$$

Player 2's gain from b_1 over b_2 when player 1 plays p:

$$D_2(p) = [pv + (1-p)x] - [pm + (1-p)z] = p(v-m) + (1-p)(x-z). \tag{2}$$

Player 1's expected payoff is $u_1(p,q) = u_1(a_2,q) + p\,D_1(q)$, linear in p, so

$$BR_1(q) = \begin{cases} \{1\} & D_1(q) > 0, \\ [0,1] & D_1(q) = 0, \\ \{0\} & D_1(q) < 0, \end{cases} \qquad BR_2(p) = \begin{cases} \{1\} & D_2(p) > 0, \\ [0,1] & D_2(p) = 0, \\ \{0\} & D_2(p) < 0. \end{cases} \tag{3}$$

(p^*, q^*) is a NE iff $p^* \in BR_1(q^*)$ and $q^* \in BR_2(p^*)$.

Step 2: a sign lemma. An affine function D on $[0,1]$ satisfies exactly one of: (i) $D(t) \geq 0$ for all $t \in [0,1]$; (ii) $D(t) \leq 0$ for all t, with $D \not\equiv 0$; or (iii) $D(0)$ and $D(1)$ are nonzero with opposite signs. In case (iii) D has a unique zero $t^* = D(0)/(D(0) - D(1)) \in (0,1)$. (If $D(0)$, $D(1)$ are both ≥ 0 or both ≤ 0, then D, a convex combination of them, keeps that sign on $[0,1]$.)

Step 3: the proof, by cases on D_1.

Case A: $D_1(q) \geq 0$ for all q. Then $p = 1 \in BR_1(q)$ for every q by (3). Let $\hat{q} \in BR_2(1)$; it exists because (3) never gives an empty set. Then $1 \in BR_1(\hat{q})$ and $\hat{q} \in BR_2(1)$: $(1, \hat{q})$ is a NE.

Case B: $D_1(q) \leq 0$ for all q. Symmetrically, $p = 0$ is a best reply to every q, and $(0, \hat{q})$ with $\hat{q} \in BR_2(0)$ is a NE.

Case C: $D_1(0)$ and $D_1(1)$ have opposite signs. Player 1 is indifferent at exactly one $q^* \in (0,1)$. Solving $D_1(q^*) = 0$ from (1):

$$q^* = \frac{y - \ell}{(u-w) + (y-\ell)}. \tag{4}$$

Now look at D_2 and apply the lemma again.

- If $D_2(p) \geq 0$ for all p: $q = 1$ is a best reply to every p; with $\hat{p} \in BR_1(1)$, $(\hat{p}, 1)$ is a NE.

- If $D_2(p) \le 0$ for all p: with $\hat{p} \in BR_1(0)$, $(\hat{p}, 0)$ is a NE.
- If $D_2(0)$, $D_2(1)$ have opposite signs: player 2 is indifferent at

$$p^* = \frac{z - x}{(v - m) + (z - x)} \in (0, 1). \tag{5}$$

  At (p^*, q^*), $D_1(q^*) = 0$ gives $BR_1(q^*) = [0, 1] \ni p^*$, and $D_2(p^*) = 0$ gives $BR_2(p^*) = [0, 1] \ni q^*$. So (p^*, q^*) is a NE, in strictly mixed strategies.

The cases A, B, C exhaust all possibilities by the lemma, and each produces a NE. Hence every game of this form has a mixed-strategy Nash equilibrium. ■

(d) Interpretation. The argument is the 2×2 version of the fixed-point proof (Tadelis pp. 121–122): each best-response correspondence is a "step" from one side of the unit square to the other, with a vertical (or horizontal) segment where the player is indifferent, and two such steps must cross. Mixing is what makes this work: in pure strategies the steps have gaps (Matching Pennies has no pure NE), and the indifference segment closes the gap. Gibbons's four cases (p. 42) are the sign patterns of $D_1(0) = \ell - y$ and $D_1(1) = u - w$.

(e) Verification. `check_A2.py` checks (1), (2), (4) and (5) with sympy, then draws 20,000 random 2×2 games (half with integer payoffs in $\{-2, \dots, 2\}$, so that ties and constant gains occur often), runs the case construction above, and confirms with the exact best-response test (3) that the profile it returns is a NE every time. *Comparison with the MWG solutions manual* (pp. 8-15–8-16, where the OCR in `Livros/` prints the number as "8.0.6"): it splits into the four strict-dominance cases ($u > w,\ \ell > y$; $u < w,\ \ell < y$; $v > m,\ x > z$; $v < m,\ x < z$) plus "all other cases", where it gives the mixed equilibrium $\alpha = (y - \ell)/(u + y - \ell - w)$ for player 2 and $\beta = (z - x)/(v + z - x - m)$ for player 1. These are our (4) and (5); the conclusion agrees. Two remarks on the manual: (i) in cases 3 and 4 it says "Player 2 will play his best response" where it means player 1 (a slip: the strategies named are a_1, a_2); (ii) its "all other cases" is not complete as written: if $u = w$ and $\ell = y$ (player 1 indifferent to everything) and player 2 has no strictly dominant strategy, the formula for α is $0/0$ (any α then works, so an equilibrium still exists, but the manual's formula does not exhibit it); the same holds for β when $v = m$ and $x = z$. The case split by the sign of D_1 above has no such gap.

(f) Common mistakes.

- Proving only the generic case. The formulas (4)–(5) divide by $(u - w) + (y - \ell)$ and $(v - m) + (z - x)$, which can be zero; cases A, B and the first two bullets of case C exist to cover exactly those games.
- Forgetting that a pure equilibrium is a mixed one. The claim is existence in $\Delta(S_1) \times \Delta(S_2)$; when one player has a dominant strategy the equilibrium found is pure, and that is enough.
- Citing Nash's theorem as the proof. It is true, but the exercise asks for a direct argument from the best-response correspondences, as the hint says.

Answer. With $p = \Pr(a_1)$, $q = \Pr(b_1)$, player 1's gain from a_1 is $D_1(q) = q(u-w)+(1-q)(\ell-y)$ and player 2's gain from b_1 is $D_2(p) = p(v - m) + (1 - p)(x - z)$, both affine. If $D_1 \ge 0$ (or ≤ 0) on $[0, 1]$, player 1 has a pure strategy that is always a best reply, and pairing it with player 2's best reply to it is a NE; the same if D_2 keeps a sign. Otherwise both change sign, at $q^* = \frac{y-\ell}{(u-w)+(y-\ell)}$ and $p^* = \frac{z-x}{(v-m)+(z-x)}$ in $(0, 1)$, and (p^*, q^*) is a NE because each player is indifferent. So a NE always exists.

Lecture 3 — Dynamic games: extensive form, backward induction, SPE

Item 1 (item 21 of the combined list)**: Tadelis Exercise 7.1** Foundations · difficulty 1/5

Source: Tadelis Exercise 7.1, page 147. The statement is reproduced verbatim in the list (item 21); it is not repeated here.

(a) What is asked. Count player i's pure strategies when he has K information sets with (a) m actions at each, (b) m_k actions at information set k.

(b) Method. A pure strategy is a complete plan: one action at *every* information set, including those the plan itself never reaches (Lecture 3, slide 25; Tadelis §7.2.1, Def. 7.4, pp. 137–138). So the set of pure strategies is the Cartesian product of the action sets, $S_i = \times_{h_i \in H_i} A_i(h_i)$, and its size is the product of their sizes (counting principle).

(c) Solution. **(a)** Label player i's information sets $h^1, \ldots, h^K$. A pure strategy is a list $s_i = (s_i(h^1), \ldots, s_i(h^K))$ with $s_i(h^k) \in A_i(h^k)$ and $|A_i(h^k)| = m$. The choice at h^1 can be made in m ways; for each of them the choice at h^2 in m ways, and so on:

$$|S_i| = \underbrace{m \cdot m \cdots m}_{K \text{ times}} = m^K.$$

(b) The same argument with $|A_i(h^k)| = m_k$:

$$|S_i| = m_1 \cdot m_2 \cdots m_K = \prod_{k=1}^{K} m_k.$$

Examples. In the sequential battle of the sexes (Lecture 3, slide 26) the follower has $K = 2$ information sets with $m = 2$ actions, so $2^2 = 4$ strategies (OO, OF, FO, FF). A player with information sets of sizes $(2, 3, 4)$ has $2 \cdot 3 \cdot 4 = 24$ pure strategies.

Mixed versus behavioral strategies. The count also measures how large the two kinds of randomization are (Lecture 3, slide 27; Tadelis §7.2.2, Defs. 7.5–7.6, pp. 139–140). A mixed strategy is a point in the simplex over $\prod_k m_k$ pure strategies, which has dimension $\prod_k m_k - 1$; a behavioral strategy picks one distribution per information set, dimension $\sum_k (m_k - 1)$. With $(m_k) = (2, 3, 4)$ these are 23 and 6. Under perfect recall the smaller space already generates every distribution over outcomes (Kuhn's theorem, Lecture 3, slide 28; Tadelis p. 142).

(d) Interpretation. The count grows exponentially in the number of information sets: what makes a dynamic game "big" is the number of contingencies, not the number of moves actually played. This is why the normal form of even a small tree (the Brexit game of Lecture 4, slides 7–8, gives the EU nine strategies) is cumbersome, and why backward induction, which works node by node, is the practical tool.

(e) Verification. `check_B_L03.py` enumerates every plan with `itertools.product` for $(m_k) = (2, 2), (3, 2, 2), (2, 3, 4), (5)$ and for $m = 3$, $K = 4$, and compares the count with the formulas; it also checks the two dimensions 23 and 6. Output line: `tadelis-7-1: m^K and prod m_k confirmed by enumeration`. Tadelis prints no answer.

(f) Common mistakes.

- Adding instead of multiplying (Km or $\sum_k m_k$): that counts *actions*, which is the dimension count of behavioral strategies plus K, not the number of plans.
- Leaving out information sets that a plan makes unreachable. A strategy specifies an action there too (the off-path parts are exactly what subgame perfection tests).
- Counting decision nodes instead of information sets: nodes in the same information set share one action choice.

Answer. (a) m^K. (b) $\prod_{k=1}^{K} m_k$.

Item 2 (item 22 of the combined list)**: Muñoz-Garcia Ch. 4, Exercise 3** Foundations · difficulty 1/5

Source: Muñoz-Garcia Ch. 4, Exercise 3, page 160. The statement is reproduced verbatim in the list (item 22); it is not repeated here.

(a) What is asked. In the sequential prisoner's dilemma of Fig. 4.9: (a) player 2's best reply to each move of player 1; (b) player 1's move and the subgame-perfect equilibrium; (c) whether play differs from the simultaneous version (Ch. 1, Fig. 1.8).

(b) Method. Backward induction in a finite game of perfect information (Lecture 3, slide 30; Tadelis §8.1, Prop. 8.1, p. 153); in such games the SPE are exactly the backward-induction profiles (Lecture 3, slide 35). An SPE is written as complete strategies: player 2's strategy names a reply at *both* of her nodes, including the one off the path (Lecture 3, slide 25; Tadelis Def. 7.4, p. 138; Gibbons pp. 125–126 on outcome vs. equilibrium). All Nash equilibria of the normal form are found with the best-reply and indifference conditions (principle of indifference, Lecture 3, slides 10–12).

(c) Solution. Notation. C = Confess, N = Not confess. Payoffs (u_1, u_2) from Fig. 4.9: $(C,C) \mapsto (-5,-5)$, $(C,N) \mapsto (0,-15)$, $(N,C) \mapsto (-15,0)$, $(N,N) \mapsto (-1,-1)$. Player 2's strategy is a pair $s_2 = s_2(C)\, s_2(N)$: her reply after C, then after N. So $S_2 = \{CC, CN, NC, NN\}$ and $S_1 = \{C, N\}$.

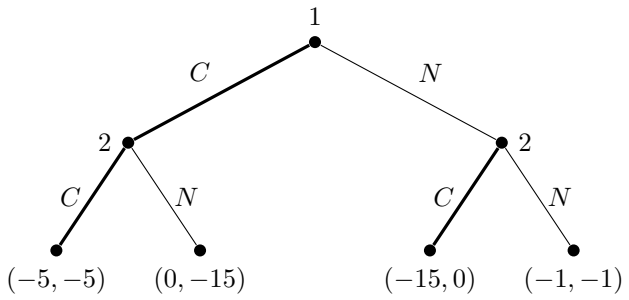


Fig. 4.9 redrawn; thick branches = backward-induction choices.

(a) Player 2's best replies.

$$\begin{aligned} &\text{after } C: \quad u_2(C,C) = -5 > -15 = u_2(C,N) \;\Rightarrow\; C; \\ &\text{after } N: \quad u_2(N,C) = 0 > -1 = u_2(N,N) \;\Rightarrow\; C. \end{aligned}$$

So $s_2^* = CC$: confess whatever player 1 did.

(b) Player 1's move and the SPE. Replacing each of player 2's nodes by its outcome,

$$u_1(C, s_2^*) = u_1(C,C) = -5 \;>\; -15 = u_1(N,C) = u_1(N, s_2^*),$$

so player 1 confesses. The SPE is

$$(s_1^*, s_2^*) = (C,\ (C \text{ after } C,\ C \text{ after } N)) = (C, CC),$$

with outcome (C,C) and payoffs $(-5,-5)$. It is unique: every comparison above is strict.

Normal form and all Nash equilibria (deepening).

	CC	CN	NC	NN
C	$\underline{-5},\ \underline{-5}$	$-5,\ \underline{-5}$	$\underline{0},\ -15$	$\underline{0},\ -15$
N	$-15,\ \underline{0}$	$\underline{-1},\ -1$	$-15,\ \underline{0}$	$-1,\ -1$

Underlined payoffs are best replies. *Pure:* only (C, CC) has both payoffs underlined. *Mixed:* let $p = \Pr_1(C)$.

- If $0 < p < 1$, player 2's payoffs are $-5p$ (CC), $-5p-(1-p)$ (CN), $-15p$ (NC), $-15p-(1-p)$ (NN); CC is the unique best reply, and against CC player 1 strictly prefers C $(-5 > -15)$, so $p = 1$. Contradiction.
- If $p = 0$, player 2's best replies are CC and NC (both reply C to N), and against either one C pays player 1 -5 or 0, more than -15. Contradiction.
- If $p = 1$, player 2's best replies are CC and CN (both reply C to C; they differ only at the unreached node). Let $w = \Pr_2(CN)$. Player 1 keeps C iff

$$u_1(C) = -5 \ \geq\ u_1(N) = -15(1-w) + (-1)w = -15 + 14w \iff w \leq \tfrac{5}{7}.$$

So the Nash equilibria are $\{(C,\ (1-w)\,CC + w\,CN) : 0 \leq w \leq \frac{5}{7}\}$, all with outcome (C, C). Only $w = 0$ is subgame perfect: for $w > 0$ player 2 plans N after N with positive probability, which is not a best reply in that subgame $(-1 < 0)$. In behavioral terms: player 2 plays C after C and, after N, plays N with probability at most $\frac{5}{7}$.

(c) Comparison. In the simultaneous game (Fig. 1.8) C is strictly dominant for each prisoner $(-5 > -15$ and $0 > -1)$, so the unique Nash equilibrium is (C, C) with $(-5, -5)$. The sequential version has the same outcome. Seeing player 1's move does not change player 2's choice, because C is her best reply to each of player 1's moves; player 1 therefore gains nothing from moving first.

(d) Interpretation. Observing the other's move matters only when the best reply depends on it. Here it does not: confessing is dominant at both of player 2's nodes, so the sequential game unravels exactly as the simultaneous one, and the Pareto-better (N, N) is still out of reach. The mixed Nash equilibria show what subgame perfection removes even here: a "promise" by player 2 to reciprocate silence (N after N) is cheap talk once player 1 has stayed silent, the Lecture 4 lesson that talk is cheap unless something makes it credible (Lecture 4, slide 11). Moving first is a commitment device only when it changes what the follower wants to do (Stackelberg, Lecture 3, slide 37).

(e) Verification. `check_B_L03.py` runs backward induction on the tree (unique result: path C, C with payoffs $(-5, -5)$), finds the pure Nash equilibria of the 2×4 normal form (only (C, CC)) and of the simultaneous game (only (C, C)), confirms on a grid of w in steps of $\frac{1}{70}$ that $(C, (1-w)CC + wCN)$ is a Nash equilibrium exactly when $w \leq \frac{5}{7}$, and that no Nash equilibrium puts weight on N. Output line: `munoz-garcia-4-3: SPE (C, CC); NE = (C, CC with prob >= 2/7)`. *Comparison with Muñoz-Garcia's answer* (pp. 160–162): it agrees on all three parts, SPNE $= \{(\text{Confess}, \text{Confess}, \text{Confess})\}$. One wording slip there: the third component is described as player 2's response "to player 1 selecting Confess (in Subgame 3)"; Subgame 3 follows Not Confess, so it is the reply to Not Confess. The book does not list the mixed Nash equilibria.

(f) Common mistakes.

- Writing the SPE as the outcome (C, C): player 2's strategy must say what she does after N too, so the SPE is (C, CC).

- Concluding that the sequential game has a different equilibrium because "player 2 can punish". Punishment after N is not a best reply, so it is not part of any SPE.
- Claiming that (C, CC) is the only Nash equilibrium: it is the only *pure* one, but player 2 may mix off the path ($w \leq \frac{5}{7}$).

Answer. (a) Player 2 confesses after Confess ($-5 > -15$) and after Not confess ($0 > -1$). (b) Player 1 confesses ($-5 > -15$); SPE $= (C, (C, C))$, outcome (C, C), payoffs $(-5, -5)$. Nash equilibria: $(C,\ (1-w)CC + wCN)$, $0 \leq w \leq \frac{5}{7}$; only $w = 0$ is subgame perfect. (c) No difference: the simultaneous game's unique equilibrium is also (C, C), since Confess is strictly dominant at every node.

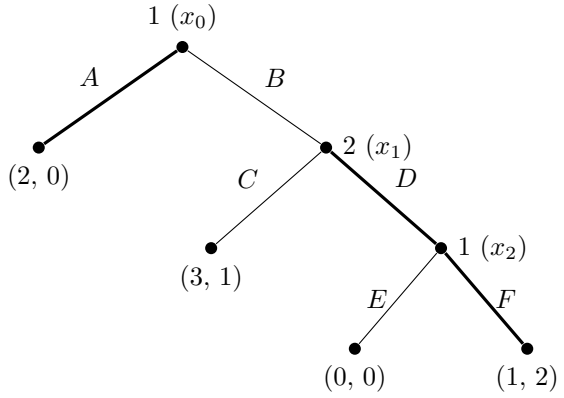
Item 3 (item 23 of the combined list)**: Tadelis Exercise 7.2** Foundations · difficulty 2/5

Source: Tadelis Exercise 7.2, page 147. The statement is reproduced verbatim in the list (item 23); it is not repeated here.

(a) What is asked. For the three-move game A/B, then C/D, then E/F: (a) its tree and type of information; (b) terminal nodes and information sets; (c) pure strategies per player; (d) with the given payoffs, all Nash equilibria and which one is more appealing.

(b) Method. Game tree and information sets (Lecture 3, slides 20–23; Tadelis §7.1, Defs. 7.1–7.2, pp. 133–135); perfect information (slide 24; Def. 7.3, p. 136); pure strategies as complete plans (slide 25; Def. 7.4, p. 138); the normal form of an extensive-form game (Tadelis §7.2.3, pp. 142–145); Nash equilibria by best-reply marking and indifference; the "appealing" one by sequential rationality and backward induction (slides 29–30; Tadelis §8.1, pp. 152–153).

(c) Solution. **(a) Tree.** Payoffs (u_1, u_2) from part (d) are written at the leaves.



Thick branches = backward-induction choices (part (d)).

Each decision node is alone in its information set (the mover always knows the history), so it is a game of **perfect information**.

(b) Counts. Terminal nodes: the leaves after A, C, E, F, so **4**. Information sets: $\{x_0\}$, $\{x_2\}$ for player 1 and $\{x_1\}$ for player 2, so **3** (two for player 1, one for player 2).

(c) Pure strategies. Player 1 picks an action at x_0 and at x_2, even when his first action (A) makes x_2 unreachable: $S_1 = \{AE, AF, BE, BF\}$, **4** strategies (2×2, item "Tadelis 7.1"). Player 2 has one information set: $S_2 = \{C, D\}$, **2** strategies.

(d) Nash equilibria. Normal form (rows player 1, columns player 2):

	C	D
AE	2, $\underline{0}$	$\underline{2}$, $\underline{0}$
AF	2, $\underline{0}$	$\underline{2}$, $\underline{0}$
BE	$\underline{3}$, $\underline{1}$	0, 0
BF	$\underline{3}$, 1	1, $\underline{2}$

Pure Nash equilibria (both payoffs underlined): (AE, D), (AF, D) and (BE, C). (BF, C) fails because against BF player 2 prefers D ($2 > 1$).

All Nash equilibria. Let $q = \Pr_2(C)$. Player 1's payoffs are

$$u_1(AE) = u_1(AF) = 2, \qquad u_1(BE) = 3q, \qquad u_1(BF) = 3q + (1-q) = 1 + 2q.$$

- $q < \frac{1}{2}$: $2 > 1 + 2q > 3q$, so player 1 plays A (any mix of AE, AF); player 2's node is not reached, she is indifferent, and any such q is a best reply. *Equilibria.*
- $q = \frac{1}{2}$: AE, AF, BF tie at 2. If BF had positive weight, player 2's node would be reached only after BF, where D is strictly better ($2 > 1$), forcing $q = 0$. So player 1 stays on $\{AE, AF\}$. *Equilibria.*
- $\frac{1}{2} < q < 1$: $1 + 2q > \max\{2, 3q\}$, so BF is the unique best reply, and then $q = 0$. Contradiction.
- $q = 1$: BE and BF tie at $3 > 2$. With $w = \Pr_1(BF)$ (the rest on BE), player 2 gets 1 from C and $2w$ from D, so C is a best reply iff $w \leq \frac{1}{2}$. *Equilibria.*

Hence the set of Nash equilibria is

$$\{(\text{any mix of } AE, AF;\ q) : 0 \leq q \leq \tfrac{1}{2}\} \ \cup\ \{((1-w)\,BE + w\,BF;\ C) : 0 \leq w \leq \tfrac{1}{2}\},$$

with outcome A and payoffs $(2, 0)$ in the first family and outcome (B, C), payoffs $(3, 1)$, in the second.

The appealing one. Backward induction: at x_2 player 1 picks F ($1 > 0$); at x_1 player 2, anticipating F, compares $C \mapsto 1$ with $D \mapsto 2$ and picks D; at x_0 player 1 compares $A \mapsto 2$ with $B \mapsto 1$ and picks A. The backward-induction profile, and the unique SPE, is (AF, D), outcome A, payoffs $(2, 0)$. It is the appealing one because every player's plan is optimal at every node, reached or not. The other two pure equilibria rely on player 1's plan to play E at x_2, which gives him $0 < 1$:

- in (BE, C) the plan E is a *threat* that scares player 2 into C; if x_2 were reached, player 1 would not carry it out;
- in (AE, D) the plan E is never tested and does not change the outcome, but it is still not sequentially rational.

(d) Interpretation. Player 1 would like to commit to E ("if you play D I will burn both our payoffs"), because that threat earns him 3 instead of 2. He cannot: at x_2 the damage is in the past and F is better. Nash equilibrium accepts the threat because x_2 is off the path; sequential rationality rejects it. This is the Brexit pattern of Lecture 4 (slides 8–13): several Nash equilibria are held up by threats nobody would execute, and only the backward-induction profile survives.

(e) Verification. `check_B_L03.py` runs backward induction on the tree (unique solution, outcome A, payoffs $(2, 0)$; continuation D then F after B), finds the three pure Nash equilibria of the 4×2 matrix, scans q on a grid of step $\frac{1}{120}$ plus $\frac{1}{2}, \frac{2}{3}$ (Nash equilibria exist exactly for $q \leq \frac{1}{2}$ and $q = 1$), and tests the $w \leq \frac{1}{2}$ condition on a grid. Output line: `tadelis-7-2: SPE (AF, D); pure NE (AE,D), (AF,D), (BE,C)`. Tadelis prints no answer.

(f) Common mistakes.

- Giving player 1 only three strategies (A, BE, BF). That is the reduced normal form; as Tadelis defines strategies, A must be completed with a plan at x_2.
- Calling (BE, C) the "best" equilibrium because it gives player 1 more. A Nash equilibrium is appealing when its threats are credible, not when it pays one player more.

- Counting 3 information sets as "3 decision nodes per player" or counting terminal nodes as information sets.

Answer. (a) Tree above; perfect information. (b) 4 terminal nodes; 3 information sets (2 for player 1, 1 for player 2). (c) Player 1: 4 (AE, AF, BE, BF); player 2: 2 (C, D). (d) Pure Nash equilibria (AE, D), (AF, D), (BE, C) (plus mixed ones: player 1 on A with $\Pr(C) \leq \frac{1}{2}$, or $(1-w)BE + wBF$ with $w \leq \frac{1}{2}$ against C). The appealing one is (AF, D), the unique SPE: the others rest on the non-credible plan E.

Item 4 (item 24 of the combined list)**: Tadelis Exercise 8.5** Exam · difficulty 3/5

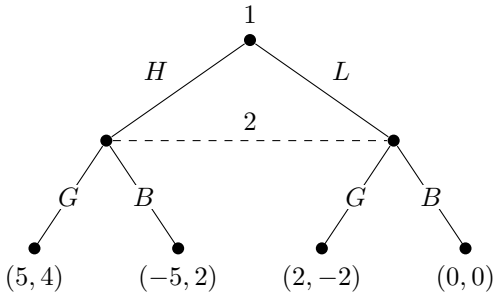
Source: Tadelis Exercise 8.5, page 170. The statement is reproduced verbatim in the list (item 24); it is not repeated here.

(a) What is asked. (a) Tree, Nash and subgame-perfect equilibria of the simultaneous task/training game; (b) tree and number of proper subgames once the CEO first decides whether to adopt (outside payoff $(1,1)$); (c) Nash and subgame-perfect equilibria of that larger game.

(b) Method. Simultaneous moves as an information set (Lecture 3, slides 22–23; Tadelis Def. 7.2, p. 135); mixed equilibria by the principle of indifference (Lecture 3, slides 10–12); proper subgames (Lecture 4, slides 14–16; Tadelis Def. 8.2, p. 154, which counts the whole game); SPE by replacing the subgame with each of its Nash equilibria and optimizing at the root (Lecture 4, slide 17; Tadelis §8.2, Def. 8.3, p. 157). The structure is the voluntary battle of the sexes (Lecture 3, slide 36; Lecture 4, slide 19; Tadelis pp. 154–159).

(c) Solution. Notation. CEO = player 1 (H or L), manager = player 2 (G or B); in (b) the CEO first chooses N (no adoption) or A (adopt). $p = \Pr_1(H)$, $q = \Pr_2(G)$ in the simultaneous game.

(a) Tree and equilibria of the simultaneous game.



Pure: best replies $BR_1(G) = H$ $(5 > 2)$, $BR_1(B) = L$ $(0 > -5)$, $BR_2(H) = G$ $(4 > 2)$, $BR_2(L) = B$ $(0 > -2)$. Pure equilibria: (H, G) with $(5, 4)$ and (L, B) with $(0, 0)$.

Mixed: the manager is indifferent when

$$u_2(G) = 4p - 2(1-p) \;=\; 2p + 0\cdot(1-p) = u_2(B) \;\Rightarrow\; 6p - 2 = 2p \;\Rightarrow\; p = \tfrac{1}{2}, \tag{1}$$

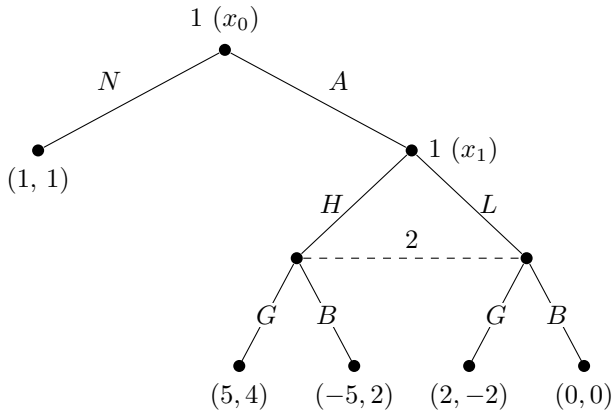
and the CEO is indifferent when

$$u_1(H) = 5q - 5(1-q) \;=\; 2q + 0\cdot(1-q) = u_1(L) \;\Rightarrow\; 10q - 5 = 2q \;\Rightarrow\; q = \tfrac{5}{8}. \tag{2}$$

Payoffs in the mixed equilibrium: $u_1 = 2q = \frac{5}{4}$, $u_2 = 2p = 1$.

Subgame perfection: the only proper subgame is the game itself (player 2's nodes share an information set, so no subgame starts at them). Every Nash equilibrium is therefore subgame perfect: (H, G), (L, B) and $(\frac{1}{2}H + \frac{1}{2}L,\ \frac{5}{8}G + \frac{3}{8}B)$.

(b) The entire game. The CEO moves first at x_0: N ends the game with $(1, 1)$; A leads to x_1, where the game of (a) is played.



Candidate roots of proper subgames are singleton information sets: x_0 and x_1. Player 2's nodes are excluded (each would cut her information set). So there are **two proper subgames**: the whole game and the adoption subgame starting at x_1 (one, if the whole game is not counted, as in Gibbons).

(c) Equilibria of the entire game. The CEO now has two information sets. His strategies are $S_1 = \{NH, NL, AH, AL\}$; the manager's are $S_2 = \{G, B\}$.

	G	B
NH	1, $\underline{1}$	$\underline{1}$, $\underline{1}$
NL	1, $\underline{1}$	$\underline{1}$, $\underline{1}$
AH	$\underline{5}$, $\underline{4}$	−5, 2
AL	2, −2	0, $\underline{0}$

Pure Nash equilibria: (AH, G), (NH, B), (NL, B).

All Nash equilibria. Given q, the CEO's payoffs are 1 (NH, NL), $5q - 5(1-q) = 10q - 5$ (AH) and $2q$ (AL). The manager's gain from G over B is 0 against $N\cdot$, $+2$ against AH and -2 against AL.

- $q < \frac{1}{2}$: $1 > 2q$ and $1 > 10q - 5$, so the CEO plays N (any mix of NH, NL); the manager is never reached and any q is a best reply. *Equilibria.*
- $q = \frac{1}{2}$: N and AL tie at 1 (AH gives 0). For $0 < q < 1$ the manager must be indifferent, i.e. $-2\Pr(AL) = 0$: the CEO stays on N. *Equilibria.*
- $\frac{1}{2} < q < \frac{5}{8}$: AL is the unique best reply ($2q > 1$, $2q > 10q - 5$); then the manager strictly prefers B, $q = 0$. Contradiction.
- $q = \frac{5}{8}$: AH and AL tie at $\frac{5}{4} > 1$; the manager is indifferent iff $2\Pr(AH) - 2\Pr(AL) = 0$, i.e. $\frac{1}{2}AH + \frac{1}{2}AL$. *Equilibrium.*
- $\frac{5}{8} < q \leq 1$: AH is the unique best reply; then the manager plays G, $q = 1$. *Equilibrium* (AH, G) at $q = 1$ only.

So the Nash equilibria are

$$\{(\text{any mix of } NH, NL;\ q) : q \leq \tfrac{1}{2}\} \cup \{(\tfrac{1}{2}AH + \tfrac{1}{2}AL;\ \tfrac{5}{8})\} \cup \{(AH;\ G)\}.$$

Subgame-perfect equilibria. In the adoption subgame play must be one of its three Nash equilibria; replace x_1 by its value and optimize at x_0:

Subgame equilibrium	Value (u_1, u_2)	CEO at x_0	SPE
(H, G)	$(5, 4)$	A $(5 > 1)$	(AH, G), payoffs $(5, 4)$
(L, B)	$(0, 0)$	N $(1 > 0)$	(NL, B), payoffs $(1, 1)$
$(\frac{1}{2}, \frac{5}{8})$	$(\frac{5}{4}, 1)$	A $(\frac{5}{4} > 1)$	$(A,\ \frac{1}{2}H + \frac{1}{2}L;\ \frac{5}{8}G)$, payoffs $(\frac{5}{4}, 1)$

Three SPE. The Nash equilibria that are not subgame perfect are (NH, B) and every profile of the first family other than (NL, B): after adoption they prescribe (H, B) or (L, q) with $q > 0$, or a mix of H and L that is not $\frac{1}{2}$–$\frac{1}{2}$ with $q \leq \frac{1}{2}$, none of which is a Nash equilibrium of the subgame. For instance, in (NH, B) the CEO's plan H against B gives him $-5 < 0$.

(d) Interpretation. Whether a new technology is adopted depends on what each side expects the other to do after adoption, not only on its potential $(5, 4)$. If the CEO expects the manager to train badly he does not adopt, and the expectation is self-fulfilling. Adoption has an option value only if the subgame is expected to coordinate on (H, G), or at least on the mixed play, which gives $\frac{5}{4} > 1$. The outside option $(1, 1)$ works like the UK's no-deal option in the Brexit case (Lecture 4, slide 11): it sets the bar that the continuation must clear, and it eliminates the (H, B) plan, which no rational CEO would carry out after adopting.

(e) Verification. `check_B_L03.py` finds the pure equilibria of the subgame, solves (1)–(2) with sympy, checks the mixed profile and its payoffs $(\frac{5}{4}, 1)$ exactly, finds the pure equilibria (NH, B), (NL, B), (AH, G) of the 4×2 matrix, scans q on a grid of step $\frac{1}{160}$ plus $\frac{1}{2}, \frac{3}{5}, \frac{5}{8}$ (equilibria exactly at $q \leq \frac{1}{2}$, $q = \frac{5}{8}$, $q = 1$), and replaces the subgame by each equilibrium value to get the three SPE. Output line: `tadelis-8-5: subgame NE (H,G), (L,B), (1/2, 5/8); three SPE`. Tadelis prints no answer.

(f) Common mistakes.

- Treating (NH, B) and (NL, B) as the same equilibrium: they are different strategies, and only (NL, B) is subgame perfect.
- Forgetting the mixed equilibrium of the subgame, and with it the third SPE (adoption with mixed play).
- Saying the game of (a) has subgames at player 2's nodes; they belong to one information set.

Answer. (a) Nash (all subgame perfect, since the only subgame is the game itself): (H, G), (L, B), $(\frac{1}{2}H + \frac{1}{2}L,\ \frac{5}{8}G + \frac{3}{8}B)$ with payoffs $(5, 4)$, $(0, 0)$, $(\frac{5}{4}, 1)$. (b) Tree above; two proper subgames (the whole game and the post-adoption game). (c) Pure Nash: (AH, G), (NH, B), (NL, B); all Nash: N (any mix of NH, NL) with $\Pr(G) \leq \frac{1}{2}$, $(\frac{1}{2}AH + \frac{1}{2}AL, \frac{5}{8})$, (AH, G). SPE: (AH, G), (NL, B) and $(A, \frac{1}{2}H + \frac{1}{2}L;\ \frac{5}{8}G)$.

Item 5 (item 25 of the combined list)**: Tadelis Exercise 8.10** Exam · difficulty 3/5

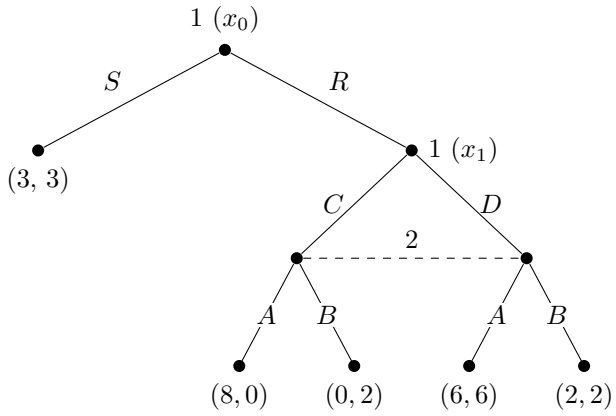
Source: Tadelis Exercise 8.10, page 172. The statement is reproduced verbatim in the list (item 25); it is not repeated here.

(a) What is asked. Player 1 either plays safe (S, payoffs $(3,3)$) or risks (R) a simultaneous 2×2 game with player 2. (a) Draw a tree and count its proper subgames; (b) say whether other trees would do; (c) write the normal form; (d) find all Nash and subgame-perfect equilibria.

(b) Method. Extensive form with an information set for the simultaneous stage (Lecture 3, slides 22–23; Tadelis §7.1, Def. 7.2, p. 135); proper subgames (Lecture 4, slides 14–16; Tadelis Def. 8.2, p. 154, which counts the whole game); strategies as complete plans (Lecture 3, slide 25; Def. 7.4, p. 138); SPE by replacing the subgame with each of its Nash equilibria (Lecture 4, slide 17; Tadelis §8.2, Def. 8.3, p. 157; the voluntary battle of the sexes, Lecture 4, slide 19, has the same structure). Mixed equilibria by the principle of indifference (Lecture 3, slides 10–12; Tadelis §6.1–6.2, pp. 101–114).

(c) Solution. Notation. After R, player 1 chooses C or D and player 2 chooses A or B without seeing it. $p = \Pr_1(C)$, $q = \Pr_2(A)$ in the subgame.

(a) Tree and subgames.



Player 2's two nodes form one information set (dashed). Candidate roots of a proper subgame are the singleton nodes x_0 and x_1; player 2's nodes fail (each would cut her information set). So the game has **two proper subgames**: the whole game and the simultaneous game that starts at x_1 (one, if, like Gibbons, one does not count the whole game).

(b) Other trees. Yes. (i) In the simultaneous stage, player 2 can be drawn as moving first at x_1 with player 1's two nodes joined in an information set; the strategies, normal form and subgames are unchanged. (ii) A different tree also represents the story: player 1 picks S, RC or RD at the root, and player 2 has one information set over the nodes after RC and RD. Its normal form is the same up to merging SC and SD (it is the *reduced* normal form), but it has only one proper subgame, the whole game, so subgame perfection there has no bite: every Nash equilibrium is subgame perfect. The choice of tree matters for SPE whenever it changes which subgames exist; tree (i) and the drawing in (a) are the faithful ones, since R is a real decision taken before the stage game.

(c) Normal form. Player 1 has two information sets, so $S_1 = \{SC, SD, RC, RD\}$ (first letter at x_0, second at x_1); $S_2 = \{A, B\}$:

	A	B
SC	3, $\underline{3}$	$\underline{3}$, $\underline{3}$
SD	3, $\underline{3}$	$\underline{3}$, $\underline{3}$
RC	$\underline{8}$, 0	0, $\underline{2}$
RD	6, $\underline{6}$	2, 2

(d) Equilibria.

Step 1: the subgame at x_1. Best replies: $BR_1(A) = C$ ($8 > 6$), $BR_1(B) = D$ ($2 > 0$), $BR_2(C) = B$ ($2 > 0$), $BR_2(D) = A$ ($6 > 2$). They cycle, so there is no pure equilibrium. Player 2 is indifferent when

$$u_2(A) = 0 \cdot p + 6(1-p) \;=\; 2p + 2(1-p) = u_2(B) \;\Rightarrow\; 6 - 6p = 2 \;\Rightarrow\; p = \tfrac{2}{3}, \tag{3}$$

and player 1 is indifferent when

$$u_1(C) = 8q \;=\; 6q + 2(1-q) = u_1(D) \;\Rightarrow\; 8q = 2 + 4q \;\Rightarrow\; q = \tfrac{1}{2}. \tag{4}$$

The unique equilibrium of the subgame is $(p, q) = (\frac{2}{3}, \frac{1}{2})$, with values

$$u_1 = 8\left(\tfrac{1}{2}\right) = 4, \qquad u_2 = 6\left(1 - \tfrac{2}{3}\right) = 2.$$

Step 2: subgame-perfect equilibrium. Replace x_1 by $(4, 2)$. At x_0, R gives $4 > 3$, so player 1 risks. Since the subgame has a unique equilibrium, the SPE is unique:

$$\sigma^{\text{SPE}}: \text{ player 1 plays } R \text{ and then } C \text{ w.p. } \tfrac{2}{3};\ \text{player 2 plays } A \text{ w.p. } \tfrac{1}{2}, \qquad \text{payoffs } (4, 2).$$

In the normal form this is player 1 mixing $\frac{2}{3}RC + \frac{1}{3}RD$ against $\frac{1}{2}A + \frac{1}{2}B$. There is *no* pure-strategy SPE.

Step 3: all Nash equilibria. Expected payoffs against q:

Player 1's strategy	u_1 given q	player 2's gain $u_2(A) - u_2(B)$ against it
SC, SD	3	0
RC	$8q$	$0 - 2 = -2$
RD	$6q + 2(1-q) = 2 + 4q$	$6 - 2 = +4$

- $q \le \frac{1}{4}$: $3 \ge 2 + 4q$ and $3 > 8q$, so S is a best reply (strictly, for $q < \frac{1}{4}$). With player 1 on $\{SC, SD\}$ in any mix, player 2 is indifferent, so any such q is a best reply. *Nash equilibria.* (At $q = \frac{1}{4}$, RD also ties at 3, but any weight on it would make A strictly better, forcing $q = 1$; so player 1 stays on S.)
- $\frac{1}{4} < q < \frac{1}{2}$: RD is the unique best reply ($2 + 4q > 3$ and $2 + 4q > 8q$); then player 2 strictly prefers A, i.e. $q = 1$. Contradiction.
- $q = \frac{1}{2}$: RC and RD tie at $4 > 3$; player 2 is indifferent only if $-2\Pr(RC) + 4\Pr(RD) = 0$, i.e. $\frac{2}{3}RC + \frac{1}{3}RD$. *The SPE above.*
- $q > \frac{1}{2}$: RC is the unique best reply; then player 2 plays B, $q = 0$. Contradiction.

So the Nash equilibria are

$$\{\,(\text{any mix of } SC, SD;\ q) : 0 \le q \le \tfrac{1}{4}\,\} \;\cup\; \{\,(\tfrac{2}{3}RC + \tfrac{1}{3}RD;\ q = \tfrac{1}{2})\,\},$$

and the *pure* Nash equilibria are (SC, B) and (SD, B), both with payoffs $(3, 3)$. None of the first family is subgame perfect: each asks player 2 to play A with probability at most $\frac{1}{4}$ in a subgame whose only equilibrium has $q = \frac{1}{2}$. The "threat" to play B deters R, but it is not credible once R has been played.

(d) Interpretation. Playing safe is supported only by player 2's threat to play B in the risky game. Once the risky game starts, both must play its (mixed) equilibrium, which gives player 1 an expected 4, more than the safe 3, so he takes the risk. Player 2 is worse off ($2 < 3$): she would like to commit to B, but cannot; a public, costly commitment (Lecture 4, slide 11: observable, sunk moves) would be her only way to keep player 1 on the safe path. The exercise also shows that subgame perfection can select a *mixed* continuation, so the credibility test applies to probabilities, not only to pure threats.

(e) Verification. `check_B_L03.py` confirms that the subgame has no pure equilibrium, solves (3)–(4) with sympy, finds the pure Nash equilibria (SC, B), (SD, B) of the 4×2 matrix, and scans q over a grid of 241 points plus the critical values $\frac{1}{4}, \frac{3}{8}, \frac{1}{2}$: a Nash equilibrium exists exactly at $q \in [0, \frac{1}{4}]$ and $q = \frac{1}{2}$, and at $q = \frac{1}{2}$ the mix $(\frac{2}{3}, \frac{1}{3})$ on (RC, RD) passes the exact Nash test while $(\frac{1}{2}, \frac{1}{2})$ fails. Output line: `tadelis-8-10: unique SPE (R; 2/3 C; 1/2 A), payoffs (4, 2).` Tadelis prints no answer.

(f) Common mistakes.

- Looking for a pure SPE: the subgame has none, so the SPE must use its mixed equilibrium.
- Swapping the indifference conditions: p (player 1's mix) is pinned down by *player 2's* indifference, and vice versa.
- Counting a subgame at each of player 2's nodes, or forgetting that (SC, B) and (SD, B) are two distinct Nash equilibria.

Answer. (a) Tree above; two proper subgames (the whole game and the simultaneous game after R). (b) Yes: player 2 can be drawn first in the stage game; a tree with root choices $\{S, RC, RD\}$ gives the same reduced normal form but no proper subgame other than the whole game. (c) Matrix above, $S_1 = \{SC, SD, RC, RD\}$. (d) Pure Nash: (SC, B), (SD, B); all Nash: player 1 on S (any mix of SC, SD) with $\Pr(A) \leq \frac{1}{4}$, plus $(\frac{2}{3}RC + \frac{1}{3}RD,\ \frac{1}{2}A + \frac{1}{2}B)$. Unique SPE: R, then C w.p. $\frac{2}{3}$; A w.p. $\frac{1}{2}$; payoffs $(4, 2)$. No pure SPE.

Item 6 (item 26 of the combined list)**: Tadelis Exercise 7.4** Exam · difficulty 3/5

Source: Tadelis Exercise 7.4, page 148. The statement is reproduced verbatim in the list (item 26); it is not repeated here.

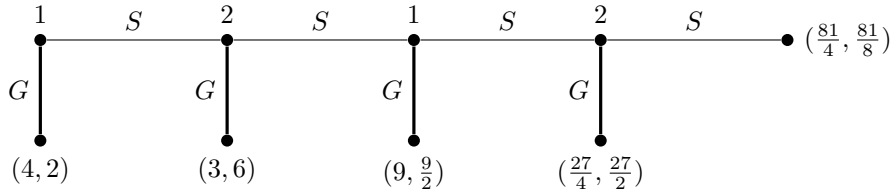
(a) What is asked. In the four-move grab/share game with a pot of \$6 growing by $\frac{3}{2}$: (a) tree and type of information; (b) terminal nodes and information sets; (c) pure strategies; (d) Nash equilibria and how many outcomes they support; (e) the same when sharing at the last node splits the pot equally.

(b) Method. Tree, information sets and perfect information (Lecture 3, slides 20–24; Tadelis §7.1, pp. 133–136); strategies as complete plans (slide 25; Def. 7.4, p. 138); the normal form and its Nash equilibria (Tadelis §7.2.3, pp. 142–145); backward induction to identify the sequentially rational one (slides 29–30; Tadelis §8.1, pp. 152–153; the centipede game itself is Tadelis §8.3.1, pp. 159–160). Nash equilibrium constrains play only on the path, so the outcome argument proceeds forward from the last node that is reached with positive probability.

(c) Solution. Pot and payoffs. The pot before move $k = 1, 2, 3, 4$ is $6 \cdot (\frac{3}{2})^{k-1}$: 6, 9, $\frac{27}{2}$, $\frac{81}{4}$; after a share at move 4 it is $\frac{243}{8}$. A grab gives the mover $\frac{2}{3}$ of the current pot and the other player $\frac{1}{3}$. Payoffs (u_1, u_2):

Terminal history	pot	u_1	u_2
G (1 grabs at move 1)	6	4	2
SG (2 grabs at move 2)	9	3	6
SSG (1 grabs at move 3)	$\frac{27}{2}$	9	$\frac{9}{2}$
$SSSG$ (2 grabs at move 4)	$\frac{81}{4}$	$\frac{27}{4}$	$\frac{27}{2}$
$SSSS$ (2 shares at move 4)	$\frac{243}{8}$	$\frac{81}{4}$	$\frac{81}{8}$

(a) Tree. Perfect information: every information set is a single node.



Thick branches = backward-induction choices (part (d)).

(b) Counts. Terminal nodes: **5** (four grabs and the final share). Information sets: **4**, two per player (moves 1 and 3 for player 1, moves 2 and 4 for player 2).

(c) Pure strategies. Each player has 2 information sets with 2 actions: $2^2 = 4$ pure strategies each. Write $s_1 = (\text{move } 1, \text{move } 3)$ and $s_2 = (\text{move } 2, \text{move } 4)$: $S_1 = S_2 = \{GG, GS, SG, SS\}$.

(d) Nash equilibria. *Backward induction first.* Move 4: player 2 grabs ($\frac{27}{2} > \frac{81}{8}$). Move 3: player 1 grabs ($9 > \frac{27}{4}$). Move 2: player 2 grabs ($6 > \frac{9}{2}$). Move 1: player 1 grabs ($4 > 3$). SPE $= (GG, GG)$, payoffs $(4, 2)$.

Normal form (rows player 1, columns player 2):

	GG	GS	SG	SS
GG	$\underline{4},\ \underline{2}$	$\underline{4},\ \underline{2}$	$4,\ \underline{2}$	$4,\ \underline{2}$
GS	$\underline{4},\ \underline{2}$	$\underline{4},\ \underline{2}$	$4,\ \underline{2}$	$4,\ \underline{2}$
SG	$3,\ \underline{6}$	$3,\ \underline{6}$	$\underline{9},\ \underline{\frac{9}{2}}$	$9,\ \underline{\frac{9}{2}}$
SS	$3,\ 6$	$3,\ 6$	$\underline{\frac{27}{4}},\ \underline{\frac{27}{2}}$	$\underline{\frac{81}{4}},\ \frac{81}{8}$

(Underlined payoffs are best replies. Outside the top-left block no cell has both payoffs underlined: in row SS player 2's best reply is SG, against which player 1 prefers SG, against which player 2 prefers to grab at move 2.) The **pure Nash equilibria** are the four profiles $\{GG, GS\} \times \{GG, GS\}$: player 1 grabs at once and player 2 plans to grab at move 2; what either plans at the later, unreached node is free. Every one gives the outcome G with $(4, 2)$.

Mixed equilibria: still one outcome. Work forward along a Nash equilibrium, using that every node reached with positive probability must be played optimally given what follows. (i) If move 4 is reached, player 2 grabs there ($\frac{27}{2} > \frac{81}{8}$). (ii) So if move 3 is reached, sharing gives player 1 $\frac{27}{4}$ against 9 from grabbing: he grabs, and move 4 is never reached. (iii) So if move 2 is reached, sharing gives player 2 $\frac{9}{2}$ against 6 from grabbing: she grabs, and move 3 is never reached. (iv) So sharing at move 1 gives player 1 $3 < 4$: he grabs. Hence **every** Nash equilibrium has player 1 grabbing at move 1: **one outcome**, $(4, 2)$. The mixed equilibria differ only off the path. With $g = \Pr_2(G \text{ at move 2})$ and $h = \Pr_2(G \text{ at move 4})$, player 1's best deviation (share, then the better of grabbing for 9 or sharing for $M = \max\{9,\ \frac{27}{4}h + \frac{81}{4}(1-h)\}$) must not beat 4:

$$3g + M(1-g) \le 4 \iff g \ge 1 - \frac{1}{M-3},$$

e.g. $g \ge \frac{5}{6}$ if $h = 1$ and $g \ge \frac{65}{69}$ if $h = 0$.

(e) Equal split at the end. Sharing at move 4 now gives each player $\frac{1}{2} \cdot \frac{243}{8} = \frac{243}{16} \approx 15.19$. Backward induction: move 4: share ($\frac{243}{16} > \frac{27}{2} = \frac{216}{16}$); move 3: share ($\frac{243}{16} > 9$); move 2: share ($\frac{243}{16} > 6$); move 1: share ($\frac{243}{16} > 4$). SPE $= (SS, SS)$ with payoffs $(\frac{243}{16}, \frac{243}{16})$. The answer to (d) changes: the pure Nash equilibria are now (SS, SS) and, as before, the four profiles $\{GG, GS\} \times \{GG, GS\}$ (player 2's plan to grab at move 2 still deters player 1, and it is never tested). So **two outcomes** are supported: grab at once, $(4, 2)$, and share to the end, $(\frac{243}{16}, \frac{243}{16})$. Grabbing at move 2, 3 or 4 is never an equilibrium outcome: the grabber at move 4 would rather share, the grabber at move 3 would be pre-empted at move 2 (player 2 compares 6 with $\frac{9}{2}$), and a grab at move 2 leaves player 1 with $3 < 4$.

(d) Interpretation. With the original split the mover at the last node always grabs, and the anticipation unravels cooperation back to the first move: the pot that could have grown to \$30 is grabbed at \$6. Changing only the *last* node's rule (an equal split) turns the last mover into someone who wants to continue, and the whole chain of reasoning reverses. Credibility at the end of the game drives behavior at the start, the backward-induction lesson of Lecture 3 and the reason why commitments that change the final payoffs (Lecture 4, slide 11) are so powerful.

(e) Verification. `check_B_L03.py` builds the tree with exact fractions (pots $6, 9, \frac{27}{2}, \frac{81}{4}, \frac{243}{8}$), runs backward induction (unique solution: grab, $(4, 2)$), builds the 4×4 normal form and finds exactly the four pure Nash equilibria, all with outcome $(4, 2)$; it checks the bound on g at $h \in \{0, \frac{1}{2}, 1\}$ on both sides ($\frac{5}{6}$, $\frac{65}{69}$); for (e) it finds the SPE all-share with $(\frac{243}{16}, \frac{243}{16})$ and the five pure Nash equilibria with two outcomes. Output line: `tadelis-7-4: (d) 4 pure NE, one outcome (4, 2); (e) SPE all-share 243/16 each; NE outcomes 2`. Tadelis prints no answer.

(f) Common mistakes.

- Answering "one Nash equilibrium" in (d): there are four pure ones (and a continuum of mixed ones); it is the *outcome* that is unique.
- Misreading the last-node rule in (d): if player 2 shares at move 4 she gets only $\frac{1}{3}$ of the pot, $\frac{81}{8} < \frac{27}{2}$, so she grabs.
- In (e), concluding that sharing is the only equilibrium outcome: the immediate grab is still a (non-subgame-perfect) Nash equilibrium.

Answer. (a) Tree above; perfect information. (b) 5 terminal nodes; 4 information sets (2 each). (c) 4 pure strategies each: GG, GS, SG, SS. (d) Pure Nash equilibria: $\{GG, GS\} \times \{GG, GS\}$; every Nash equilibrium (mixed ones too) has player 1 grabbing at once: one outcome, $(4, 2)$; SPE (GG, GG). (e) Yes: SPE (SS, SS) with $(\frac{243}{16}, \frac{243}{16})$; pure Nash equilibria (SS, SS) and $\{GG, GS\} \times \{GG, GS\}$; two outcomes.

Item 7 (item 27 of the combined list)**: Tadelis Exercise 7.9** Exam · difficulty 4/5

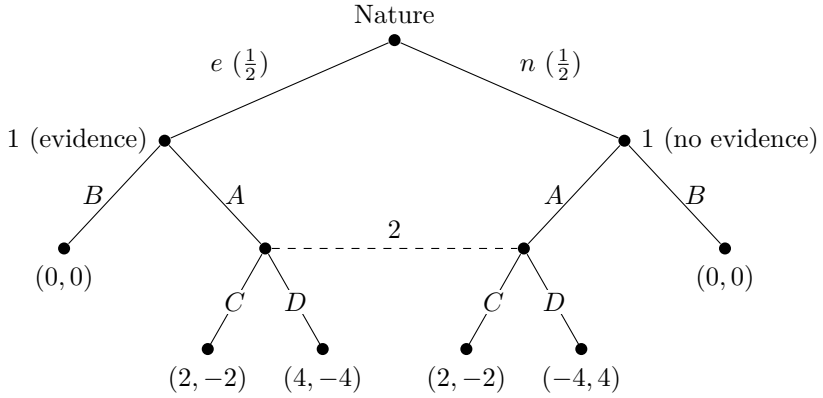
Source: Tadelis Exercise 7.9, page 149–150. The statement is reproduced verbatim in the list (item 27); it is not repeated here.

(a) What is asked. The dean learns whether he has evidence (probability $\frac{1}{2}$), then accuses (A) or bounces (B); an accused student, who does not know whether there is evidence, confesses (C) or denies (D). (a) Game tree; (b) normal-form matrix; (c) Nash equilibria.

(b) Method. A move of Nature at the root and information sets for what a player does not observe (Lecture 3, slides 21–23; Tadelis §7.1, Def. 7.2, p. 135; Nature as in the Brexit tree, Lecture 4, slides 5–6). Strategies as plans over information sets (Lecture 3, slide 25), with expected payoffs over Nature's move in the normal form (Tadelis §7.2.3, pp. 142–145). Nash equilibria by dominance, best replies and the principle of indifference (Lecture 3, slides 10–12).

(c) Solution. Notation. Nature draws e (evidence) or n (no evidence), each with probability $\frac{1}{2}$. The dean (player 1) observes it. Payoffs (u_1, u_2) = (dean, student): $B \mapsto (0,0)$; $A, C \mapsto (2,-2)$; A, D with evidence $\mapsto (4,-4)$; A, D without evidence $\mapsto (-4,4)$. Note that $u_1 + u_2 = 0$ at every leaf: the game is zero-sum.

(a) Tree. The student's two nodes (after eA and after nA) form one information set: he knows he was accused, not whether there is evidence.



(b) Normal form. The dean has two information sets, so a strategy is a pair (action with evidence, action without): $S_1 = \{AA, AB, BA, BB\}$; $S_2 = \{C, D\}$. Each cell is $\frac{1}{2}\times$(payoff after e) $+\frac{1}{2}\times$(payoff after n). For example, $u(AB, D) = \frac{1}{2}(4,-4) + \frac{1}{2}(0,0) = (2,-2)$ and $u(AA, D) = \frac{1}{2}(4,-4) + \frac{1}{2}(-4,4) = (0,0)$.

	C	D
AA	$\underline{2}$, -2	0, $\underline{0}$
AB	1, $\underline{-1}$	$\underline{2}$, -2
BA	1, -1	-2, $\underline{2}$
BB	0, $\underline{0}$	0, $\underline{0}$

(c) Nash equilibria.

Step 1: no pure equilibrium. No cell has both payoffs underlined: against C the dean plays AA, against which the student denies; against D the dean plays AB, against which the student confesses. The best replies cycle.

Step 2: dominated strategies. BA is strictly dominated by AA ($2 > 1$, $0 > -2$): accusing only when there is no evidence is never sensible. BB is strictly dominated by AB ($1 > 0$, $2 > 0$). So in any equilibrium the dean accuses whenever he has evidence and mixes only between AA and AB, i.e. over what he does *without* evidence.

Step 3: indifference. Let $q = \Pr_2(C)$ and $p = \Pr_1(AA)$ (with $1-p$ on AB). The dean is indifferent when

$$u_1(AA) = 2q \;=\; q + 2(1-q) = u_1(AB) \;\Rightarrow\; 2q = 2 - q \;\Rightarrow\; q = \tfrac{2}{3}, \tag{5}$$

and the student is indifferent when

$$u_2(C) = -2p - (1-p) \;=\; 0 \cdot p - 2(1-p) = u_2(D) \;\Rightarrow\; -1 - p = -2 + 2p \;\Rightarrow\; p = \tfrac{1}{3}. \tag{6}$$

At $q = \frac{2}{3}$ the dean's payoffs are $AA, AB : \frac{4}{3}$, $BA : 3q - 2 = 0$, $BB : 0$, so mixing over AA and AB is optimal. The unique Nash equilibrium is

$$\sigma_1^* = \tfrac{1}{3}\,AA + \tfrac{2}{3}\,AB, \qquad \sigma_2^* = \tfrac{2}{3}\,C + \tfrac{1}{3}\,D, \qquad (u_1, u_2) = (\tfrac{4}{3},\ -\tfrac{4}{3}).$$

Uniqueness. For $q < \frac{2}{3}$ only AB is a best reply and the student would then confess ($-1 > -2$), so $q = 1$; for $q > \frac{2}{3}$ only AA is a best reply and the student would deny ($0 > -2$), so $q = 0$; at $q = \frac{2}{3}$ only $p = \frac{1}{3}$ keeps the student indifferent.

Behavioral reading. With evidence the dean always accuses; without evidence he accuses (bluffs) with probability $\frac{1}{3}$. An accused student therefore faces evidence with probability $\frac{1/2}{1/2+1/2\cdot 1/3} = \frac{3}{4}$, and at that belief C and D both give -2 (the check: $\frac{3}{4}(-4) + \frac{1}{4}(4) = -2$).

(d) Interpretation. The dean must bluff sometimes: if he accused only with evidence, every accusation would reveal evidence and the student would confess, which in turn makes accusing without evidence profitable. The equilibrium bluffing rate keeps the student exactly indifferent. Because the game is zero-sum, $\frac{4}{3}$ is its value: the dean cannot guarantee more, the student cannot hold him to less. The structure (Nature, a privately informed first mover, a responder who sees only the action) is the one Lecture 4 used for the Brexit negotiation, and it is the bridge to games of incomplete information later in the course.

(e) Verification. `check_B_L03.py` builds the matrix from the tree with exact fractions (it equals the one above), checks the zero-sum property and that there is no pure equilibrium, scans q on a grid of step $\frac{1}{300}$ (an equilibrium exists only at $q = \frac{2}{3}$), verifies $(\frac{1}{3}AA + \frac{2}{3}AB,\ \frac{2}{3}C + \frac{1}{3}D)$ with the exact Nash test, and solves (6) with sympy. Output line: `tadelis-7-9: unique NE dean 1/3 AA + 2/3 AB, student C w.p. 2/3; value 4/3`. Tadelis prints no answer.

(f) Common mistakes.

- Giving the dean only two strategies (A, B): he chooses after observing Nature, so a strategy is a pair of actions.
- Drawing the student's two nodes as separate information sets: then he would know whether there is evidence, and the game would change completely.
- Using payoffs conditional on one state in the matrix instead of the $\frac{1}{2}$–$\frac{1}{2}$ expectation.

Answer. (a) Tree above, with the student's nodes in one information set. (b) Matrix above, rows AA, AB, BA, BB (action with evidence, without). (c) No pure equilibrium; unique Nash equilibrium $(\frac{1}{3}AA + \frac{2}{3}AB,\ \frac{2}{3}C + \frac{1}{3}D)$: the dean always accuses with evidence and bluffs w.p. $\frac{1}{3}$ without it; the student confesses w.p. $\frac{2}{3}$; payoffs $(\frac{4}{3}, -\frac{4}{3})$.

Item 8 (item 28 of the combined list)**: Tadelis Exercise 7.10** Exam · difficulty 4/5

Source: Tadelis Exercise 7.10, page 150. The statement is reproduced verbatim in the list (item 28); it is not repeated here.

(a) What is asked. In the game of Figure 7.14: (a) pure-strategy sets; (b) every behavioral strategy of player 1 has an outcome-equivalent mixed strategy; (c) the same for player 2; (d) whether (c) fails once player 2's two bottom information sets are merged.

(b) Method. Pure, mixed and behavioral strategies (Lecture 3, slides 25 and 27; Tadelis §7.2.1–7.2.2, Defs. 7.4–7.6, pp. 137–140); perfect recall and Kuhn's equivalence theorem (Lecture 3, slide 28; Tadelis Def. 7.7 and the discussion on pp. 141–142). Outcome equivalence is checked node by node: two strategies of a player are equivalent if, against every pure strategy of the opponent (and then, by linearity, every mixed one), they put the same probability on each terminal node.

(c) Solution. The game. Player 2 moves first (a or b). After a, player 1 chooses L or R; after b, he chooses W or E. Player 2 then moves again without seeing player 1's choice: at information set $h_a = \{aL, aR\}$ with actions t, z, and at $h_b = \{bW, bE\}$ with actions x, y. Payoffs (u_1, u_2): $(2,1)$ after Lt and Wx, $(1,2)$ after Rz and Ey, $(0,0)$ otherwise.

(a) Pure strategies. Player 1 has two information sets (after a, after b), each with two actions:

$$S_1 = \{L, R\} \times \{W, E\} = \{LW, LE, RW, RE\} \quad (4).$$

Player 2 has three information sets (the root, h_a, h_b):

$$S_2 = \{a, b\} \times \{t, z\} \times \{x, y\} \quad (2^3 = 8),$$

e.g. (a,t,x), (a,t,y), …, (b,z,y).

(b) Player 1: behavioral ⇒ mixed. A behavioral strategy is $\beta_1 = (\lambda, \omega)$, with $\lambda = \Pr(L)$ at the left node and $\omega = \Pr(W)$ at the right node. Define the mixed strategy as the product

$$\sigma_1(LW) = \lambda\omega, \quad \sigma_1(LE) = \lambda(1-\omega), \quad \sigma_1(RW) = (1-\lambda)\omega, \quad \sigma_1(RE) = (1-\lambda)(1-\omega).$$

Fix any pure $s_2 = (r, m_a, m_b)$. If $r = a$, the play reaches player 1's left node and ends at (a, L, m_a) or (a, R, m_a). Under β_1 these have probabilities λ and $1-\lambda$. Under σ_1, (a, L, m_a) occurs iff the drawn plan starts with L:

$$\sigma_1(LW) + \sigma_1(LE) = \lambda\omega + \lambda(1-\omega) = \lambda,$$

and (a, R, m_a) has probability $1-\lambda$. The case $r = b$ is symmetric with ω. All other terminal nodes have probability 0 under both. So β_1 and σ_1 induce the same distribution against every pure s_2, hence (expected values are linear in σ_2) against every mixed σ_2 too. Since player 1 is called once per play, only the marginals matter: any σ_1 with $\sigma_1(LW) = \mu$, $\sigma_1(LE) = \lambda - \mu$, $\sigma_1(RW) = \omega - \mu$, $\sigma_1(RE) = 1 - \lambda - \omega + \mu$, $\mu \in [\max\{0, \lambda+\omega-1\}, \min\{\lambda, \omega\}]$, also works.

(c) Player 2: behavioral ⇒ mixed. A behavioral strategy is $\beta_2 = (\alpha, \tau, \xi)$: $\Pr(a)$ at the root, $\Pr(t)$ at h_a, $\Pr(x)$ at h_b. Define the product mixed strategy

$$\sigma_2(r, m_a, m_b) = \Pr_\alpha(r) \cdot \Pr_\tau(m_a) \cdot \Pr_\xi(m_b),$$

e.g. $\sigma_2(a,t,x) = \alpha\tau\xi$. Fix any pure $s_1 = (\ell, m)$ with $\ell \in \{L, R\}$, $m \in \{W, E\}$. Under β_2, terminal node (a, ℓ, t) has probability $\alpha\tau$. Under σ_2 it occurs iff the drawn plan has $r = a$ and $m_a = t$ (any m_b):

$$\sum_{m_b \in \{x,y\}} \sigma_2(a, t, m_b) = \alpha\tau\xi + \alpha\tau(1-\xi) = \alpha\tau.$$

The same computation gives $\alpha(1-\tau)$ for (a,ℓ,z), $(1-\alpha)\xi$ for (b,m,x) and $(1-\alpha)(1-\xi)$ for (b,m,y); nodes off s_1's choices have probability 0 under both. So the two strategies are outcome-equivalent against every s_1, hence against every σ_1. What makes this work is *perfect recall*: at h_a player 2 remembers she chose a, so her bottom move can be drawn independently of, and after, her root move, and the product of independent draws reproduces it.

(d) Imperfect recall. Merge h_a and h_b into one information set h. Actions at the nodes of an information set must be the same, so identify $x \equiv t$ and $y \equiv z$. Player 2 now has $S_2' = \{a,b\}\times\{t,z\}$ (4 pure strategies); a behavioral strategy is (α,τ); she has *forgotten* her own first move.

The claim of (c) survives. The product mixed strategy $\sigma_2(r,m) = \Pr_\alpha(r)\Pr_\tau(m)$ still reproduces (α,τ): each play passes through h at most once, so the probability of a terminal node (r,ℓ,m) is $\Pr_\alpha(r)\Pr_\tau(m)$ under both. No example can break (c) in this game. (Behavioral $\Rightarrow$ mixed fails only when one play can cross the same information set twice, as in the absent-minded driver, Tadelis p. 142.)

What fails is the converse (mixed $\Rightarrow$ behavioral, the other half of Kuhn's theorem). Take

$$\sigma_2 = \tfrac{1}{2}\,(a,t) + \tfrac{1}{2}\,(b,z)$$

("go left and then play t, or go right and then play z": the plan coordinates the bottom move with the root move she no longer remembers). Against $s_1 = LW$ it gives $\Pr(a,L,t) = \frac{1}{2}$, $\Pr(b,W,z) = \frac{1}{2}$, $\Pr(a,L,z) = 0$. A behavioral (α,τ) gives

$$\Pr(a,L,t) = \alpha\tau, \qquad \Pr(a,L,z) = \alpha(1-\tau), \qquad \Pr(b,W,z) = (1-\alpha)(1-\tau).$$

$\Pr(a,L,z) = 0$ forces $\alpha = 0$ or $\tau = 1$; the first makes $\Pr(a,L,t) = 0$, the second makes $\Pr(b,W,z) = 0$, contradicting $\frac{1}{2}$ in either case. So this mixed strategy has no outcome-equivalent behavioral strategy. It also matters for payoffs: against LW, σ_2 gives player 2 $\frac{1}{2}\cdot 1 + \frac{1}{2}\cdot 0 = \frac{1}{2}$, and against RE it gives $\frac{1}{2}\cdot 0 + \frac{1}{2}\cdot 2 = 1$; behavioral strategies cannot replicate this coordination.

(d) Interpretation. Mixed strategies are plans chosen before the game; behavioral strategies are choices made at each information set. With perfect recall the two are interchangeable, which is why the course (and Tadelis) can analyze dynamic games with behavioral strategies and define SPE with them (Lecture 3, slide 34). Without recall a player who randomizes "in the moment" cannot correlate her later move with an earlier one she has forgotten, while a plan made in advance can.

(e) Verification. `check_B_L03.py` computes with sympy, for each pure strategy of the opponent and each of the eight terminal nodes, the probability under the behavioral strategy and under the product mixed strategy, for player 1 (b) and player 2 (c); all differences simplify to 0. For (d) it confirms the product construction still works and that the system $\alpha\tau = \frac{1}{2}$, $\alpha(1-\tau) = 0$, $(1-\alpha)(1-\tau) = \frac{1}{2}$ has no solution. Output line: `tadelis-7-10: (b), (c) equivalences hold symbolically; (d) mixed->behavioral fails`. Tadelis prints no answer. **Note on the statement:** read literally, (d) asks for a failure of behavioral $\Rightarrow$ mixed, which cannot fail here (each play crosses the merged set once); the example above shows the failure of the converse, which is the equivalence that perfect recall protects in this game.

(f) Common mistakes.

- Giving player 2 only $2\times 2 = 4$ strategies in (a): with perfect recall she has two separate bottom information sets, so $2^3 = 8$.
- Proving (b) only against one strategy of player 2: equivalence must hold against all of them.

- In (d), claiming that the product construction breaks. It does not; the coordination of a mixed plan is what behavioral strategies lose.

Answer. (a) $S_1 = \{LW, LE, RW, RE\}$; $S_2 = \{a, b\} \times \{t, z\} \times \{x, y\}$ (8 strategies). (b), (c) The product of the behavioral probabilities is a mixed strategy with the same distribution over terminal nodes against every opponent strategy. (d) Behavioral $\Rightarrow$ mixed still holds (each play crosses the merged set once); the equivalence that fails is mixed $\Rightarrow$ behavioral: $\frac{1}{2}(a, t) + \frac{1}{2}(b, z)$ has no behavioral equivalent.

Item 9 (item 29 of the combined list)**: MWG Exercise 9.B.4** Advanced · difficulty 4/5

Source: MWG Exercise 9.B.4, page 302. The statement is reproduced verbatim in the list (item 29); it is not repeated here.

(a) What is asked. Prove that in a finite zero-sum game of perfect information all subgame-perfect Nash equilibria give the same payoffs.

(b) Method. Backward induction and Zermelo's theorem (MWG §9.B, Prop. 9.B.1, p. 272; Lecture 3, slide 30, whose corollary gives uniqueness only when no player has equal payoffs at two terminal nodes); subgames and SPNE (MWG Defs. 9.B.1–9.B.2, pp. 274–275; Lecture 3, slides 32–34); in a finite game of perfect information the SPNE are exactly the backward-induction profiles (MWG Prop. 9.B.2; Lecture 3, slide 35). The proof is an induction on subgames that tracks *values*, not strategies: ties may create many SPNE, and the point is that they cannot create different payoffs.

(c) Solution. Setting. Two players; a finite tree in which every information set is a single node; $u_1(z) + u_2(z) = 0$ at every terminal node z, so we can track $v(z) = u_1(z)$ and $u_2(z) = -v(z)$. For a decision node x, Γ_x is the subgame starting at x (every node starts one, by perfect information), and $C(x)$ is the set of its immediate successors. An SPNE may use behavioral strategies.

Claim. For every node x there is a number $V(x)$ such that *every* SPNE of Γ_x gives player 1 the payoff $V(x)$ (and player 2 $-V(x)$).

Proof by induction on the length of the longest path from x to a terminal node.

Base. If x is terminal, $V(x) = v(x)$.

Step. Let x be a decision node of player i and suppose the claim holds for every $y \in C(x)$. Take any SPNE σ of Γ_x.

(i) For each $y \in C(x)$, the restriction of σ to Γ_y is an SPNE of Γ_y (a subgame of Γ_y is a subgame of Γ_x). By the induction hypothesis it gives player 1 exactly $V(y)$. So if play moves from x to y, the continuation pays $(V(y), -V(y))$, whatever σ prescribes inside Γ_y.

(ii) The restriction of σ to Γ_x is a Nash equilibrium of Γ_x. Player i could change only his action at x, keeping the rest of σ; moving to y would give him $V(y)$ if $i = 1$, or $-V(y)$ if $i = 2$. Therefore every action played with positive probability at x must lead to a maximizer:

$$i = 1: \quad y \in \arg\max_{y' \in C(x)} V(y'), \qquad i = 2: \quad y \in \arg\max_{y' \in C(x)} \big(-V(y')\big) = \arg\min_{y' \in C(x)} V(y').$$

(iii) All the successors in the support therefore have the same value, so the payoff of σ in Γ_x is

$$V(x) = \max_{y \in C(x)} V(y) \ \text{ if } i = 1, \qquad V(x) = \min_{y \in C(x)} V(y) \ \text{ if } i = 2,$$

a number that does not depend on which SPNE σ we took.

Applying the claim at the root: every SPNE of the game gives $(V(x_0), -V(x_0))$. SPNE exist by Zermelo's theorem, so the set of SPNE payoffs is exactly $\{(V(x_0), -V(x_0))\}$. □

Where zero-sum is used. In step (iii). When player i is indifferent among several successors, they give *him* the same payoff, and in a two-player zero-sum game they then give the opponent the same payoff too. Without zero-sum, the opponent's payoff can differ across the tied successors. Counterexample: player 1 chooses $l \mapsto (0, 1)$ or $r \mapsto (0, 2)$; both l and r are SPNE, with different payoffs for player 2.

Two remarks.

- *Two players are needed.* With three players and $u_1 + u_2 + u_3 = 0$ the result fails: player 1 choosing between $(0, 1, -1)$ and $(0, -1, 1)$ has two SPNE with different payoffs for players 2 and 3. "Zero-sum" in the exercise means two-player zero-sum.
- *Chance moves.* If the tree has moves of Nature with known probabilities, define $V(x)$ at a chance node as the expected value of its successors' values; the same induction goes through. By the minimax theorem $V(x_0)$ is also the value of the game, and every Nash equilibrium, not only every SPNE, yields it.

(d) Interpretation. In a pure conflict, ties are harmless: if I am indifferent between two moves, so are you, so my tie-breaking cannot be used to help or hurt you. That is why chess-like games have a well-defined value even though the backward-induction strategies are far from unique. In the non-zero-sum games of Lecture 4 (entry, Brexit) a follower's indifference *is* strategically important: how she breaks it can change the leader's payoff (compare MWG 9.B.5, item "MWG 9.B.5", part (e)).

(e) Verification. `check_B_L03.py` draws 400 random two-player zero-sum trees (depth up to 4, two or three moves per node, integer payoffs in $\{-2, \ldots, 2\}$, fixed seed), computes the set of pure SPNE payoff vectors over all tie-breakings, and finds exactly one in each tree; 151 of the trees have more than one SPNE, so ties do occur. It also confirms the two counterexamples (non-zero-sum and three-player zero-sum) have two SPNE payoff vectors. Output line: `mwg-9-b-4: 400 random zero-sum trees, unique SPNE payoffs (151 with several SPNE)`. *Comparison with the MWG solutions manual* (Exercise 9.B.4, p. 9-2 of the manual; label read as "9.8.4" in the OCR): it argues that if a player has equal payoffs at two terminal nodes, the zero-sum property makes the other player indifferent too, so "there may exist many different ... SPNE ... but all of them will yield the same payoffs". **Agreement** on the result; the manual's argument speaks of terminal nodes only, and the induction above extends it to ties between subgame values, which is where it is actually needed.

(f) Common mistakes.

- Proving that the SPNE is unique: it need not be (ties); only the payoffs are.
- Using Zermelo's theorem's uniqueness clause, which requires no ties at all, instead of an argument that allows ties.
- Forgetting that SPNE may be mixed at tied nodes; the argument must cover any probability on the maximizers, which (ii)–(iii) do.

Answer. By induction on subgames: if every subgame after x has a unique SPNE value $V(y)$ for player 1, then at x any SPNE puts weight only on successors that maximize the mover's payoff, $\max_y V(y)$ for player 1 or $\min_y V(y)$ for player 2, so $V(x)$ is unique. At the root all SPNE pay $(V(x_0), -V(x_0))$. Zero-sum (with two players) is what makes the mover's ties irrelevant for the opponent.

Item 10 (item 30 of the combined list)**: Gibbons Problem 2.1** Advanced · difficulty 5/5

Source: Gibbons Problem 2.1, page 130. The statement is reproduced verbatim in the list (item 30); it is not repeated here.

(a) What is asked. Prove Becker's Rotten Kid theorem: when a child's action A produces incomes $I_C(A)$, $I_P(A)$ and an altruistic parent then chooses a bequest B, the backwards-induction outcome has the child choosing the A that maximizes $I_C(A) + I_P(A)$.

(b) Method. Backward induction in a two-stage game of perfect information with continuous actions (Lecture 3, slides 29–30; Gibbons §2.1.A, pp. 57–61; Tadelis §8.3.2): solve the parent's problem for every pair of incomes, then the child's problem given the parent's reply. The key step is that the parent's reply depends on the incomes only through their sum, and raises the child's consumption when the sum rises.

(c) Solution. Notation and assumptions. $A \geq 0$ the child's action; I_C, I_P strictly concave with maximizers $A_C, A_P > 0$; $B \in \mathbb{R}$ the bequest; U, V increasing and strictly concave; $k > 0$. Write $Y = I_C + I_P$ (family income) and $x = I_C + B$ (the child's consumption), so the parent consumes $I_P - B = Y - x$. As is standard for this problem, U and V are differentiable and the parent's problem has a solution.

Step 1: the parent's problem depends only on Y. Given (I_C, I_P) the parent solves

$$\max_{B \in \mathbb{R}} \; V(I_P - B) + k\,U(I_C + B).$$

Because B is unrestricted in sign, choosing B is the same as choosing $x = I_C + B \in \mathbb{R}$, and the problem becomes

$$\max_{x \in \mathbb{R}} \; W(x; Y) = V(Y - x) + k\,U(x). \tag{7}$$

W is strictly concave in x (a sum of strictly concave functions), so the maximizer $x^*(Y)$ is unique and satisfies the first-order condition

$$V'(Y - x^*) = k\,U'(x^*). \tag{8}$$

The bequest is $B^*(I_C, I_P) = x^*(I_C + I_P) - I_C$: it depends on both incomes, but the child's consumption depends only on their sum.

Step 2: x^* is strictly increasing in Y. Take $Y' > Y$ and suppose $x' = x^*(Y') \leq x = x^*(Y)$. Then $Y' - x' > Y - x$. Strict concavity makes V' strictly decreasing and U' decreasing, so

$$V'(Y' - x') < V'(Y - x) \qquad \text{and} \qquad k\,U'(x') \geq k\,U'(x).$$

Applying (8) at Y and at Y':

$$V'(Y' - x') = k\,U'(x') \;\geq\; k\,U'(x) = V'(Y - x) \;>\; V'(Y' - x'),$$

a contradiction. Hence $x^*(Y') > x^*(Y)$. (With U, V twice differentiable, $dx^*/dY = V''/(V'' + kU'') \in (0, 1)$: the parent passes part, not all, of an income gain to the child.)

Step 3: the child's problem. Anticipating B^*, the child chooses $A \geq 0$ to maximize

$$U\big(I_C(A) + B^*(I_C(A), I_P(A))\big) = U\big(x^*(I_C(A) + I_P(A))\big) = U\big(x^*(Y(A))\big).$$

$U \circ x^*$ is strictly increasing (Steps 1–2 and U increasing), so A maximizes the child's payoff if and only if it maximizes $Y(A) = I_C(A) + I_P(A)$.

Step 4: the maximizer exists and is unique. Y is strictly concave. For $A < \min\{A_C, A_P\}$ both incomes are increasing, and for $A > \max\{A_C, A_P\}$ both are decreasing, so Y attains its maximum on the compact interval $[\min\{A_C, A_P\}, \max\{A_C, A_P\}] \subset (0, \infty)$, at a unique point A^*. The backwards-induction outcome is

$$(A^*,\ B^*(I_C(A^*), I_P(A^*))), \qquad A^* = \arg\max_{A \geq 0} \left[\, I_C(A) + I_P(A) \,\right]. \qquad \square$$

Worked example (exact). $U = V = \ln$, $k = 2$, $I_C = 4A - A^2$ ($A_C = 2$), $I_P = 10A - A^2$ ($A_P = 5$). The parent's condition (8) reads $\frac{1}{Y-x} = \frac{2}{x}$, so $x^*(Y) = \frac{2Y}{3}$ (in general $x^* = \frac{kY}{1+k}$). The child maximizes $\ln(\frac{2}{3}(14A - 2A^2))$: $14 - 4A = 0$, $A^* = \frac{7}{2}$, between $A_C = 2$ and $A_P = 5$. There $Y = \frac{49}{2}$, $x^* = \frac{49}{3}$, $I_C = \frac{7}{4}$, so the bequest is $B^* = \frac{49}{3} - \frac{7}{4} = \frac{175}{12}$. Had the child chosen its own income maximizer $A_C = 2$, family income would be $Y(2) = 20 < \frac{49}{2}$ and its consumption only $\frac{40}{3} < \frac{49}{3}$.

(d) Interpretation. The child is selfish but faces a parent who "taxes" any loss of family income: if the child raised I_C at a larger cost to I_P, the parent, who cares about the split, would cut the bequest by more than the child gained. Altruism on one side, plus the parent's last move, aligns the first mover's incentives with joint surplus. Each assumption has a role:

- *the parent moves last*: it is the parent's credible, sequentially rational reply that disciplines the child (Lecture 4's theme: what matters is what a player will actually do when the moment comes);
- *B unrestricted*: if $B \geq 0$ were imposed and binding, the child's consumption would be I_C alone over a range and Step 1 fails;
- *the child moves first and cannot commit the parent*: if the bequest were fixed before A, the child would simply choose A_C.

(e) Verification. `check_B_L03.py` checks with sympy that $dx^*/dY = V''/(V'' + kU'') \in (0, 1)$ for several negative (V'', U'') and positive k; derives $x^* = kY/(1+k)$ for log utility (second derivative negative) and $x^* = (Y + \ln k)/2$ for CARA utility $-e^{-x}$; solves the child's problem in the log example ($A^* = \frac{7}{2}$), confirms on an exact grid of step $\frac{1}{20}$ that the maximizer of $\frac{2}{3}Y(A)$ is $\frac{7}{2}$ while I_C alone peaks at 2, and recomputes $B^* = \frac{175}{12}$. Output line: `gibbons-2-1: x*(Y) increasing; child picks A* = 7/2 = argmax(I_C + I_P), not A_C = 2`. Gibbons prints no answer.

(f) Common mistakes.

- Assuming the child maximizes I_C: its payoff depends on the bequest, which reacts to A.
- Treating the bequest as a fixed number instead of a function $B^*(I_C, I_P)$ (a strategy of the parent in the second stage).
- Proving only that x^* is weakly increasing, or skipping existence/uniqueness of A^*; the strict monotonicity in Step 2 is what makes "maximize $U(x^*(Y))$" equivalent to "maximize Y".

Answer. The parent's best bequest gives the child consumption $x^*(I_C + I_P)$ that solves $V'(Y - x) = kU'(x)$; it depends only on family income Y and is strictly increasing in it. So the child's payoff $U(x^*(Y(A)))$ is a strictly increasing function of $Y(A)$, and the child chooses $A^* = \arg\max_A [I_C(A) + I_P(A)]$, which exists and is unique by strict concavity.

Lecture 4 — Credible threats and subgame perfection

Item 1 (item 31 of the combined list)**: Gibbons Problem 2.12** Foundations · difficulty 1/5

Source: Gibbons Problem 2.12, page 135. The statement is reproduced verbatim in the list (item 31); it is not repeated here.

(a) What is asked. Define a strategy, a subgame and a subgame-perfect Nash equilibrium in a repeated game.

(b) Method. The definitions of Gibbons §2.3.B (pp. 93–95), stated for the finitely repeated game $G(T)$ and the infinitely repeated game $G(\infty,\delta)$; the matching definitions in the slides (Lecture 4, slides 21–22: repeated game, $G(T,\delta)$, "a strategy says what to play in each period after each possible history", "each period starts a new subgame"; SPE, slide 17) and in Tadelis §9.3–9.4 (pp. 178–182). Each definition is illustrated with the two-stage Prisoners' Dilemma of Gibbons Fig. 2.3.1 (p. 82).

(c) Solution. Setting. A stage game $G = \{A_1,\dots,A_n;\ u_1,\dots,u_n\}$ is played in periods $t = 1,2,\dots$; before each period all players observe all past actions. The *history* through period t is $h^t = (a^1,\dots,a^t)$, with $a^s = (a_1^s,\dots,a_n^s)$ the action profile of period s; $h^0 = \varnothing$. H^t is the set of histories through t.

1. Strategy. A strategy of player i specifies the action he will take in each stage, for each possible history of play through the previous stage (Gibbons p. 94):

$$s_i = (s_i^1, s_i^2, \dots), \qquad s_i^t : H^{t-1} \to A_i.$$

It is a complete contingent plan, including histories that the plan itself would never produce. *Example.* In the two-stage Prisoners' Dilemma (actions L_i, R_i) there are four first-stage outcomes, so a strategy has five instructions (v,w,x,y,z): a first-stage action and a second-stage action after each of (L_1,L_2), (L_1,R_2), (R_1,L_2), (R_1,R_2) (Gibbons p. 93). Each player has $2^5 = 32$ pure strategies. In general, with a 2×2 stage game played T times, player i has $1+4+\cdots+4^{T-1} = \frac{4^T-1}{3}$ information sets and $2^{(4^T-1)/3}$ pure strategies.

2. Subgame. A subgame is the piece of the game that remains to be played from any point at which the complete history so far is common knowledge (Gibbons p. 94). In $G(T)$ a subgame beginning at stage $t+1$ is the repeated game in which G is played $T-t$ times, $G(T-t)$, following one particular history h^t; there is one such subgame for each $h^t \in H^t$. In $G(\infty,\delta)$ each subgame beginning at stage $t+1$ is identical to $G(\infty,\delta)$ itself, and again there is one for each history through t. *Not* a subgame: stage t taken on its own, for $t < T$; a subgame includes all the moves that follow its starting point (Gibbons p. 95). *Example.* The two-stage Prisoners' Dilemma has four proper subgames after period 1 (one per first-stage outcome), each a one-shot Prisoners' Dilemma, plus the whole game.

3. Subgame-perfect Nash equilibrium. A Nash equilibrium is subgame-perfect if the players' strategies constitute a Nash equilibrium in every subgame (Selten 1965; Gibbons p. 95; Lecture 4, slide 17 states it for general extensive forms: the restriction of the profile to every proper subgame is a Nash equilibrium of that subgame). In a repeated game this means: after *every* history h^t, on or off the equilibrium path, the continuation strategies $s|_{h^t}$ form a Nash equilibrium of the continuation game.

Example (computed). In the two-stage Prisoners' Dilemma ($u(L,L) = (1,1)$, $u(L,R) = (5,0)$, $u(R,L) = (0,5)$, $u(R,R) = (4,4)$, payoffs summed over the two stages), enumerating all 32×32 profiles gives 16 Nash equilibria; all of them have the outcome (L_1, L_2) in both stages, payoffs $(2,2)$, but they differ in what is planned after unreached first-stage outcomes. Exactly one is subgame perfect: each player plays L in stage 1 and L after every history, (L, L, L, L, L), because each of the four second-stage subgames is a one-shot Prisoners' Dilemma whose only Nash equilibrium is (L_1, L_2).

(d) Interpretation. Three ideas carry over from Lectures 3–4 to repeated games. A strategy must say what happens after deviations, because those plans are what make promises and threats; a subgame starts after each observed history; and subgame perfection requires the punishments and rewards to be ones players would actually carry out. This is the toolkit for the carrots-and-sticks and folk-theorem arguments of Lecture 5: cooperation can be sustained only by continuation play that is itself an equilibrium.

(e) Verification. `check_B_L04.py` enumerates the 32 strategies (v, w, x, y, z) of each player in the two-stage Prisoners' Dilemma, computes all Nash equilibria of the 32×32 game and keeps those that play a stage Nash equilibrium in every second-stage subgame; it also checks the count $\frac{4^T-1}{3}$ for $T = 1, 2, 3$ and the 3^{10} strategies of the 3×3 two-stage game of Gibbons Fig. 2.3.3. Output line: `gibbons-2-12: 32 strategies each, 16 NE profiles, 1 SPE (L,L,L,L,L) each`. Gibbons prints no separate answer; the definitions are those of pp. 94–95.

(f) Common mistakes.

- Defining a strategy as a sequence of actions, one per period ("play b, then c"): it must be contingent on the history.
- Calling the stage game played in period t a subgame: a subgame includes all later periods.
- Checking Nash behavior only on the equilibrium path: subgame perfection is about every history, including those after deviations.

Answer. A strategy assigns an action in every period to every possible history of play through the previous period. A subgame is the remainder of the repeated game after a given history (in $G(T)$, the game $G(T-t)$ after one history h^t; in $G(\infty, \delta)$, a copy of the whole game). A subgame-perfect Nash equilibrium is a strategy profile whose restriction to every subgame, after every history, is a Nash equilibrium of that subgame.

Item 2 (item 32 of the combined list)**: Tadelis Exercise 7.6** Foundations · difficulty 1/5

Source: Tadelis Exercise 7.6, page 148–149. The statement is reproduced verbatim in the list (item 32); it is not repeated here.

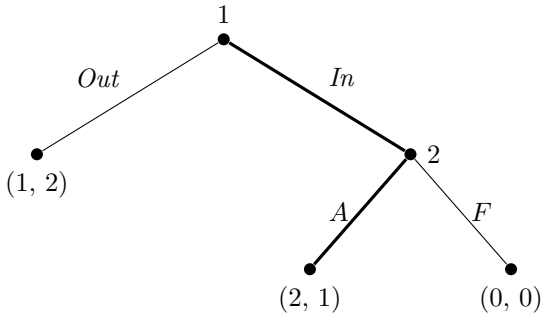
(a) What is asked. (a) A game tree with payoffs that represent the stated rankings; (b) the number of pure strategies of each player; (c) all Nash equilibria.

(b) Method. Extensive form and game trees (Tadelis §7.1, pp. 130–137); pure strategies as complete plans (§7.2.1, p. 137); the normal-form representation of a game tree (§7.2.3, p. 143); Nash equilibrium and paths of play, on and off the equilibrium path (§7.3 and Def. 7.8, pp. 145–146). The question stops at Nash equilibrium; which equilibrium survives sequential rationality is Tadelis §8.1 (Def. 8.1, p. 152) and Lecture 4, slides 13 and 17 (the entry-deterrence logic of the Brexit case, slides 9–11).

(c) Solution. (a) Tree. Ordinal payoffs suffice, since only the rankings are given. Write *In* and *Out* for player 1's choice and A (accommodate), F (fight) for player 2's. The rankings are

$$\text{player 1: } (In, A) \succ Out \succ (In, F), \qquad \text{player 2: } Out \succ (In, A) \succ (In, F).$$

Player 1's middle outcome must be *Out* and player 2's must be (In, A), the only outcomes not named as best or worst. Payoffs $2, 1, 0$ in that order give



Payoffs (u_1, u_2); thick branches = backward-induction choices (used below).

Any numbers with the same rankings give the same answers to (b) and (c).

(b) Pure strategies. Each player has one information set with two actions, so $S_1 = \{In, Out\}$ and $S_2 = \{A, F\}$: two pure strategies each. Player 2's strategy is a plan for the node after entry, and it is part of the profile even when player 1 stays out.

(c) Nash equilibria. The normal form, with best replies underlined:

	A	F
In	$\underline{2}, \underline{1}$	$0, 0$
Out	$1, \underline{2}$	$\underline{1}, \underline{2}$

Pure. (In, A) and (Out, F). Against A, *In* gives $2 > 1$; against F, *Out* gives $1 > 0$. Against *In*, A gives $1 > 0$; against *Out*, player 2 gets 2 with either action, so F is a (weak) best reply.

Mixed. Let $p = \Pr(In)$ and $f = \Pr(F)$.

- If $p > 0$, then $u_2(A) - u_2(F) = p \cdot (1 - 0) > 0$, so $f = 0$, and then *In* is player 1's unique best reply $(2 > 1)$: $p = 1$. This is (In, A).

- If $p = 0$, player 2 is indifferent, so any f is a best reply. *Out* is a best reply for player 1 iff

$$u_1(Out) = 1 \;\geq\; u_1(In) = 2(1-f) + 0 \cdot f \iff f \geq \tfrac{1}{2}.$$

So the set of all Nash equilibria is

$$\{(In, A)\} \;\cup\; \{(Out,\ \sigma_2) : \sigma_2(F) \geq \tfrac{1}{2}\},$$

a single entry equilibrium plus a continuum of deterrence equilibria, (Out, F) among them.

Which are sequentially rational. At player 2's node, A gives $1 > 0$. Every equilibrium of the second family puts weight $f \geq \frac{1}{2}$ on F at a node that is off the equilibrium path (Def. 7.8), where F is not a best response. Backward induction (thick branches) keeps only (In, A), which is therefore the unique subgame-perfect equilibrium (Tadelis Fact, p. 158; slide 17).

(d) Interpretation. "Fight if you enter" deters entry only as long as nobody tests it. Once the entrant is in, the price war hurts the incumbent ($0 < 1$), so the threat is not credible, and an entrant who reasons backward enters. Nash equilibrium cannot see this: in (Out, F) the war is never fought, so planning it costs player 2 nothing. To deter entry, the incumbent needs a commitment made before the entry decision that makes fighting its best response after entry (excess capacity, a sunk investment, a contract). The Brexit case uses the same logic: the UK's sunk preparation makes rejecting a bad deal credible (slides 9–11).

(e) Verification. `check_D.py` builds the tree, enumerates the 2×2 pure profiles, tests every one for Nash and for subgame perfection by definition, and tests every mixed profile on a $\frac{1}{100}$ grid in both probabilities with exact fractions. Output: "Tadelis 7.6: 2 strategies each; pure NE (In,A), (Out,F); all NE = (In,A) and Out with Pr(F) >= 1/2; SPE (In,A) only". Tadelis prints no answer.

(f) Common mistakes.

- Giving player 1 the payoff ordering of a "most-preferred / least-preferred" list without placing *Out* in the middle. The middle outcome is implied, not stated.
- Answering (c) with (In, A) only. The question asks for Nash equilibria, and (Out, F) is one, as are all (Out, σ_2) with $\sigma_2(F) \geq \frac{1}{2}$.
- Calling (Out, F) not Nash because "player 2 would not fight". Player 2 is indifferent on the path; the flaw is off the path, which is what subgame perfection adds.

Answer. (a) Tree above, with $u(Out) = (1, 2)$, $u(In, A) = (2, 1)$, $u(In, F) = (0, 0)$. (b) Two pure strategies each: $\{In, Out\}$ and $\{A, F\}$. (c) Pure Nash equilibria (In, A) and (Out, F); with mixing, all Nash equilibria are (In, A) and (Out, σ_2) with $\sigma_2(F) \geq \frac{1}{2}$. Only (In, A) is subgame perfect: the threat to fight is not credible.

Item 3 (item 33 of the combined list)**: Tadelis Exercise 8.8** Foundations · difficulty 2/5

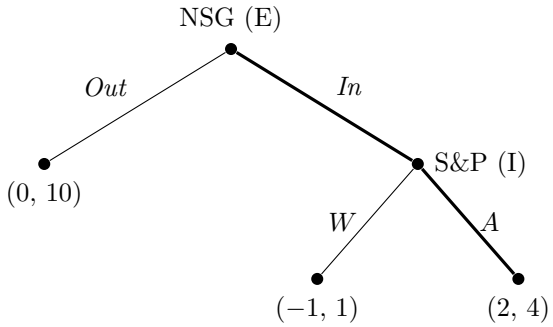
Source: Tadelis Exercise 8.8, page 171–172. The statement is reproduced verbatim in the list (item 33); it is not repeated here.

(a) What is asked. Set up the game tree of the entry story (NSG enters or not, then S&P fights or accommodates) and find the subgame-perfect equilibrium outcome.

(b) Method. Backward induction in a finite game of perfect information (Lecture 3, slide 30; Tadelis §8.1, Def. 8.1 and Prop. 8.1, pp. 152–153). With perfect information the subgame-perfect equilibria are exactly the profiles that survive backward induction (Lecture 4, slide 17; Tadelis §8.2, p. 157). Non-credible threats in Nash equilibrium: Lecture 4, slides 8 and 13. Same structure as the entry game of Cabral §7.2 (`Notas/L03/L03-06`).

(c) Solution. Notation. Players: NSG (the entrant, E) and S&P (the incumbent, I). E chooses *In* or *Out*; after *In*, I chooses W (price war) or A (accommodate). Payoffs (u_E, u_I) in \$ million of net present value. The entry cost of 2 is sunk once E enters, so it is subtracted from every payoff that follows *In*:

$$\begin{aligned}(u_E, u_I)(Out) &= (0,\ 10),\\ (u_E, u_I)(In, W) &= (1-2,\ 1) = (-1,\ 1),\\ (u_E, u_I)(In, A) &= (4-2,\ 4) = (2,\ 4).\end{aligned}$$



Payoffs (u_E, u_I); thick branches = backward-induction choices.

Step 1: the incumbent's node. After entry, S&P compares

$$u_I(In, A) = 4 > 1 = u_I(In, W),$$

so it accommodates. The price war is not a credible response.

Step 2: the entrant's node. Replacing S&P's node by its outcome $(2, 4)$, NSG compares

$$u_E(In) = 2 > 0 = u_E(Out),$$

so it enters. SPE: (In, A), outcome entry and accommodation, payoffs $(2, 4)$. The game has two proper subgames (the whole game and the one at S&P's node), and the SPE plays a Nash equilibrium in both.

Step 3: all Nash equilibria. Normal form (rows: E's strategies; columns: I's strategies):

	W	A
Out	$\underline{0},\ \underline{10}$	$0,\ \underline{10}$
In	$-1,\ 1$	$\underline{2},\ \underline{4}$

Pure Nash equilibria: (In, A) and (Out, W). With $w = \Pr_I(W)$, NSG's payoff from entering is $-w + 2(1-w) = 2 - 3w$, so Out is a best reply iff $0 \geq 2 - 3w$, i.e. $w \geq \frac{2}{3}$; S&P is not reached and is indifferent. The full set is

$$\{(In, A)\} \ \cup \ \{(Out, \ w\,W + (1-w)A) : w \geq \tfrac{2}{3}\}.$$

Every equilibrium in the second family fails sequential rationality at the node S&P never reaches (Tadelis Def. 8.1, p. 152): the threat "war with probability at least $\frac{2}{3}$" costs nothing because it is never carried out, and it would not be carried out ($1 < 4$).

Step 4: what would make the threat credible (deepening). Suppose S&P could, before NSG decides, commit to a penalty b that it pays if it accommodates (a public contract, a capacity investment, a "most-favored-customer" clause). Then at its node S&P fights iff $1 > 4 - b$, i.e. $b > 3$; NSG, anticipating war, stays out ($-1 < 0$), and S&P earns 10 without ever paying b. A credible commitment is worth up to $10 - 4 = 6$ to S&P. Independently, an entry cost above 4 would deter entry even with accommodation ($4 - \text{cost} < 0$).

(d) Interpretation. S&P's prediction of a price war is cheap talk: once NSG has sunk its \$2 million, a war gives S&P 1 instead of 4, and NSG knows it. These are the three lessons of the Brexit case (Lecture 4, slide 11) in miniature: *talk is cheap* (announcing war without changing payoffs is not credible); *sunk investments can be strategic* (only a sunk, payoff-changing move such as the penalty b makes war credible); *observability matters* (the commitment works only if NSG sees it before deciding). The outcome does not depend on how the entry cost is booked: if "each firm earns \$1 million" were already net of it, NSG would still enter ($2 > 0$, and even a war would pay $1 > 0$).

(e) Verification. `check_B_L04.py` recomputes both backward-induction steps, the two pure Nash equilibria of the 2×2 normal form, scans w on a grid of step $\frac{1}{300}$ (Nash equilibria exactly at $w = 0$ with entry and $w \geq \frac{2}{3}$ with Out), and checks the commitment thresholds $b > 3$ and the value 6. Output line: `tadelis-8-8: SPE (In, A); NE (Out, Pr(W) >= 2/3); commitment worth up to 6`. Tadelis prints no answer.

(f) Common mistakes.

- Forgetting to subtract the sunk entry cost from NSG's payoffs after entry.
- Reporting (Out, W) as an SPE because it is a Nash equilibrium.
- Giving the outcome only ("NSG enters") when the SPE is a strategy profile: S&P's plan A must be stated.

Answer. Tree above. Subgame-perfect equilibrium: NSG plays In, S&P plays A (accommodate); outcome: entry and accommodation, payoffs $(2, 4)$ in \$ million (NSG net of the entry cost). The other Nash equilibria, Out with $\Pr(W) \geq \frac{2}{3}$, rest on a non-credible threat.

Item 4 (item 34 of the combined list)**: Tadelis Exercise 8.6** Exam · difficulty 3/5

Source: Tadelis Exercise 8.6, page 170–171. The statement is reproduced verbatim in the list (item 34); it is not repeated here.

(a) What is asked. Firm 1 may pay F to cut its unit cost from 10 to 5 before a Cournot duopoly with $p = 100 - q$: (a) the threshold F^* below which the unique SPE has firm 1 investing; (b) for $F > F^*$, a Nash equilibrium that is not subgame perfect.

(b) Method. A two-stage game: an observed investment decision, then a simultaneous Cournot game (Gibbons §2.2.A, pp. 71–73; Tadelis §8.2). SPE: solve the Cournot subgame after each investment decision, then compare at the root (Lecture 4, slide 17; Tadelis Def. 8.3, p. 157). Cournot with asymmetric linear costs: first-order conditions, as in Lecture 2 and Gibbons §1.2.A. Non-SPE Nash equilibria by off-path threats (Lecture 4, slides 8 and 13).

(c) Solution. Notation. Firm 1 chooses I (invest) or N; both firms observe it, then choose $q_1, q_2 \geq 0$. Profits $\pi_i = (100 - q_1 - q_2 - c_i)q_i$, minus F for firm 1 after I.

Step 1: Cournot subgame with costs (c_1, c_2)**.** The first-order conditions are

$$100 - 2q_1 - q_2 - c_1 = 0, \qquad 100 - q_1 - 2q_2 - c_2 = 0,$$

so the best replies are $q_1 = \frac{100 - c_1 - q_2}{2}$, $q_2 = \frac{100 - c_2 - q_1}{2}$, and

$$q_1 = \frac{100 - 2c_1 + c_2}{3}, \qquad q_2 = \frac{100 - 2c_2 + c_1}{3}, \qquad \pi_i = q_i^2. \tag{9}$$

(The last equality uses $p - c_i = q_i$ from the first-order condition.) The subgame equilibrium is unique (the best replies have slope $-\frac{1}{2}$ and cross once).

Subgame	q_1	q_2	p	(π_1, π_2) before F
N: $c_1 = c_2 = 10$	30	30	40	(900, 900)
I: $c_1 = 5$, $c_2 = 10$	$\frac{100}{3}$	$\frac{85}{3}$	$\frac{115}{3}$	$(\frac{10000}{9}, \frac{7225}{9})$

Check for I: $q_1 = \frac{100-10+10}{3} = \frac{100}{3}$, $q_2 = \frac{100-20+5}{3} = \frac{85}{3}$, $p = 100 - \frac{185}{3} = \frac{115}{3}$, $p - 5 = \frac{100}{3} = q_1$.

(a) The threshold. Replace each subgame by its value. At the root firm 1 compares $\frac{10000}{9} - F$ (invest) with 900 (not):

$$F^* = \frac{10000}{9} - 900 = \frac{10000 - 8100}{9} = \frac{1900}{9} \approx 211.1.$$

For $F < F^*$ the unique SPE is: firm 1 invests; in each subgame both firms play the Cournot quantities of that subgame. For $F > F^*$ the unique SPE has no investment; at $F = F^*$ both choices are optimal and there are two SPE.

Decomposition (deepening). If firm 2 kept $q_2 = 30$, firm 1's best reply with cost 5 would be $\frac{95-30}{2} = 32.5$, profit $32.5^2 = \frac{4225}{4}$: a *direct* gain of $\frac{4225}{4} - 900 = \frac{625}{4} = 156.25$. Firm 2 in fact cuts its output to $\frac{85}{3}$, adding a *strategic* gain of $\frac{10000}{9} - \frac{4225}{4} = \frac{1975}{36} \approx 54.9$. So $F^* = \frac{625}{4} + \frac{1975}{36} = \frac{1900}{9}$: about a quarter of the value of the investment comes from its effect on the rival.

(b) A Nash equilibrium that is not subgame perfect, for $F > F^*$**.** A strategy of firm 1 is (invest?, q_1^N, q_1^I); of firm 2, (q_2^N, q_2^I). Consider

$$\text{firm 1: } (I,\ q_1^N = 0,\ q_1^I = \tfrac{100}{3}), \qquad \text{firm 2: } (q_2^N = 90,\ q_2^I = \tfrac{85}{3}).$$

Firm 2 threatens to flood the market if firm 1 does not invest.

- *Firm 2:* on the path (I) it plays its best reply $\frac{85}{3}$ to $\frac{100}{3}$; its plan after N is never reached, so changing it does not change its payoff.
- *Firm 1:* on the path it earns $\frac{10000}{9} - F$. If it deviates to N, it faces $q_2 = 90$, and its best reply earns $\max_{q_1}(100 - 90 - q_1 - 10)q_1 = \max_{q_1}(-q_1^2) = 0$. So the profile is a Nash equilibrium whenever $\frac{10000}{9} - F \geq 0$.

For $F \in (F^*, \frac{10000}{9}]$ this is a Nash equilibrium in which firm 1 invests although the SPE has no investment. It is not subgame perfect: in the subgame after N, $q_2 = 90$ is not a best reply (against $q_1 = 0$ firm 2's best reply is 45), so the flood is not credible.

For *any* $F > F^*$ (including $F > \frac{10000}{9}$) another example works: firm 1 plays $(N,\ 30,\ 0)$, firm 2 plays $(30,\ 90)$, i.e. it threatens to flood if firm 1 invests. On the path both play Cournot $(30, 30)$; deviating to I would earn at most $\max_{q_1}(100 - 90 - q_1 - 5)q_1 - F = \frac{25}{4} - F < 900$. It is Nash, but the subgame after I is played at $(0, 90)$, which is not a Nash equilibrium of that subgame.

(d) Interpretation. The investment is a sunk, observable commitment, the Brexit lesson of Lecture 4 (slide 11): it pays partly because it changes the rival's behavior (firm 2 retreats from 30 to $\frac{85}{3}$), not only because it lowers firm 1's cost. Because firm 2 cannot credibly promise to respond differently, firm 1's decision depends only on the Cournot continuation values; the non-SPE equilibria show how a non-credible threat by the *rival* could otherwise push firm 1 into, or away from, investment.

(e) Verification. `check_B_L04.py` solves both Cournot subgames with sympy (quantities, profits 900 and $\frac{10000}{9}$, $\frac{7225}{9}$), computes $F^* = \frac{1900}{9}$ and its decomposition $\frac{625}{4} + \frac{1975}{36}$, and for $F = \frac{1000}{3} \in (F^*, \frac{10000}{9}]$ checks the threat profile: firm 1's best payoff after N against $q_2 = 90$ is 0 (grid of step $\frac{1}{4}$), investing is a best reply, firm 2's on-path output is its best reply, and its best reply after N would be 45, not 90; for the second example, firm 1's best payoff after investing against $q_2 = 90$ is $\frac{25}{4}$ before F. Output line: `tadelis-8-6: F* = 1900/9; flood threat supports investing for F in (F*, 10000/9]`. Tadelis prints no answer.

(f) Common mistakes.

- Computing F^* with firm 2's output held at 30 ($F^* = 156.25$): firm 2 re-optimizes in the new subgame.
- Writing strategies without the off-path quantities: firm 2's strategy must say what it produces after I *and* after N, or no threat can even be stated.
- Giving as a "non-SPE Nash equilibrium" a profile in which a firm is not best-replying on the path (e.g. firm 1 investing when $F > \frac{10000}{9}$).

Answer. (a) $F^* = \frac{10000}{9} - 900 = \frac{1900}{9} \approx 211.1$: for $F < F^*$ the unique SPE has firm 1 investing, then $(q_1, q_2) = (\frac{100}{3}, \frac{85}{3})$; without investment the Cournot play is $(30, 30)$. (b) For $F \in (F^*, \frac{10000}{9}]$: firm 1 invests and plays $\frac{100}{3}$ (and 0 after N); firm 2 plays $\frac{85}{3}$ after I and threatens $q_2 = 90$ after N. Nash, not subgame perfect. (For any $F > F^*$: no investment, Cournot $(30, 30)$, with firm 2 threatening 90 after investment.)

Item 5 (item 35 of the combined list)**: Muñoz-Garcia Ch. 4, Exercise 11**Exam · difficulty 3/5

Source: Muñoz-Garcia Ch. 4, Exercise 11, page 180. The statement is reproduced verbatim in the list (item 35); it is not repeated here.

(a) What is asked. Stackelberg duopoly with $p = 1 - Q$ and costs $c_L \le c_F < 1$: (a) the follower's best response; (b) the SPNE strategies; (c) when both outputs are positive; (d) with $c_L = c_F = c$, total output and price, compared with monopoly, Cournot and Bertrand.

(b) Method. Backward induction in a perfect-information game with a continuum of actions: solve the follower's problem for *every* q_L, substitute into the leader's problem (Tadelis §8.3.2, "Stackelberg Competition", pp. 160–163; SPE Def. 8.3, p. 157). The Cournot benchmark is Tadelis §5.2.3, p. 87. Lecture 4: SPE definition and the backward-induction fact, slide 17; sunk, observable commitments, slide 11.

(c) Solution. **(a) Follower.** Given q_L, firm F solves $\max_{q_F \ge 0} (1 - q_L - q_F)q_F - c_F q_F$. The objective is strictly concave (second derivative -2), and the first-order condition $1 - q_L - 2q_F - c_F = 0$ gives

$$q_F(q_L) \;=\; \max\Big\{\frac{1 - c_F - q_L}{2},\; 0\Big\} \;=\; \frac{1 - c_F}{2} - \frac{q_L}{2} \quad \text{for } q_L \le 1 - c_F. \tag{10}$$

The best response falls with the leader's output (slope $-\frac{1}{2}$) and shifts down when c_F rises. For $q_L \ge 1 - c_F$ the residual demand no longer covers the follower's cost, and it produces nothing.

(b) Leader. On $q_L \le 1 - c_F$, substituting (10), the price is $1 - q_L - \frac{1 - c_F - q_L}{2} = \frac{1 + c_F}{2} - \frac{q_L}{2}$, and the leader solves

$$\max_{q_L} \Big(\frac{1 + c_F}{2} - \frac{q_L}{2} - c_L\Big) q_L \quad \Longrightarrow \quad \frac{1 + c_F}{2} - q_L - c_L = 0 \quad \Longrightarrow \quad q_L^* = \frac{1 + c_F - 2c_L}{2}.$$

The objective is strictly concave (second derivative -1). Then

$$q_F^* = \frac{1 - c_F}{2} - \frac{1 + c_F - 2c_L}{4} = \frac{1 - 3c_F + 2c_L}{4}, \qquad p^* = 1 - q_L^* - q_F^* = \frac{1 + c_F + 2c_L}{4}.$$

The SPNE is the *pair of strategies*

$$\Big(q_L^* = \tfrac{1 + c_F - 2c_L}{2},\;\; q_F(q_L) = \max\big\{\tfrac{1 - c_F - q_L}{2}, 0\big\} \text{ for every } q_L \ge 0\Big),$$

not just the two numbers: the follower's plan covers every output the leader could choose, on and off the path. Each firm's output falls with its own cost and rises with the rival's.

When the formula applies. $q_L^* \le 1 - c_F$ iff $c_L \ge \frac{3c_F - 1}{2}$, i.e. exactly when $q_F^* \ge 0$. Otherwise the leader's optimum is not interior and the follower is out of the market. For $q_L \ge 1 - c_F$ the leader is a monopolist with profit $(1 - q_L - c_L)q_L$, maximized at $\frac{1 - c_L}{2}$. Then:

- $c_L > \frac{3c_F - 1}{2}$: interior, as above, both firms produce;
- $2c_F - 1 < c_L \le \frac{3c_F - 1}{2}$: *limit output*, $q_L^* = 1 - c_F$, $q_F^* = 0$ (the leader's profit rises up to $1 - c_F$ on the left piece and falls beyond it on the right piece, because $\frac{1 - c_L}{2} < 1 - c_F$);
- $c_L \le 2c_F - 1$: *blockaded entry*, $q_L^* = \frac{1 - c_L}{2}$ (monopoly), $q_F^* = 0$.

For example $(c_L, c_F) = (\frac{1}{4}, \frac{1}{2})$ gives limit output $q_L^* = \frac{1}{2}$, and $(\frac{1}{10}, \frac{7}{10})$ gives blockaded entry $q_L^* = \frac{9}{20}$.

(c) Both outputs positive.

$$q_F^* > 0 \iff 1 - 3c_F + 2c_L > 0 \iff c_L > \frac{3c_F - 1}{2}.$$

The leader's condition $q_L^* > 0 \iff c_L < \frac{1+c_F}{2}$ never binds: with $c_L \le c_F < 1$, $q_L^* \ge \frac{1-c_F}{2} > 0$. So both firms produce iff

$$\frac{3c_F - 1}{2} < c_L \le c_F,$$

a non-empty band because $\frac{3c_F-1}{2} < c_F \iff c_F < 1$. If $c_F \le \frac{1}{3}$ the lower bound is not positive, so every $c_L \in (0, c_F]$ works. The leader's cost advantage must not be too large.

(d) Symmetric costs, $c_L = c_F = c$. The condition in (c) holds ($c > \frac{3c-1}{2}$), so the interior formulas apply:

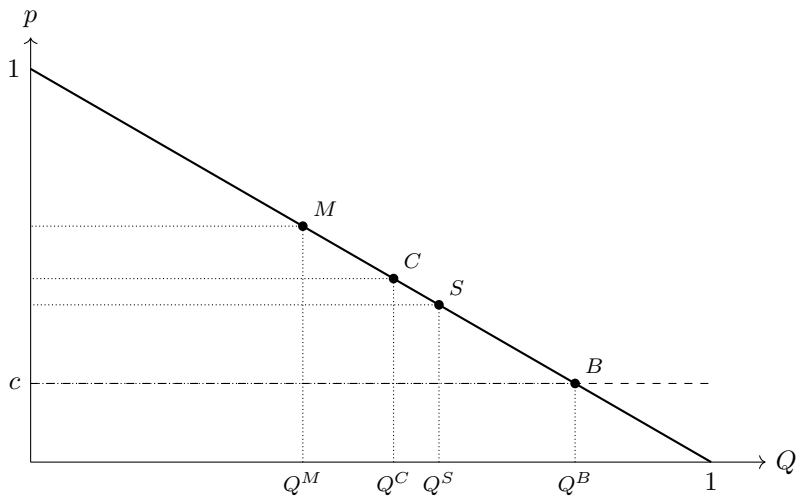
$$q_L^* = \frac{1-c}{2}, \qquad q_F^* = \frac{1-c}{4}, \qquad Q^S = \frac{3(1-c)}{4}, \qquad p^S = 1 - Q^S = \frac{1+3c}{4}.$$

The follower produces half of the leader's output. Benchmarks:

- *Monopoly*: $\max_Q (1 - Q - c)Q$ gives $Q^M = \frac{1-c}{2}$, $p^M = \frac{1+c}{2}$.
- *Cournot*: $q_i = \frac{1-c-q_j}{2}$ for both firms gives $q_i = \frac{1-c}{3}$, $Q^C = \frac{2(1-c)}{3}$, $p^C = \frac{1+2c}{3}$.
- *Bertrand* (= perfect competition with equal costs): $p^B = c$, $Q^B = 1 - c$.

Because $\frac{1}{2} < \frac{2}{3} < \frac{3}{4} < 1$,

$$Q^M < Q^C < Q^S < Q^B \qquad \text{and} \qquad p^M > p^C > p^S > p^B.$$



Demand $p = 1 - Q$ drawn for $c = 0.2$: $Q^M = 0.4$, $Q^C \approx 0.53$, $Q^S = 0.6$, $Q^B = 0.8$; prices $\frac{1+c}{2}$, $\frac{1+2c}{3}$, $\frac{1+3c}{4}$, c.

Profits: the leader earns $(p^S - c)q_L^* = \frac{(1-c)^2}{8}$, more than its Cournot profit $\frac{(1-c)^2}{9}$; the follower earns $\frac{(1-c)^2}{16}$, less.

(d) Interpretation. The leader's advantage comes from commitment, not from information: its output is observed and cannot be revised, so the follower takes it as given and contracts. The leader expands past its Cournot output ($\frac{1-c}{2} > \frac{1-c}{3}$) because it knows the follower will respond by cutting back. The follower would like to threaten "I produce $\frac{1-c}{3}$ whatever you do". Against that plan, $q_L = \frac{1-c}{3}$ is a best reply, so the Cournot outcome is a Nash equilibrium of the sequential game (Tadelis p. 162). The threat is not credible: once q_L is sunk, $\frac{1-c}{3}$ is not the follower's best

response to any $q_L \neq \frac{1-c}{3}$. Subgame perfection removes it (slide 17). The cost regimes in (b) carry the same idea: a large enough cost advantage lets the leader commit to the output $1 - c_F$, which leaves the follower no room to produce (limit output). This is the quantity version of entry deterrence through an observable, irreversible commitment (slide 11). For consumers, sequential quantity competition is more competitive than Cournot: total output rises from $\frac{2}{3}$ to $\frac{3}{4}$ of the competitive level.

(e) Verification. `check_D.py` derives BR_F, q_L^*, q_F^*, p^* and the cutoff $c_L = \frac{3c_F-1}{2}$ with sympy. For six cost pairs covering the three regimes it finds the leader's best output by exact search over $q_L \in \{k/2000\}$, using the truncated best response, and confirms the regime formula each time. It also checks the four symmetric totals, prices and profits, the strict ranking at four values of c, and that the Cournot outcome is a Nash but not a subgame-perfect profile. Output: "Munoz-Garcia 4.11: BR_F = (1-cF-qL)/2; qL* = (1+cF-2cL)/2, qF* = (1-3cF+2cL)/4; both > 0 iff cL > (3cF-1)/2; Q: monopoly (1-c)/2 < Cournot 2(1-c)/3 < Stackelberg 3(1-c)/4 < Bertrand 1-c (book: same)". *Comparison with Muñoz-Garcia's answer* (pp. 181–185): the same best response, SPNE, cutoffs (the book's C_A: $c_L > -\frac{1}{2} + \frac{3}{2}c_F$; C_B: $c_L < \frac{1}{2} + \frac{1}{2}c_F$, not binding below the 45° line), symmetric quantities and the ranking in its Fig. 4.34. The book omits the truncation at $q_F = 0$, so its best response and SPNE formula are valid only inside the region of part (c). The limit-output and blockaded regimes above are not in the book.

(f) Common mistakes.

- Reporting the SPNE as the numbers (q_L^*, q_F^*). The follower's strategy is the whole function $q_F(q_L)$; the numbers are the equilibrium outcome.
- Solving both first-order conditions simultaneously, which gives the Cournot quantities $\frac{1-c}{3}$, instead of substituting the follower's response into the leader's profit first.
- Using the interior formula $q_F^* = \frac{1-3c_F+2c_L}{4}$ when it is negative. When it is, the follower produces zero and the leader's problem changes.

Answer. (a) $q_F(q_L) = \max\{\frac{1-c_F-q_L}{2}, 0\}$. (b) SPNE: $q_L^* = \frac{1+c_F-2c_L}{2}$ and the follower plays $q_F(q_L)$ after every q_L; outcome $q_F^* = \frac{1-3c_F+2c_L}{4}$, $p^* = \frac{1+c_F+2c_L}{4}$ (valid when $q_F^* \geq 0$). (c) Both produce iff $c_L > \frac{3c_F-1}{2}$; $q_L^* > 0$ always. (d) $Q^S = \frac{3(1-c)}{4}$, $p^S = \frac{1+3c}{4}$; $Q^M = \frac{1-c}{2} < Q^C = \frac{2(1-c)}{3} < Q^S < Q^B = 1 - c$, and prices in the reverse order (book: same).

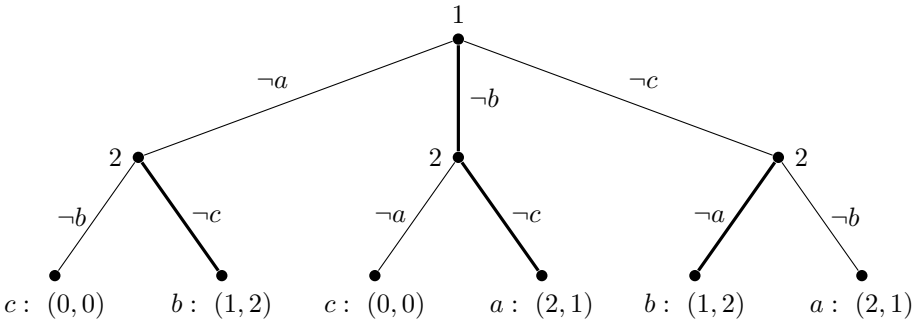
Item 6 (item 36 of the combined list)**: Tadelis Exercise 8.11** Exam · difficulty 3/5

Source: Tadelis Exercise 8.11, page 172–173. The statement is reproduced verbatim in the list (item 36); it is not repeated here.

(a) What is asked. (a) The game tree of the two-veto procedure with payoffs 2/1/0; (b) the SPE and whether it is unique; (c) Nash equilibria that are not subgame perfect; (d) whether player 2 would use a pre-game e-mail to remove a candidate, and which one.

(b) Method. Backward induction in a finite game of perfect information (Tadelis §8.1, Def. 8.1 and Prop. 8.1, pp. 152–153). Every decision node starts a subgame, so the SPE are exactly the backward-induction profiles (Def. 8.3, p. 157, and the Fact on p. 158). Nash equilibria through the normal form of the tree (§7.2.3, p. 143) and the on/off-path distinction (Def. 7.8, p. 146). Lecture 4: non-credible threats, slides 8 and 13; subgames, slide 14; SPE, slide 17.

(c) Solution. **(a) Tree.** Payoffs: $u_1(a,b,c) = (2,1,0)$ and $u_2(a,b,c) = (1,2,0)$. Player 1 vetoes one candidate, player 2 vetoes one of the two left, and the last one is assigned. "$\neg j$" means "veto j"; leaves show the assigned candidate and (u_1, u_2).



Thick branches = backward-induction choices.

Player 1 has 3 pure strategies. Player 2 has $2^3 = 8$: write $s_2 = (x_a, x_b, x_c)$, where x_j is the candidate she vetoes after player 1 vetoed j.

(b) Backward induction. Player 2's three nodes:

- after $\neg a$ ($\{b, c\}$ left): $\neg c$ leaves b, $u_2 = 2 > 0$, so $x_a = c$;
- after $\neg b$ ($\{a, c\}$ left): $\neg c$ leaves a, $u_2 = 1 > 0$, so $x_b = c$;
- after $\neg c$ ($\{a, b\}$ left): $\neg a$ leaves b, $u_2 = 2 > 1$, so $x_c = a$.

Player 1 anticipates the outcomes b, a, b of $\neg a$, $\neg b$, $\neg c$, worth $1, 2, 1$ to him, and vetoes b. The SPE is

$$\big(\ \neg b,\ \ s_2 = (c,\ c,\ a)\ \big), \qquad \text{outcome } a, \quad (u_1, u_2) = (2, 1).$$

It is unique: every comparison above is strict, so backward induction never has to break a tie. Player 1 vetoes the rival's favourite, b, not his own least preferred, c. Player 2 will veto c herself, so a veto spent on c is wasted.

(c) Nash equilibria that are not subgame perfect. Enumerating the $3 \times 8 = 24$ pure profiles gives 7 Nash equilibria:

Player 1	Player 2 (x_a, x_b, x_c)	outcome, (u_1, u_2)	SPE?
$\neg b$	(c, c, a)	a, $(2, 1)$	yes
$\neg b$	(b, c, a), (c, c, b), (b, c, b)	a, $(2, 1)$	no
$\neg a$	(c, a, a)	b, $(1, 2)$	no
$\neg c$	(c, a, a), (b, a, a)	b, $(1, 2)$	no

So yes, there are six. Two kinds:

- *Same outcome, different off-path plans.* Player 1 vetoes b and player 2 vetoes c on the path. Off the path she plans to veto b after $\neg a$ (leaving c: 0 instead of 2) or b after $\neg c$ (leaving a: 1 instead of 2). These nodes are never reached, so the plans cost her nothing, but they are not best responses in the subgames that start there.
- *A different outcome, supported by a threat.* With $x_b = a$, player 2 announces: "if you veto b, I veto a, and c gets the job." Facing it, $\neg b$ gives player 1 0, so he vetoes a or c and player 2 gets her favourite b. The threat is not credible: after $\neg b$ her choice is between a (1) and c (0), and she would veto c. Check of one of them, $(\neg c,\ (c, a, a))$: player 1 gets 1 against 0 from $\neg b$ and 1 from $\neg a$; player 2 gets 2, her maximum. The threatened node is off the path, so the threat is never tested.

No Nash equilibrium elects c: on the path, player 2 could always veto c instead and gain.

(d) The alienating e-mail. Read as follows: the e-mail comes first and its effect is seen by both. Then player 1 vetoes one of the two remaining candidates, and the last one is assigned (player 2's veto has nothing left to act on). In each two-candidate subgame player 1 keeps his preferred candidate:

Player 2's first move	candidates left	player 1 vetoes	outcome, (u_1, u_2)
no e-mail	$\{a, b, c\}$	b (part (b))	a, $(2, 1)$
e-mail a	$\{b, c\}$	c	b, $(1, 2)$
e-mail b	$\{a, c\}$	c	a, $(2, 1)$
e-mail c	$\{a, b\}$	b	a, $(2, 1)$

Player 2 gets 2 only by e-mailing a, so she does. In the SPE of the extended game, she e-mails a, player 1 vetoes c, and b is assigned, $(u_1, u_2) = (1, 2)$. The strategy profile also specifies the original SPE after "no e-mail" and player 1's vetoes after the other two e-mails.

(d) Interpretation. The order of moves decides the outcome. Moving first, player 1 removes the rival's favourite. Player 2 then cannot hurt him, because the only threat she has ("I'll leave you c") hurts her too. The e-mail works where the threat fails because it is an irreversible move made before player 1 acts, not a promise about later behaviour. Player 2 removes a, her own second choice. That looks self-defeating, but it takes away player 1's best option, and with only $\{b, c\}$ left he prefers b. A smaller choice set helps the player who controls it. This is the same commitment logic as the Brexit preparation (slide 11): an observable, sunk action changes what the opponent can do, while announcing future vetoes changes nothing ("talk is cheap", slide 11; non-credible threats, slide 13).

(e) Verification. `check_D.py` builds the tree, runs backward induction (asserting no ties), enumerates all 24 pure profiles and tests each one for Nash and for subgame perfection by definition (Nash in every subgame). It then builds the extended game with the e-mail move and

solves it backward. Output: "Tadelis 8.11: SPE veto b, then (bc: veto c, ac: veto c, ab: veto a), outcome a (2,1), unique; 7 pure NE (4 with outcome a, 3 with outcome b), 6 not SPE" and "Tadelis 8.11 (d): player 2 e-mails candidate a; player 1 then vetoes c; outcome b (1,2)". (In the script, node "bc" is player 2's node after $\neg a$, and so on.) Tadelis prints no answer.

(f) Common mistakes.

- Having player 1 veto his least-preferred candidate c. Player 2 would veto c anyway; the useful veto removes her favourite.
- Writing player 2's strategy as one veto. She has three information sets, so a strategy is a triple, and it is the off-path entries that separate the Nash equilibria from the SPE.
- In (d), removing c or b ("remove the worst" or "remove the rival's favourite" by analogy). Only removing a changes player 1's choice.

Answer. (a) Tree above; player 1 has 3 strategies, player 2 has 8. (b) Unique SPE: player 1 vetoes b; player 2 vetoes c after $\neg a$, c after $\neg b$, a after $\neg c$; a is selected, payoffs $(2, 1)$. (c) Yes: 7 pure Nash equilibria, 6 not subgame perfect. Three of them elect b, payoffs $(1, 2)$, using player 2's non-credible threat to veto a after $\neg b$. (d) Yes: she e-mails a; player 1 then vetoes c and b is selected, payoffs $(1, 2)$.

Item 7 (item 37 of the combined list)**: Tadelis Exercise 8.4** Exam · difficulty 4/5

Source: Tadelis Exercise 8.4, page 170. The statement is reproduced verbatim in the list (item 37); it is not repeated here.

(a) What is asked. Leader firm 1 chooses q_1; firms 2 and 3 observe it and choose q_2, q_3 simultaneously; $P = a - Q$, marginal cost c. (a) Number of proper subgames; (b) perfect or imperfect information; (c) the SPE and its uniqueness; (d) a Nash equilibrium that is not subgame perfect.

(b) Method. Proper subgames and perfect information (Lecture 4, slides 14–16; Lecture 3, slide 24; Tadelis Defs. 7.3 and 8.2, pp. 136 and 154). Two-stage game with a simultaneous second stage (Gibbons §2.2.A, pp. 71–73): for each q_1 solve the followers' Cournot game, then let firm 1 optimize (Stackelberg logic, Lecture 3, slide 37; Tadelis §8.3.2, pp. 160–163). Non-SPE Nash equilibria by off-path threats (Lecture 4, slides 8 and 13).

(c) Solution. Notation. $a > c \geq 0$; $q_i \geq 0$; profit $\pi_i = (a - Q - c)\, q_i$ with $Q = q_1 + q_2 + q_3$. A strategy of follower $j \in \{2, 3\}$ is a function $q_j(\cdot)$ of the observed q_1.

(a) Proper subgames. After each $q_1 \geq 0$ the game reaches a node at which firm 2 moves; that node is a singleton information set (everyone knows q_1), and the simultaneous choice of firm 3 lies in an information set contained in what follows it. So every q_1 starts a proper subgame: there is a **continuum** of them (one for each $q_1 \in [0, \infty)$), plus the whole game. No other node starts one: firm 3's nodes belong to an information set that pools all of firm 2's quantities.

(b) Imperfect information. Firms 2 and 3 move simultaneously, so one of them (say firm 3) does not observe the other's choice: its information set contains many nodes. The game has **imperfect information** (complete information, but not perfect).

(c) SPE and uniqueness.

Step 1: the subgame after q_1. If $q_1 < a - c$, the followers' first-order conditions are

$$\frac{\partial \pi_2}{\partial q_2} = a - c - q_1 - 2q_2 - q_3 = 0, \tag{11}$$

$$\frac{\partial \pi_3}{\partial q_3} = a - c - q_1 - q_2 - 2q_3 = 0. \tag{12}$$

Subtracting (12) from (11) gives $q_3 = q_2$; substituting, $a - c - q_1 - 3q_2 = 0$, so

$$q_2(q_1) = q_3(q_1) = \frac{a - c - q_1}{3}.$$

Uniqueness in the subgame. Each profit is strictly concave in the firm's own output, so best replies are $q_j = \max\{0, \frac{a-c-q_1-q_k}{2}\}$. The system (11)–(12) has determinant $2 \cdot 2 - 1 = 3 \neq 0$, so the interior solution is unique; a corner with $q_2 = 0 < q_3$ would need $q_3 = \frac{a-c-q_1}{2}$ and $0 \geq \frac{a-c-q_1-q_3}{2} = \frac{a-c-q_1}{4} > 0$, impossible. The best-reply map has slope $-\frac{1}{2}$, a contraction, so there is no other (also no mixed) equilibrium: the subgame Cournot game is dominance solvable. If $q_1 \geq a - c$, both followers' unique best reply is 0.

Step 2: the leader. For $q_1 < a - c$,

$$\pi_1 = \Big(a - q_1 - 2 \cdot \frac{a - c - q_1}{3} - c\Big) q_1 = \frac{a - c - q_1}{3}\, q_1,$$

strictly concave, with

$$\frac{d\pi_1}{dq_1} = \frac{a - c - 2q_1}{3} = 0 \;\Rightarrow\; q_1^* = \frac{a - c}{2}.$$

For $q_1 \geq a - c$ the price is at most c, so $\pi_1 \leq 0$; the interior choice is the unique global maximum.

Step 3: the SPE.

$$q_1^* = \frac{a-c}{2}, \qquad q_2^*(q_1) = q_3^*(q_1) = \max\Big\{0,\ \frac{a-c-q_1}{3}\Big\}.$$

On the path $q_2 = q_3 = \frac{a-c}{6}$, $Q^* = \frac{5(a-c)}{6}$, $P^* = \frac{a+5c}{6}$, $\pi_1^* = \frac{(a-c)^2}{12}$, $\pi_2^* = \pi_3^* = \frac{(a-c)^2}{36}$. *Uniqueness:* every subgame has a unique Nash equilibrium (Step 1), so in any SPE the followers must use exactly these functions; given them, firm 1's problem has a unique solution (Step 2).

(d) A Nash equilibrium that is not subgame perfect. Let $\hat{q} = \frac{a-c}{4}$ and consider

$$q_1 = \hat{q}, \qquad q_2(q_1) = q_3(q_1) = \begin{cases} \hat{q}, & q_1 = \hat{q}, \\ a - c, & q_1 \neq \hat{q}. \end{cases}$$

- *Followers:* on the path $q_1 = \hat{q}$, each best reply to $(\hat{q}, \hat{q})$ is $\frac{a-c-2\hat{q}}{2} = \frac{a-c}{4} = \hat{q}$. Their plans after other q_1 are off the path and do not affect their payoffs.
- *Leader:* on the path the outcome is three-firm Cournot, $\pi_1 = \frac{(a-c)^2}{16} > 0$. Any $q_1 \neq \hat{q}$ triggers $q_2 = q_3 = a - c$, so $Q \geq 2(a-c)$ and $P - c \leq -(a-c) < 0$: profit ≤ 0. So $\hat{q}$ is a best reply.

It is a Nash equilibrium with outcome $(\frac{a-c}{4}, \frac{a-c}{4}, \frac{a-c}{4})$, not the SPE outcome. It is not subgame perfect: after any $q_1 \neq \hat{q}$ (with $q_1 < a - c$) the pair $(a - c, a - c)$ is not a Nash equilibrium of the subgame (each follower's best reply to $a - c$ is 0).

(d) Interpretation. Moving first and being observed is a commitment: firm 1 produces the monopoly output $\frac{a-c}{2}$ and the followers' joint reply (slope $-\frac{2}{3}$) absorbs part of it, so the leader earns $\frac{(a-c)^2}{12}$ instead of the Cournot $\frac{(a-c)^2}{16}$, and total output rises. The non-SPE equilibrium in (d) shows how the followers could "undo" leadership with a threat to flood the market, but the threat is not credible once q_1 is sunk: this is the Lecture 4 contrast between Nash equilibria held up by threats and the subgame-perfect prediction.

(e) Verification. `check_B_L04.py` solves (11)–(12) and the leader's problem with sympy (determinant 3; $q_1^* = \frac{a-c}{2}$, $q_j^* = \frac{a-c}{6}$, second derivative negative); with $a = 13$, $c = 1$ it iterates the followers' best replies from three starting points for six values of q_1 (including $q_1 \geq a - c$) and always converges to $\max\{0, \frac{a-c-q_1}{3}\}$; it checks the threat profile of (d): on-path profit 9, every deviation on a grid of step $\frac{1}{4}$ earns ≤ 0, the followers' on-path reply is 3, and flooding is not a best reply. Output line: `tadelis-8-4: SPE q1 = (a-c)/2, q2 = q3 = (a-c)/6; flood-threat NE gives Cournot (a-c)/4 each`. Tadelis prints no answer. The SPE outcome agrees with the 01-04 solution of Gibbons 2.6 (same game).

(f) Common mistakes.

- Counting "two" subgames (one per stage): every value of q_1 starts its own subgame.
- Giving the followers' equilibrium *quantities* $\frac{a-c}{6}$ as their strategies: an SPE needs the reply functions $q_j(q_1)$.
- Solving the followers' problem as a Stackelberg chain (firm 2 before firm 3): here they move simultaneously, so the second stage is a Cournot duopoly on residual demand $a - q_1$.

Answer. (a) A continuum: one proper subgame for each $q_1 \geq 0$, plus the whole game. (b) Imperfect information (firms 2 and 3 move simultaneously). (c) Unique SPE: $q_1^* = \frac{a-c}{2}$, $q_2(q_1) = q_3(q_1) = \max\{0, \frac{a-c-q_1}{3}\}$; outcome $(\frac{a-c}{2}, \frac{a-c}{6}, \frac{a-c}{6})$, $P = \frac{a+5c}{6}$. (d) $q_1 = \frac{a-c}{4}$; followers play $\frac{a-c}{4}$ if $q_1 = \frac{a-c}{4}$ and $a - c$ otherwise: Nash, not subgame perfect.

Item 8 (item 38 of the combined list)**: Tadelis Exercise 8.13** Exam · difficulty 4/5

Source: Tadelis Exercise 8.13, page 173–174. The statement is reproduced verbatim in the list (item 38); it is not repeated here.

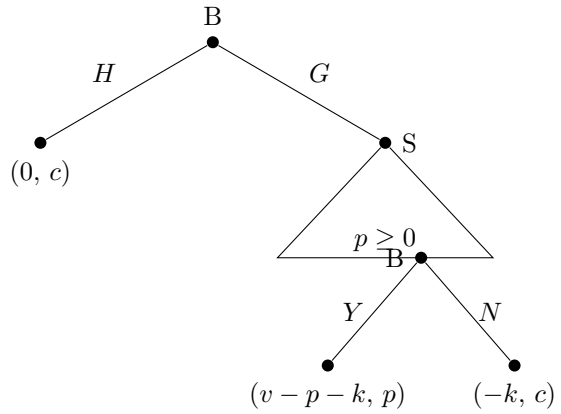
(a) What is asked. A buyer pays a commuting cost k to reach a seller, who then names a take-it-or-leave-it price. (a) Extensive form and the buyer's reply at the acceptance node; (b) the SPE, its uniqueness and Pareto optimality; (c) Nash equilibria better for both; (d) whether the seller would pay ε to mail a postcard that commits him to a price.

(b) Method. Backward induction with a continuum of actions (Lecture 3, slides 30 and 37; Tadelis §8.1 and §8.3.2, the Stackelberg logic, pp. 160–163); the ultimatum-game argument for ties (Tadelis Prop. 11.2, p. 223: the responder must accept at indifference, or the proposer has no best offer). Nash equilibria sustained by off-path threats (Lecture 4, slides 8 and 13). Commitment as a move added before the game (Lecture 4, slide 11; `Notas/L03/L03-06`, Cabral §7.2).

(c) Solution. Notation. Buyer (B) and seller (S). B chooses G (go to the store) or H (stay home); then S names $p \geq 0$; then B accepts (Y) or walks out (N). Data: $c > 0$, $k > 0$, $v > c + k$. Payoffs (u_B, u_S):

$$H:\ (0,\ c); \qquad G, p, Y:\ (v - p - k,\ p); \qquad G, p, N:\ (-k,\ c).$$

(a) Extensive form and the buyer's reply.



The triangle stands for the continuum of prices; one typical price p is drawn.

At the node after price p, the commuting cost is already sunk; it appears in both payoffs. B accepts if

$$v - p - k \ \geq\ -k \iff p \leq v.$$

So the best reply is: accept every $p < v$, reject every $p > v$, and either action at $p = v$.

(b) SPE, uniqueness, efficiency.

Step 1: the acceptance rule at $p = v$. Suppose B rejected $p = v$. Then S would get p for every $p < v$ and $c < v$ otherwise, so his payoff approaches v as $p \uparrow v$ but never reaches it; S would have no best offer, so the profile cannot be an SPE. Hence in every SPE B accepts if and only if $p \leq v$.

Step 2: the seller. Given that rule, S's payoff is p for $p \leq v$ and $c < v$ for $p > v$; the unique maximizer is $p^* = v$.

Step 3: the buyer's commuting decision. Going yields

$$u_B(G) = v - p^* - k = v - v - k = -k \;<\; 0 = u_B(H),$$

so B stays home. Every step had a unique optimum, so the SPE is unique:

B: H, and accept iff $p \le v$; S: offer $p = v$. Outcome: no trade, payoffs $(0,\ c)$.

Not Pareto optimal. Trade at any price $p \in (c,\ v-k)$ (nonempty since $v > c+k$) gives $u_B = v - p - k > 0$ and $u_S = p > c$: both strictly better. The lost surplus is $v - c - k > 0$. This is a hold-up: the commuting cost is sunk when the price is set, so S extracts all of v and B, anticipating it, never makes the trip.

(c) Better Nash equilibria. Yes. Fix $\hat{p} \in (c,\ v-k)$ and consider

B: G, accept iff $p \le \hat{p}$; S: offer $\hat{p}$.

S's best reply: offers $p \le \hat{p}$ are accepted and give $p \le \hat{p}$; offers above are rejected and give $c < \hat{p}$. So $\hat{p}$ is optimal. *B's best reply:* given $\hat{p}$, going and accepting gives $v - \hat{p} - k > 0$, more than staying (0) or going and rejecting $(-k)$; B's rule at prices other than $\hat{p}$ is off the path and does not affect his payoff. Payoffs $(v - \hat{p} - k,\ \hat{p})$ beat $(0, c)$ for both. The same construction is a Nash equilibrium for every $\hat{p} \in [c, v-k]$ (the endpoints give one player only a weak gain), so the set of Nash payoffs that Pareto-dominate the SPE is $\{(v - \hat{p} - k, \hat{p}) : \hat{p} \in (c, v-k)\}$. None is subgame perfect: after an off-path offer $p \in (\hat{p}, v)$, rejecting gives $-k < v - p - k$, so the threat to walk out is not credible.

(d) The postcard. Add a first move: S either mails a postcard committing to $\hat{p}$ (at cost ε) or does not. Without the postcard the game is the one in (b): payoff c to S. With it, B buys at $\hat{p}$ if he goes and $\hat{p} \le v$, so he goes iff $v - \hat{p} - k \ge 0$, i.e. $\hat{p} \le v - k$. By the argument of Step 1, B goes at indifference. Then S's payoff from the postcard is

$$u_S(\hat{p}) = \begin{cases} \hat{p} - \varepsilon, & \hat{p} \le v - k, \\ c - \varepsilon, & \hat{p} > v - k, \end{cases} \qquad \Rightarrow \qquad \hat{p}^* = v - k, \quad u_S = v - k - \varepsilon.$$

Compared with not mailing,

$$(v - k - \varepsilon) - c = (v - c - k) - \varepsilon > 0 \quad \text{since } \varepsilon < v - c - k.$$

So in the SPE the seller mails the postcard with price $v-k$; B goes and buys; payoffs $(0,\ v-k-\varepsilon)$. Trade now happens; the total surplus is $v - c - k - \varepsilon$, all of it to the seller.

(d) Interpretation. The commitment in (c) has to come from the buyer and is not credible; the commitment in (d) comes from the seller, who limits his own future options, and it is credible because the coupon binds him. These are the three Lecture 4 lessons (slide 11) in one example: the buyer's *sunk* commuting cost is what exposes him to hold-up; "I will only pay $\hat{p}$" is *cheap talk*; and the postcard works because it is *observable* before the buyer decides and changes what the seller can do afterwards. "Junk mail" has value even when it contains no information. The seller still captures the whole surplus; the coupon only removes the inefficiency, the same logic as a bond in the entry game (Cabral §7.2).

(e) Verification. `check_B_L04.py` uses $c = 2$, $v = 10$, $k = 1$, $\varepsilon = \frac{1}{2}$ and prices on a 0.25 grid: the seller's best offer is $p = 10 = v$ and the buyer stays home; under "reject at v" the best grid offer is $v - 0.25$ (no maximum on a continuum); thresholds $\hat{p} \in \{2.5, 5, 8.5\}$ give Nash equilibria that beat $(0, c)$ for both; with the postcard, $\hat{p}^* = 9 = v - k$ and $8.5 > 2$; sympy confirms the gain $(v - c - k) - \varepsilon$. Output line: `tadelis-8-13: SPE p = v, no trade; postcard at v - k is bought, seller nets v - k - eps`. Tadelis prints no answer.

(f) Common mistakes.

- Letting the buyer reject at $p = v$ in the SPE: then the seller has no optimal offer, so no SPE exists with that rule.
- Including k in the acceptance decision ("accept iff $p \leq v - k$"): k is sunk at that node.
- Pricing the postcard at v: the buyer would then not commute; the coupon must leave him $v - \hat{p} - k \geq 0$.

Answer. (a) Tree above; at the acceptance node the buyer accepts iff $p \leq v$ (the sunk k does not matter). (b) Unique SPE: seller offers $p = v$, buyer accepts iff $p \leq v$ and therefore stays home; payoffs $(0, c)$; not Pareto optimal, since trade at any $p \in (c, v - k)$ helps both. (c) Yes: buyer goes and accepts only $p \leq \hat{p}$, seller offers $\hat{p}$, for any $\hat{p} \in (c, v - k)$; Nash but not subgame perfect. (d) Yes: the seller mails a coupon at $\hat{p} = v - k$, the buyer comes and buys, seller earns $v - k - \varepsilon > c$.

Item 9 (item 39 of the combined list)**: Tadelis Exercise 8.7** Advanced · difficulty 4/5

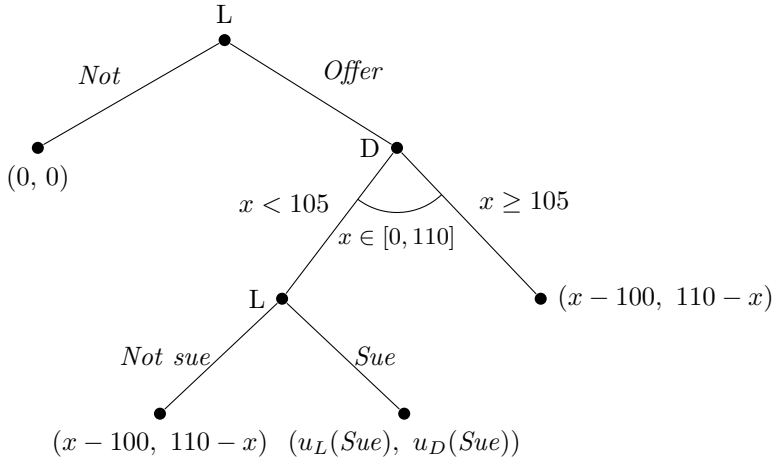
Source: Tadelis Exercise 8.7, page 171. The statement is reproduced verbatim in the list (item 39); it is not repeated here.

(a) What is asked. For the loan game with no courts, free courts and costly courts: the trees and SPE, with uniqueness ((a), (b), (d)); Nash equilibria that are not SPE ((c), (e)); what the lender would pay for a "loser pays the legal fees" law (f); and whether a planner should adopt it (g).

(b) Method. Backward induction with a continuum of actions: for every repayment x, solve the lender's suit decision; then the debtor's choice of x; then the lender's offer (Tadelis §8.1, Def. 8.1, p. 152; SPE Def. 8.3, p. 157; the Fact on p. 158). Nash equilibria that are not SPE rely on plans at nodes off the equilibrium path (Def. 7.8, p. 146; §8.2, pp. 157–158). Lecture 4: credible and non-credible threats, slides 8, 11 and 13; SPE, slide 17.

(c) Solution. Set-up and reading of the statement. Player 1 (lender, L) chooses *Offer* or *Not*. After *Offer*, player 2 (debtor, D) repays $x \in [0, 110]$. Repaying a negative amount is not an option, so $x \geq 0$. Without a suit, $(u_L, u_D) = (x - 100,\ 110 - x)$. When courts exist and $x < 105$, the lender then chooses *Sue* or *Not sue*. A suit brings the total repayment to 105, with the x already paid counting toward it. The debtor then pays the fine or the fee on top. The three legal regimes differ only in the payoffs after *Sue*:

	(b) free courts, fine 5	(d) lender pays fee 105	(f) debtor pays fee 105
$u_L(Sue)$	$105 - 100 = 5$	$105 - 100 - 105 = -100$	$105 - 100 = 5$
$u_D(Sue)$	$110 - 105 - 5 = 0$	$110 - 105 - 5 = 0$	$110 - 105 - 105 = -100$



Payoffs (u_L, u_D). The arc marks the continuum of repayments x. The lender's node after $x < 105$ exists only in (b), (d) and (f); in (a) every x ends the game with $(x - 100, 110 - x)$.

A pure strategy of the lender is an offer decision plus a suit rule $sue(x)$ for every $x < 105$: a function, not a single choice. A pure strategy of the debtor is a number x.

(a) No legal system. *Debtor:* $110 - x$ is strictly decreasing, so $x^* = 0$, giving $(u_L, u_D) = (-100, 110)$. *Lender: Not* gives $0 > -100$. SPE: (*Not*; $x = 0$), payoffs $(0, 0)$. It is unique, since both maximizers are unique.

(b) Costless courts. *Lender after a default $x < 105$: Sue* gives $5 > x - 100$, so he sues after every default. *Debtor:* any $x < 105$ is followed by a suit and gives 0; $x \in [105, 110]$ gives $110 - x \leq 5$, with the maximum at $x = 105$. Since $5 > 0$, $x^* = 105$. *Lender at the root: Offer* gives $105 - 100 = 5 > 0$. SPE:

$$\big(\textit{Offer},\ \textit{sue}(x) = \textit{Sue}\ \forall x < 105;\quad x = 105\big), \qquad (u_L, u_D) = (5, 5).$$

It is unique: every comparison is strict and $x = 105$ is the only maximizer.

(c) Nash equilibria of (b) that are not SPE: none in pure strategies.

- *No pure NE without lending.* Whatever x the debtor plans, the lender can switch to *Offer* and to a suit rule that is best at that x. He then gets 5 if $x < 105$ (sue) or $x - 100 \geq 5$ if $x \geq 105$, more than 0. So *Not* is never a best reply. Unlike a threat against the opponent, a lender's plan "I would not sue" cannot keep him out: he controls it.
- *With lending, the profile is forced.* If $x > 105$ the debtor gains by moving to 105. If $x < 105$ the lender gains by suing (if he does not) or the debtor gains by moving to 105 (if he does: $5 > 0$). So $x = 105$. If the suit rule skipped even one $x' < 105$, the debtor would move there and get $110 - x' > 5$. So the rule must be "sue after every $x' < 105$", which is the SPE.

So in pure strategies the only Nash equilibrium is the SPE. With behavioural strategies the answer changes, though not the outcome. The lender may randomize at unreached defaults and sue after x' with probability $s(x')$, provided the debtor still prefers 105: $(1 - s(x'))(110 - x') \leq 5$, i.e. $s(x') \geq 1 - \frac{5}{110 - x'}$ (e.g. $\frac{1}{6}$ at $x' = 104$). Any $s(x') < 1$ is not sequentially rational, since suing is strictly better. These are Nash equilibria that are not SPE, with the same path and payoffs $(5, 5)$.

(d) Costly courts (lender pays the fee). *Lender after a default $x < 105$: Sue* gives -100 and *Not sue* gives $x - 100$. Not suing is strictly better for every $x > 0$, and at $x = 0$ the two tie ($-100 = -100$). *Debtor:* if the lender does not sue at $x = 0$, the debtor's payoff $110 - x$ on $[0, 105)$ is maximized at $x = 0$, giving 110, which beats $110 - x \leq 5$ on $[105, 110]$. Could the lender instead sue at $x = 0$, with probability $s > 0$? The debtor would get $110(1 - s) < 110$ at $x = 0$ and $110 - x$ at every small $x > 0$. The supremum 110 is then not attained, so the debtor has no best reply. The tie at 0 must therefore be broken toward *Not sue*. *Lender at the root: Offer* gives $-100 < 0$. The unique SPE is

$$\big(\textit{Not},\ \textit{sue}(x) = \textit{Not sue}\ \forall x < 105;\quad x = 0\big), \qquad (u_L, u_D) = (0, 0),$$

the outcome of (a). A legal system that is too expensive to use is the same as none.

(e) Nash equilibria of (d) that are not SPE: yes.

- *Lending supported by a non-credible threat:*

  $$\big(\textit{Offer},\ \textit{sue}(x) = \textit{Sue}\ \forall x < 105;\quad x = 105\big), \qquad (u_L, u_D) = (5, 5).$$

  It is a NE. The debtor gets 5 at $x = 105$ and 0 at any default, since he is sued. The lender gets $5 > 0$, and no suit node is reached, so the suit rule costs him nothing. It is not an SPE. In the subgame after any default $x > 0$, suing gives $-100 < x - 100$. As in (c), this is the only NE with lending in pure strategies: one gap in the suit rule invites the debtor to default there. It is better for *both* players than the SPE $(0, 0)$. The threat is what would make lending possible, but it is not credible.
- *No lending, with other plans: Not*, any planned $x \leq 100$ and any suit decision at that x. Offering would give the lender at most $\max\{-100,\ x - 100\} = x - 100 \leq 0$. All of these except the SPE are Nash equilibria that are not subgame perfect, e.g. (*Not*, never sue; $x = 50$), because $x = 50$ is not the debtor's best reply after *Offer*.

(f) Loser pays the fees. After a default $x < 105$, *Sue* gives the lender $105 - 100 = 5 > x - 100$: the suit is now credible. The debtor gets -100 from any default and $110 - x$ from $x \in [105, 110]$,

so $x = 105$ ($5 > -100$). Offering gives $5 > 0$. Unique SPE: (*Offer*, *Sue* $\forall x < 105$; $x = 105$), payoffs $(5, 5)$. Compared with (d) the lender's equilibrium payoff rises from 0 to 5. Anything he pays for the reform is sunk before the game starts and does not change the SPE, so he gains iff the price is below \$5. He would pay *up to \$5*.

(g) The planner. Total surplus is 0 without the law (no loan) and $110 - 100 = 10$ with it. On the equilibrium path nobody defaults, so nobody is sued and the \$105 fee is never paid. The reform moves both players from $(0, 0)$ to $(5, 5)$, a Pareto improvement, so the planner should adopt it if enacting it costs less than \$10 (the debtor too would pay up to \$5). The harsh off-path penalty (-100 for the debtor) has no social cost in this model because it is never imposed. It would have one if default could happen by bad luck, which the model rules out.

(d) Interpretation. The project creates \$10 of value, but the lender can capture it only if the debtor believes default will be punished. In (a) and (d) the debtor's promise to repay is not credible, and in (d) neither is the lender's threat to sue, because a suit costs more than it recovers. Lending collapses even though both would gain. The Nash equilibrium in (e) shows the gains a credible threat would deliver. The SPE shows why the threat by itself does not work. The reform in (f) does not change what happens on the path; it changes who pays for enforcement *off* the path. That makes suing the lender's best reply to default, so the threat becomes credible and is never used. This is the Brexit lesson in a credit market (slides 11 and 13): the institution that makes a threat credible creates value, while announcing the threat creates none. Free courts in (b) and loser-pays in (f) give the same outcome.

(e) Verification. `check_D.py` encodes the payoffs of the four regimes exactly and works on the repayment grid $x \in \{k/8\}$, which contains 0, 100, 104.5 and 105. It computes the sequentially rational suit choice at every default, showing the tie at $x = 0$ in (d) and strict preferences elsewhere. It then computes the debtor's best x and the lender's offer. It tests the claimed Nash equilibria with unilateral deviations: the lender's joint offer/suit deviations and every debtor x. It confirms that a single gap in the suit rule breaks the threat equilibria and that no no-lending NE exists in (b). It also checks the behavioural bound $s(x') \geq 1 - \frac{5}{110 - x'}$. Output: "Tadelis 8.7: (a) no loan, x = 0, (0,0); (b) loan, x = 105, sue every x < 105, (5,5); (c) pure NE = SPE; (d) no loan, x = 0, never sue (tie at x = 0), (0,0); (e) NE with loan and threat to sue, (5,5), and no-loan NE with x <= 100; (f) lender pays up to 5; (g) surplus +10". Tadelis prints no answer.

(f) Common mistakes.

- Reading (c) as "the lender threatens not to sue, so he does not lend". The lender's own off-path plan cannot discipline the lender: he can change his whole strategy, offer and suit rule together. In (b) every Nash equilibrium in pure strategies lends.
- In (d), leaving the tie at $x = 0$ unresolved, or claiming an SPE where the lender sues only at $x = 0$. Then the debtor has no best reply, so that profile cannot be part of an equilibrium.
- In (f) and (g), counting the \$105 fee as a cost of the reform. It is paid only after a default, which never happens in equilibrium.

Answer. (a) SPE (*Not*; $x = 0$), payoffs $(0, 0)$, unique. (b) SPE: offer, repay $x = 105$, sue after every $x < 105$; payoffs $(5, 5)$, unique. (c) None in pure strategies: the only pure Nash equilibrium is the SPE (behavioural NE with off-path suit probability $s(x') \in [1 - \frac{5}{110-x'}, 1)$ exist, with the same outcome). (d) SPE: no loan, $x = 0$, never sue; payoffs $(0, 0)$, unique (the tie at $x = 0$ must go to not suing). (e) Yes: (offer, sue after every default; $x = 105$), payoffs $(5, 5)$, rests on a non-credible threat to sue; so do the other no-loan profiles with planned $x \leq 100$. (f) Yes, up to \$5 (his SPE payoff goes from 0 to 5). (g) Yes: surplus rises from 0 to 10, both gain \$5, and the fee is never paid in equilibrium.

Item 10 (item 40 of the combined list)**: MWG Exercise 9.B.5** Advanced · difficulty 5/5

Source: MWG Exercise 9.B.5, page 302. The statement is reproduced verbatim in the list (item 40); it is not repeated here.

(a) What is asked. For a finite two-player game played simultaneously or with player 1 moving first and observed: (a), (b) count strategies; (c) multiple SPNE imply a payoff tie; (d) without ties for player 2, moving first guarantees player 1 at least any pure Nash payoff of the simultaneous game, in every SPNE (and whether in every Nash equilibrium); (e) examples where (d) fails with ties for player 2 or with a mixed Nash equilibrium.

(b) Method. Strategies as complete plans (MWG §7.D; Lecture 3, slide 25); SPNE and backward induction in finite games of perfect information (MWG §9.B, Prop. 9.B.1 (Zermelo), p. 272, and Prop. 9.B.2; Lecture 3, slides 30 and 35; Lecture 4, slide 17). The sequential game in (b) has perfect information, so its SPNE are the backward-induction profiles; with no ties they are unique. Non-credible threats in Nash equilibrium: Lecture 4, slides 8 and 13.

(c) Solution. Notation. As in the statement, m_i is both the number of actions and (from (c) on) a typical action; write $|M_i|$ for the number to avoid confusion. In the sequential game a strategy of player 2 is a function $s_2 : M_1 \to M_2$.

(a) Simultaneous moves. Each player has one information set, so player i has $|M_i| = m_i$ pure strategies (and the mixed strategies $\Delta(M_i)$, a simplex of dimension $m_i - 1$).

(b) Player 1 first, observed. Player 1 still has one information set: m_1 pure strategies. Player 2 has one information set for each of player 1's moves, each with m_2 actions, so $m_2^{m_1}$ pure strategies (the functions $M_1 \to M_2$; item "Tadelis 7.1").

(c) Multiple SPNE imply a tie. Suppose, to the contrary, that no player is indifferent between any two distinct pairs: $\phi_i(m_1, m_2) \neq \phi_i(m_1', m_2')$ for $i = 1, 2$ whenever $(m_1, m_2) \neq (m_1', m_2')$.

- *Player 2's subgames.* After each m_1, the subgame is a one-player decision; any SPNE must play a best reply there. Since $\phi_2(m_1, \cdot)$ takes m_2 distinct values, the best reply $b(m_1)$ is unique and played with probability 1. So player 2's SPNE strategy is the unique function b.
- *Player 1.* Given b, choosing m_1 yields $\phi_1(m_1, b(m_1))$. The pairs $(m_1, b(m_1))$ differ across m_1 (their first components differ), so by assumption these m_1 payoffs are distinct, and the maximizer m_1^* is unique; any SPNE puts probability 1 on it.

So the SPNE is unique, (m_1^*, b), contradicting multiplicity. Hence if there are multiple SPNE, some player has a tie: (i) or (ii) holds for two distinct pairs. (In fact more is shown: either player 2 has a tie *within* some $\phi_2(m_1, \cdot)$, or player 1 has a tie between two pairs with different m_1.) □

(d) Moving first does not hurt player 1 (in SPNE). Assume (ii) never holds: player 2 has no ties. Let $(\hat{m}_1, \hat{m}_2)$ be a pure Nash equilibrium of the simultaneous game with $\pi_1 = \phi_1(\hat{m}_1, \hat{m}_2)$.

(i) In the simultaneous game $\hat{m}_2$ is a best reply to $\hat{m}_1$: $\phi_2(\hat{m}_1, \hat{m}_2) \geq \phi_2(\hat{m}_1, m_2)$ for all m_2. With no ties, it is the *unique* best reply, so $b(\hat{m}_1) = \hat{m}_2$.

(ii) In any SPNE of the sequential game, player 2 plays $b(m_1)$ after every m_1 (with probability 1, since best replies are unique). In particular she plays $\hat{m}_2$ after $\hat{m}_1$.

(iii) An SPNE is a Nash equilibrium, so player 1's equilibrium payoff is at least what he gets by deviating to $\hat{m}_1$, which is $\phi_1(\hat{m}_1, b(\hat{m}_1)) = \phi_1(\hat{m}_1, \hat{m}_2) = \pi_1$.

So in every SPNE player 1 gets at least π_1 (for every pure Nash equilibrium of (a), hence at least the largest such π_1). □

Not for every Nash equilibrium of (b). A Nash equilibrium of the sequential game may use a non-credible threat by player 2 after moves that player 1 does not make. Example (no ties for player 2; rows player 1, columns player 2, both choose O or F):

	O	F
O	2, 1	0, 0
F	0, −1	1, 2

Player 2's payoffs $1, 0, -1, 2$ are all distinct. The simultaneous game has the pure equilibrium (O, O) with $\pi_1 = 2$ (and (F, F) with 1). In the sequential game the unique SPNE is $(O,\ s_2 = (O \text{ after } O, F \text{ after } F))$, payoff 2 to player 1, as (d) predicts. But $(F,\ s_2 = (F, F))$ is also a Nash equilibrium: against "always F", player 1 gets 0 from O and 1 from F; against F, player 2's on-path choice F is optimal ($2 > -1$). Player 1 gets $1 < 2 = \pi_1$. The threat "F after O" ($0 < 1$ for player 2) is not credible.

(e) Two ways the conclusion of (d) fails.

(e.1) Player 2 has ties. Take the game of the MWG manual, with player 2's payoff depending only on player 1's move:

	L	R
U	0, 1	1, 1
D	0, 0	1, 0

(U, R) is a pure Nash equilibrium of the simultaneous game with $\pi_1 = 1$ (player 1: $1 \geq 1$; player 2: $1 \geq 1$). In the sequential game player 2 is indifferent at both nodes, so every reply is sequentially rational, and $(U, (L \text{ after } U, L \text{ after } D))$ is an SPNE (player 1 gets 0 from either move). Player 1's SPNE payoff is $0 < 1 = \pi_1$.

(e.2) A mixed Nash equilibrium instead of a pure one. Matching pennies:

	L	R
U	1, 0	0, 1
D	0, 1	1, 0

Player 2 has no ties within either row. The simultaneous game has no pure equilibrium; its mixed equilibrium $(\frac{1}{2}, \frac{1}{2})$ gives player 1 $\frac{1}{2}$. In the sequential game player 2 replies R after U and L after D, so player 1 gets 0 whatever he does: the SPNE are $(U, (R, L))$ and $(D, (R, L))$, both with payoff $0 < \frac{1}{2}$. Moving first and being observed is a *disadvantage* when player 1 wants to be unpredictable.

(d) Interpretation. Moving first helps when it is a commitment the follower must respond to (Stackelberg, entry deterrence); (d) makes this precise: if the follower's reply to each move is pinned down, the leader can always reproduce any pure Nash outcome of the simultaneous game by playing his part of it, so he cannot lose by moving first. The two failures in (e) mark the limits: if the follower is indifferent, her tie-breaking can punish the leader at no cost to herself; and if the leader's simultaneous-move advantage comes from being unpredictable (mixing), observation destroys it. The Nash-but-not-SPNE example in (d) is the Lecture 4 point once more: threats off the path can make the leader look worse off, but they are not credible.

(e) Verification. `check_B_L04.py` enumerates all pure SPNE (a strategy $s_2 : M_1 \to M_2$ that best replies in every subgame, and a best m_1) of 3000 random games with $m_1, m_2 \in \{2, 3\}$ and integer payoffs in $\{0, \ldots, 4\}$ (fixed seed). In each of the 1795 games with several SPNE a tie (i) or (ii) is present; in each of the 166 games without ties for player 2, every SPNE gives player 1 at least the payoff of every pure Nash equilibrium of the simultaneous game. It confirms the counterexample in (d) (unique SPNE pays 2, the Nash equilibrium $(F, (F, F))$ pays 1), (e.1) (SPNE $(U, (L, L))$ pays 0 against $\pi_1 = 1$) and (e.2) (SPNE $(U, (R, L))$, $(D, (R, L))$ pay 0; mixed value $\frac{1}{2}$). Output line: `mwg-9-b-5: (c) on 1795 random games with multiple SPNE, (d) on 166 games; examples (d), (e1), (e2) confirmed`. *Comparison with the MWG solutions manual* (Exercise 9.B.5, pp. 9-2 to 9-4; label OCR'd as "9.8.5"): (a) m_i pure strategies; (b) m_1 and $m_2^{m_1}$; (c) proof by contradiction, as here; (d) the same argument, and "low payoffs for player 1 could be sustained in a NE with incredible threats by player 2"; (e) the same two examples (the indifferent follower of Fig. 9.B.5(e.1) and matching pennies, Fig. 9.B.5(e.2)). **Agreement on every part.** One slip in the manual (or its OCR): the SPNE of the pennies game are written "$(U, (R, L))$ and $(L, (R, L))$"; the second must be $(D, (R, L))$, since L is not a move of player 1.

(f) Common mistakes.

- Giving player 2 $m_1 m_2$ strategies in (b) (pairs) instead of $m_2^{m_1}$ (functions).
- In (d), forgetting that the argument needs player 2's best reply to $\hat{m}_1$ to be *unique*: with a tie she might answer $\hat{m}_1$ differently in the sequential game, which is exactly example (e.1).
- Claiming that moving first is always an advantage: (e.2) shows it can be strictly worse.

Answer. (a) m_1 and m_2. (b) m_1 for player 1, $m_2^{m_1}$ for player 2. (c) Without ties player 2's reply to each move is unique and then player 1's best move is unique, so the SPNE is unique; multiple SPNE therefore require (i) or (ii). (d) In any SPNE player 2 answers $\hat{m}_1$ with her unique best reply $\hat{m}_2$, so player 1 can secure π_1 by playing $\hat{m}_1$; not for every Nash equilibrium (e.g. $(F, (F, F))$ in the example above gives $1 < 2$). (e) Indifferent follower: $(U, (L, L))$ is an SPNE with payoff $0 < 1$; matching pennies: SPNE payoff $0 <$ mixed Nash payoff $\frac{1}{2}$.

Lecture 5 — Repeated games and bargaining

Item 1 (item 41 of the combined list)**: Cabral Exercise 7.13** Foundations · difficulty 1/5

Source: Cabral Exercise 7.13, page 182. The statement is reproduced verbatim in the list (item 41); it is not repeated here.

(a) What is asked. Prove that when Cabral's Prisoner's Dilemma (Figure 7.1) is repeated a known number T of times, the only subgame-perfect equilibrium (SPE) has (B, R) played in every period.

(b) Method. Backward induction on the finitely repeated game $G(T, \delta)$ (Lecture 4, slide 22: each period starts a new subgame; slide 17: an SPE is a Nash equilibrium in every subgame; slides 9–10: backward induction). The result is Tadelis Prop. 10.1 (§10.1, p. 192), a special case of Prop. 9.3 (§9.4, p. 181): if the stage game has a unique Nash equilibrium, the finitely repeated game has a unique SPE, the stage equilibrium in every period after every history. Cabral sets up the repeated Prisoner's Dilemma in §7.3 (pp. 172–174). The proof is an induction on the number of periods left.

(c) Solution. Notation. Periods $t = 1, \ldots, T$; h^{t-1} is the history of action profiles before period t; player i's payoff is $\sum_{t=1}^{T} \delta^{t-1} u_i(a^t)$ with $\delta \in (0, 1]$ (slide 22). Stage payoffs (Figure 7.1, player 1 first):

	L	R
T	5, 5	3, $\underline{6}$
B	$\underline{6}$, 3	$\underline{4}$, $\underline{4}$

Step 1: the stage game. For player 1, B beats T against both columns: $6 > 5$ against L and $4 > 3$ against R. So B is strictly dominant; by symmetry R is strictly dominant for player 2. The stage game therefore has exactly one Nash equilibrium, (B, R) with payoffs $(4, 4)$, and no mixed equilibrium (a strictly dominated action gets zero weight in any equilibrium).

Step 2: the last period, $t = T$. Take any history h^{T-1}. The payoffs already earned, $\sum_{t<T} \delta^{t-1} u_i(a^t)$, are fixed and nothing follows period T. So the subgame after h^{T-1} is the stage game with every payoff multiplied by $\delta^{T-1} > 0$ and a constant added. Neither operation changes best replies, so the only Nash equilibrium of that subgame is (B, R). An SPE must therefore play (B, R) in period T after *every* history.

Step 3: the induction step. Suppose that in every SPE, (B, R) is played in periods $k+1, \ldots, T$ after every history. Take any history h^{k-1}. Whatever is played in period k, the continuation from $k+1$ on is (B, R) in each of the $T - k$ remaining periods, worth

$$W \;=\; \sum_{t=k+1}^{T} \delta^{t-1} \cdot 4,$$

the same constant for both players and independent of the period-k action. Period k's subgame therefore reduces to the stage game scaled by δ^{k-1} plus the constant W (plus the sunk past payoffs). Its unique Nash equilibrium is again (B, R). The claim holds for period k after every history.

Step 4: conclusion. By induction from $k = T$ down to $k = 1$, the unique SPE is: each player plays B (player 1) and R (player 2) in every period after every history. Its payoff is $4 \sum_{t=1}^{T} \delta^{t-1}$

for each player. The argument uses neither the value of δ nor the value of T, only that T is finite and known.

One-stage deviation check. At any history in period k, a single deviation to T (or L) changes only the period-k payoff, from 4 to 3 if the opponent plays R, while the continuation stays (B, R). The deviation loses δ^{k-1}, so no one-stage deviation pays (Tadelis Thm. 9.1, p. 185).

A stronger statement (Nash equilibrium paths). Even without subgame perfection, every Nash equilibrium of the repeated game has the *path* (B, R) in every period. Player 1 can always play B, which gives him at least 4 per period (row B pays 6 or 4), and 4 is also what (B, R) pays. Going backwards along the path: if the path had T (or L) in some period, the player could switch to B (or R) there and in every later period, earn one more in that period, and get at least 4 later, which is what the path gives later by the induction. The punishment threats that could deter this are worth nothing, because the minimax payoff, 4, equals the equilibrium payoff. `check_C_L05.py` confirms this for $T = 2$ by brute force.

(d) Interpretation. Cooperation in a finitely repeated game needs a threat for the future, and here there is none. In the last period there is no future, so the players defect. In the second-to-last period the last period's outcome is already fixed, so there is again nothing to protect, and the argument unravels back to period 1. Cabral's point in §7.3 is that an *indefinitely* repeated game, one that ends after each period with probability $1 - \alpha$ (Cabral's notation, p. 172) and so has no known last period, is needed for retaliation to sustain (T, L). A known end date, however distant, removes it.

(e) Verification. `check_C_L05.py` checks that B and R are strictly dominant and that (B, R) is the unique Nash equilibrium of the stage game and of the folded game with any added constant. For $T = 2$ it enumerates all $32 \times 32 = 1024$ pure strategy profiles (a period-1 action and a map from the four period-1 outcomes to a period-2 action) at $\delta \in \{\frac{1}{10}, \frac{1}{2}, 1\}$. It finds exactly one SPE (always B, always R), and every Nash equilibrium has the path $(B, R), (B, R)$. Output: "Cabral 7.13: unique stage NE (B,R); T=2 brute force (1024 profiles x 3 deltas): 1 SPE, all NE paths (B,R),(B,R)". Cabral prints no answer; the result matches Tadelis Prop. 10.1.

(f) Common mistakes.

- Showing only the outcome, "(B, R) every period", and not the strategy "(B, R) after *every* history". An SPE prescribes play off the path too, and the induction must cover every history.
- Arguing that δ large enough or T long enough allows cooperation. With a known last period and a unique stage equilibrium, neither matters.
- Folding the game forward ("if they cooperate today they will cooperate tomorrow") instead of starting at period T. Backward induction runs from the end.

Answer. B and R are strictly dominant, so (B, R) is the unique stage equilibrium. In period T every subgame is the stage game, so (B, R) is played after every history; given that, the continuation from any period k on does not depend on period-k actions, so period k is again the stage game plus a constant and (B, R) is played. By induction the unique SPE plays (B, R) in every period after every history, for any T and any δ.

Item 2 (item 42 of the combined list)**: Tadelis Exercise 10.2** Foundations · difficulty 2/5

Source: Tadelis Exercise 10.2, page 215. The statement is reproduced verbatim in the list (item 42); it is not repeated here.

(a) What is asked. In the infinitely repeated 3×3 Prisoner's Dilemma variant, find the discount factors for which (M, C) in every period, and then (T, L) in every period, can be sustained, and explain why the two answers differ.

(b) Method. Grim-trigger strategies and the one-stage deviation principle (Tadelis §10.3, pp. 197–200: grim trigger p. 198, Prop. 10.3 p. 198, condition (10.3) p. 200), and the general threshold (10.5), p. 202:

$$\delta \;\geq\; \delta^* = \frac{v_i^d - v_i^c}{v_i^d - v_i^*},$$

with v_i^c the cooperative stage payoff, v_i^d the best one-period deviation payoff, and v_i^* the stage Nash payoff used as punishment (`Notas/L05/L05-05`). Lecture 5 has no slides yet; the repeated-game set-up is Lecture 4, slides 21–22, and the SPE definition is slide 17.

(c) Solution. Notation. Present value $v_i = \sum_{t\geq1}\delta^{t-1}u_i(a^t)$ (Tadelis Def. 10.2, p. 193). Best replies are underlined:

	L	C	R
T	$6,\ 6$	$-1,\ 7$	$-2,\ \underline{8}$
M	$7,\ -1$	$4,\ 4$	$-1,\ \underline{5}$
B	$\underline{8},\ -2$	$\underline{5},\ -1$	$\underline{0},\ \underline{0}$

Step 1: the stage equilibrium. Row B beats T ($8>6$, $5>-1$, $0>-2$) and M ($8>7$, $5>4$, $0>-1$) against every column, so B is strictly dominant for player 1. The game is symmetric, so R is strictly dominant for player 2. The unique stage Nash equilibrium is (B, R), with $v_i^* = 0$. It is also each player's minimax payoff: player 2 can hold player 1 to $\min_c \max_r u_1(r, c) = \min\{8, 5, 0\} = 0$ by playing R, and B guarantees player 1 at least $\min\{8, 5, 0\} = 0$.

Step 2: the strategies. For a target profile (a_1, a_2), grim trigger is:

- *cooperation phase*: in period 1, and after any history in which (a_1, a_2) was played in every period, player 1 plays a_1 and player 2 plays a_2;
- *punishment phase*: after any other history, player 1 plays B and player 2 plays R forever.

By Prop. 10.3 this is an SPE if and only if no player gains from a single deviation at a single history. There are two kinds of histories, so there are two checks.

Step 3: punishment phase (both parts). The continuation is (B, R) forever whatever is played today, so a one-period deviation changes only today's payoff. Against R, player 1 gets 0 with B, -1 with M and -2 with T. Deviating loses, for every δ, and player 2 is symmetric.

(a) (M, C)**.** Cooperation phase. Staying gives 4 every period:

$$v_1^{\text{coop}} = \frac{4}{1-\delta}.$$

The best one-period deviation against C is B ($5>4>-1$), and it starts the punishment phase, which pays 0 forever:

$$v_1^{\text{dev}} = 5 + \delta\cdot\frac{0}{1-\delta} = 5.$$

No deviation pays iff

$$\frac{4}{1-\delta} \geq 5 \iff 4 \geq 5-5\delta \iff 5\delta \geq 1 \iff \delta \geq \frac{1}{5}.$$

Check with (10.5): $\delta^* = \frac{5-4}{5-0} = \frac{1}{5}$. Player 2's problem is the same by symmetry. So (M, C) in every period is an SPE outcome iff $\delta \geq \frac{1}{5}$.

(b) (T, L)**.** Staying gives $\frac{6}{1-\delta}$. Against L, player 1's best deviation is B (payoff 8), not M (payoff 7). It is followed by 0 forever:

$$\frac{6}{1-\delta} \geq 8 \iff 6 \geq 8-8\delta \iff 8\delta \geq 2 \iff \delta \geq \frac{1}{4}.$$

Check: $\delta^* = \frac{8-6}{8-0} = \frac{1}{4}$. The same holds for player 2. So (T, L) in every period is an SPE outcome iff $\delta \geq \frac{1}{4}$.

These thresholds cannot be beaten. The punishment (B, R) forever holds the deviator to 0, his minimax payoff, and he can guarantee 0 in every period by playing B. So in *any* equilibrium (any punishment, even a Nash equilibrium that is not subgame perfect) a deviation from the stationary path is worth at least 5 in (a) and 8 in (b). The thresholds $\frac{1}{5}$ and $\frac{1}{4}$ are therefore necessary as well as sufficient.

Why the answers differ. Write (10.5) as $\delta^* = g/(g+\ell)$, with $g = v^d - v^c$ the one-period gain and $\ell = v^c - v^*$ the per-period loss once punished:

target	g	ℓ	g/ℓ	δ^*
(M, C)	$5-4=1$	$4-0=4$	1/4	1/5
(T, L)	$8-6=2$	$6-0=6$	1/3	1/4

Moving from (M, C) to (T, L) raises the value of cooperating (the loss ℓ rises from 4 to 6), but it raises the temptation more in proportion (the gain g doubles, from 1 to 2). The ratio g/ℓ rises, so more patience is needed. For $\delta \in [\frac{1}{5}, \frac{1}{4})$ the moderate outcome (M, C) can be sustained and the efficient (T, L) cannot.

(d) Interpretation. More ambitious cooperation is harder to sustain when it also creates a bigger temptation. (T, L) is better for both players than (M, C), but at (T, L) each player gives up a lot today by not grabbing the 8. Between moderately patient partners the self-enforcing agreement is the modest one. The same trade-off runs through collusion (Tadelis §10.4): higher collusive prices raise both the gain from undercutting and the value of the cartel.

(e) Verification. `check_C_L05.py` confirms that B and R are strictly dominant, that (B, R) is the unique stage equilibrium with payoff 0 = minimax, and that the best deviations are B against C (5) and B against L (8, versus 7 for M). Using exact fractions it tests the cooperation-phase inequality at δ^*, $\delta^* \pm \frac{1}{1000}$ and on the grid $\delta = k/1000$: it holds exactly for $\delta \geq \frac{1}{5}$ in (a) and $\delta \geq \frac{1}{4}$ in (b). The punishment phase has no profitable deviation. Output: "Tadelis 10.2: stage NE (B,R) (0,0); (a) delta* = 1/5, (b) delta* = 1/4; grid k/1000 and +-1/1000 around each cutoff pass". Tadelis prints no answer.

(f) Common mistakes.

- Using M as the deviation from (T, L), because it is "next door". That gives $\frac{7-6}{7-0} = \frac{1}{7}$, which is wrong: the deviation to use is the *best* reply, B.

- Checking only the cooperation phase. Prop. 10.3 needs the punishment phase too; here it holds because (B, R) is a stage equilibrium, and that must be said.
- Reading "(T, L) is better for both" as "(T, L) is easier to sustain". The threshold depends on the ratio of temptation to loss, not on how good the outcome is.

Answer. Unique stage equilibrium (B, R), payoff $(0, 0)$ (also the minimax). With grim trigger (revert to (B, R) forever): (a) (M, C) is sustained iff $\frac{4}{1-\delta} \geq 5$, i.e. $\delta \geq \frac{1}{5}$; (b) (T, L) is sustained iff $\frac{6}{1-\delta} \geq 8$, i.e. $\delta \geq \frac{1}{4}$. Since the punishment is the minimax, no other strategies do better. The answers differ because $\delta^* = g/(g + \ell)$: at (T, L) the temptation doubles ($g = 2$ against 1) while the loss from punishment rises only from 4 to 6.

Item 3 (item 43 of the combined list)**: Muñoz-Garcia Ch. 6 Exercise 1** Foundations · difficulty 2/5

Source: Muñoz-Garcia Ch. 6 Exercise 1, page 274. The statement is reproduced verbatim in the list (item 43); it is not repeated here.

(a) What is asked. In the infinitely repeated Prisoner's Dilemma of Table 6.1, can (NC, NC) be an SPE outcome when a deviation is punished by (C, C) for (a) one period, (b) two periods, and (c) for ten periods, what is the threshold δ?

(b) Method. Temporary Nash reversion ("length-k punishment") and the one-stage deviation principle (Tadelis Prop. 10.3, p. 198; grim trigger and (10.3), pp. 198–200; the general threshold (10.5), p. 202). Compare the stream from cooperating with the stream from deviating once and then being punished for k periods; the two streams coincide from period $k+2$ on. Lecture 5 has no slides yet; the repeated-game set-up is Lecture 4, slides 21–22. Notation as in `Notas/L05/L05-04-grim-trigger-spe.md`.

(c) Solution. Notation. Stage payoffs (player 1 first): $(C,C) = (2,2)$, $(C,NC) = (6,0)$, $(NC,C) = (0,6)$, $(NC,NC) = (3,3)$. Present values $\sum_{t\geq 0} \delta^t u_i^t$, measured from the period in which a player considers deviating. A length-k punishment strategy has two phases:

- *cooperation*: play NC in period 1 and as long as no deviation from the prescribed actions has occurred;
- *punishment*: after a deviation, play C for k periods, then return to cooperation (a deviation during the punishment restarts it).

Step 1: the stage game. C is strictly dominant ($2 > 0$ against C, $6 > 3$ against NC), so (C,C), with $(2,2)$, is the unique stage equilibrium. The deviation gain is $g = 6 - 3 = 3$; each punishment period costs $\ell = 3 - 2 = 1$.

Step 2: the general condition. Cooperating gives $3 + 3\delta + \dots$. Deviating gives 6 now, 2 in each of the next k periods, then 3 forever. From period $k+1$ on (counting the current period as 0) the streams are equal, so the comparison is over periods $0, \dots, k$:

$$3 + 3\sum_{j=1}^{k} \delta^j \;\geq\; 6 + 2\sum_{j=1}^{k} \delta^j \iff \sum_{j=1}^{k} \delta^j \;\geq\; 3 \iff \frac{\delta(1-\delta^k)}{1-\delta} \;\geq\; 3.$$

In words: the one-period gain, 3, must be covered by k discounted losses of 1 each. Since $\sum_{j=1}^{k} \delta^j < k$ for $\delta < 1$, no punishment of length $k \leq 3$ can ever work.

(a) One period ($k = 1$). The condition is $\delta \geq 3$: impossible for $\delta < 1$. Explicitly, $3 + 3\delta \geq 6 + 2\delta \iff \delta \geq 3$. Cooperation cannot be supported.

(b) Two periods ($k = 2$). The condition is

$$\delta + \delta^2 \geq 3 \iff \delta^2 + \delta - 3 \geq 0.$$

The roots are $\delta = \frac{-1 \pm \sqrt{13}}{2} \approx -2.30,\ 1.30$; the quadratic is negative on $(0,1)$ (at most $1+1-3 = -1$). Cooperation cannot be supported.

(c) Ten periods ($k = 10$). The condition is

$$\delta + \delta^2 + \dots + \delta^{10} \geq 3.$$

The left side rises from 0 to 10 on $[0,1]$, so there is exactly one root in $(0,1)$. Numerically (bisection or a polynomial root-finder),

$$\delta_{10} \approx 0.7627.$$

Cooperation is an SPE outcome iff $\delta \geq \delta_{10}$. For comparison, the thresholds for other lengths are $\delta_4 \approx 0.888$, $\delta_5 \approx 0.834$, $\delta_6 \approx 0.805$, $\delta_8 \approx 0.775$, $\delta_{20} \approx 0.7506$, and grim trigger ($k = \infty$) needs $\frac{\delta}{1-\delta} \geq 3$, i.e. $\delta \geq \frac{3}{4}$ ((10.5): $\frac{6-3}{6-2} = \frac{3}{4}$).

Step 3: the punishment phase. In a punishment period the prescribed action is C, a stage best reply to C. Deviating to NC pays $0 < 2$ today and either leaves the future unchanged or, with the restart rule, lengthens the punishment, which pays $2 < 3$ per period. Either way the deviation loses, for every δ. So the only binding constraint is Step 2's, and the profile is an SPE exactly when that holds.

(d) Interpretation. A punishment deters only if the deviator loses more than he grabs. Here the grab is 3 and each period of punishment costs only 1, so at least four periods are needed, and even then only very patient players are deterred. Longer punishments lower the threshold, with grim trigger as the limit. Short, forgiving punishments are attractive (mistakes are not punished forever), but they need a punishment phase that hurts much more per period than reverting to the Nash equilibrium does here.

(e) Verification. `check_C_L05.py` confirms that (C,C) is the unique stage equilibrium and derives the condition $\sum_{j=1}^{k} \delta^j \geq 3$ symbolically for $k = 1,2,3,4,10$. It shows (a), (b), and also $k = 3$ fail on the grid $\delta = j/1000$. For (c) it finds the single root in $(0,1)$, $0.762700984\ldots$, and brackets it in exact fractions: the condition fails at 0.762700, holds at 0.762701, fails at 0.76 and holds at 0.77. It also checks that the thresholds decrease in k towards $\frac{3}{4}$. Output: "Munoz-Garcia 6.1: k=1 and k=2 never work (k=3 needs delta=1); k=10: delta >= 0.762701 (book: 0.76, which fails); grim: 3/4; k=4: 0.8882". *Comparison with Muñoz-Garcia's answer* (pp. 275–278): the same conditions and conclusions for (a) and (b) ($\delta \geq 3$; $\delta^2 + \delta - 3 \geq 0$ with roots -2.3 and 1.3). For (c) the book reports $\delta \geq 0.76$, which is the threshold rounded *down*: at $\delta = 0.76$ the sum is about $2.963 < 3$, so cooperation fails. The correct statement is $\delta \geq 0.7627$. Two slips in the book's text for (a): it says the punisher finds "NC a best response to player i playing NC", meaning C against C; and it calls (NC, NC) the "competitive outcome" where it means the cooperative one. The book does not check the punishment phase.

(f) Common mistakes.

- Calling these strategies "tit-for-tat". True tit-for-tat copies the opponent's last move, which is not the same as "both play C for k periods". Tit-for-tat proper is generally *not* subgame perfect, because after a deviation it requires a player to punish while being punished. The book uses the name loosely.
- Comparing the streams only up to period 1 ("$6 + 2\delta$ versus $3 + 3\delta$") when $k > 1$. All k punishment periods enter the comparison.
- Rounding the threshold to 0.76. A threshold must be rounded up, or stated as the exact root.

Answer. With a k-period reversion to (C,C), cooperation is an SPE iff $\sum_{j=1}^{k} \delta^j \geq 3$ (the punishment phase never binds). (a) $k = 1$: $\delta \geq 3$, impossible. (b) $k = 2$: $\delta^2 + \delta - 3 \geq 0$, root $\frac{-1+\sqrt{13}}{2} \approx 1.30 > 1$, impossible. (c) $k = 10$: $\delta + \delta^2 + \cdots + \delta^{10} \geq 3$, i.e. $\delta \geq \delta_{10} \approx 0.7627$ (the book's 0.76 is slightly too low). Grim trigger would need $\delta \geq \frac{3}{4}$.

Item 4 (item 44 of the combined list)**: Muñoz-Garcia Ch. 6 Exercise 9** Exam level · difficulty 3/5

Source: Muñoz-Garcia Ch. 6 Exercise 9, page 305. The statement is reproduced verbatim in the list (item 44); it is not repeated here.

(a) What is asked. For the infinitely repeated 2×2 game of Table 6.6: (a) the stage Nash equilibrium; (b) the feasible payoff set and the normalization used; (c) the feasible, individually rational set; (d) a Nash equilibrium that gives $(9, 9)$ every period, and the condition it needs on δ.

(b) Method. Stage equilibrium by strict dominance (Tadelis §4.1). Feasible payoffs as the convex hull of the stage payoffs (Tadelis Defs. 10.6–10.7, p. 209–210) with payoffs measured as average discounted payoffs (Def. 10.3, p. 194). Individual rationality relative to the minimax payoff, and the folk theorem (Theorem 10.1, p. 211; MWG Prop. 12.AA.5, p. 422, for the minimax version). Part (d) uses grim trigger and the threshold (10.5), p. 202, checked by the one-stage deviation principle (Prop. 10.3, p. 198). Lecture 5 has no slides yet; the repeated-game set-up is Lecture 4, slides 21–22. Notation as in `Notas/L05/L05-06-folk-theorem.md`.

(c) Solution. Notation. $q = \Pr(L)$ in a mixed strategy of player 2, $p = \Pr(U)$ for player 1. The average payoff of a stream is $\bar{v}_i = (1-\delta)\sum_{t\geq 1} \delta^{t-1} u_i(a^t)$. Best replies are underlined:

	L	R
U	$9,\ 9$	$1,\ \underline{10}$
D	$\underline{10},\ 1$	$\underline{7},\ \underline{7}$

(a) Stage equilibrium. For player 1, D beats U against both columns ($10 > 9$, $7 > 1$), so D is strictly dominant; by symmetry R is strictly dominant for player 2. The only outcome consistent with Nash equilibrium play is (D, R), payoffs $(7, 7)$. There is no mixed equilibrium: for every q, $u_1(D, q) = 10q + 7(1-q) = 7 + 3q$ exceeds $u_1(U, q) = 9q + 1(1-q) = 1 + 8q$, because $(7 + 3q) - (1 + 8q) = 6 - 5q > 0$ on $[0, 1]$. The game is a Prisoner's Dilemma: (U, L) gives both $9 > 7$.

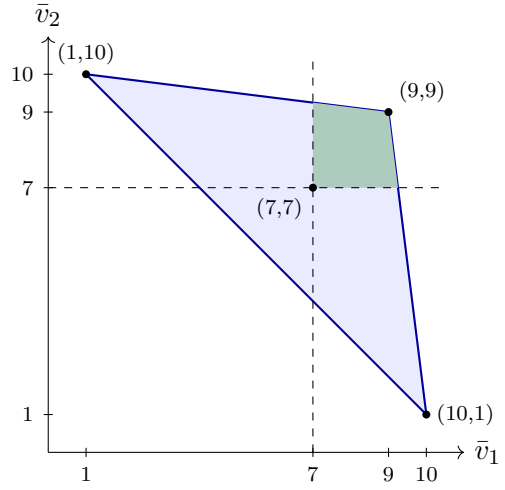
(b) Feasible payoffs. *Normalization.* Repeated-game payoffs are measured as average discounted payoffs, $\bar{v}_i = (1-\delta)v_i$. With this scaling a constant stream x is worth x (Tadelis (10.2), p. 195), so repeated-game payoffs are in the same units as stage payoffs and can be drawn in the same plane. *Convexification.* The four pure payoff vectors are $(9, 9)$, $(1, 10)$, $(10, 1)$, $(7, 7)$. Every point of their convex hull is reached by playing the outcomes with the corresponding long-run frequencies (exactly with a public randomization device, as in the book, or approximately for δ near 1 with deterministic sequences). The feasible set is

$$V = \text{CoHull}\{(9, 9), (1, 10), (10, 1), (7, 7)\}.$$

Its shape. $(7, 7)$ lies strictly inside the triangle with vertices $(1, 10)$, $(10, 1)$, $(9, 9)$, since

$$(7, 7) = \tfrac{3}{7}(9, 9) + \tfrac{2}{7}(1, 10) + \tfrac{2}{7}(10, 1),$$

with all weights positive. So V is that triangle. Its edges are: $v_1 + v_2 = 11$ from $(1, 10)$ to $(10, 1)$; $v_2 = 10 - \frac{v_1 - 1}{8}$ from $(1, 10)$ to $(9, 9)$; and $v_1 = 10 - \frac{v_2 - 1}{8}$ from $(10, 1)$ to $(9, 9)$.



Feasible set V (triangle) and the FIR set (shaded, without its edges on $\bar{v}_1 = 7$ and $\bar{v}_2 = 7$).

(c) Individually rational payoffs. Player 1's minimax payoff is

$$\underline{v}_1 = \min_{q\in[0,1]} \max\{u_1(U,q),\, u_1(D,q)\} = \min_{q} (7+3q) = 7 \quad (q=0,\ \text{i.e. } R),$$

because D is the best reply for every q. By symmetry $\underline{v}_2 = 7$ (player 1 plays D). Here the minimax payoff equals the stage Nash payoff. The feasible, individually rational (FIR) set is

$$\{\bar{v} \in V : \bar{v}_1 > 7,\ \bar{v}_2 > 7\}.$$

Cutting the triangle with $\bar{v}_1 = 7$ (on the edge $(1,10)$–$(9,9)$: $\bar{v}_2 = 10 - \frac{6}{8} = \frac{37}{4}$) and with $\bar{v}_2 = 7$ (on the edge $(10,1)$–$(9,9)$: $\bar{v}_1 = \frac{37}{4}$) gives the quadrilateral with corners

$$(7,7), \quad (\tfrac{37}{4},7), \quad (9,9), \quad (7,\tfrac{37}{4}),$$

without its two sides on $\bar{v}_1 = 7$ and $\bar{v}_2 = 7$. Its area is $\frac{9}{2}$.

(d) A Nash equilibrium with $(9,9)$ every period. Grim trigger:

- player 1 plays U in period 1 and as long as every past outcome was (U,L); after any other history, D forever;
- player 2 plays L in period 1 and as long as every past outcome was (U,L); after any other history, R forever.

Cooperation phase. Staying gives $9 + 9\delta + 9\delta^2 + \cdots = \frac{9}{1-\delta}$. The best deviation is D (10 against L), followed by (D,R) forever: $10 + \frac{7\delta}{1-\delta}$. No deviation pays iff

$$\frac{9}{1-\delta} \geq 10 + \frac{7\delta}{1-\delta} \iff 9 \geq 10(1-\delta) + 7\delta = 10 - 3\delta \iff \delta \geq \frac{1}{3}.$$

(In average payoffs: $9 \geq (1-\delta)10 + \delta\, 7$.) By (10.5), $\delta^* = \frac{10-9}{10-7} = \frac{1}{3}$. Player 2's problem is identical by symmetry. *Punishment phase.* (D,R) is a stage equilibrium and the continuation does not depend on today's action, so deviating to U (or L) loses $7 - 1 = 6$ today and changes nothing later. Both checks pass, so the profile is a subgame-perfect equilibrium, and therefore a Nash equilibrium, for $\delta \geq \frac{1}{3}$.

The restriction is necessary. In any Nash equilibrium with path (U,L) forever, player 1 can deviate to D in period 1 and keep playing D. Since $u_1(D,\cdot) \geq 7$, this gives at least $10 + \frac{7\delta}{1-\delta}$ whatever player 2 does. So $\delta \geq \frac{1}{3}$ is needed for every Nash equilibrium, not just for grim trigger: the punishment already pushes the deviator down to his minimax payoff of 7.

(d) Interpretation. $(9, 9)$ is feasible and gives each player more than the minimax 7, so the folk theorem says it can be sustained by patient enough players. The exercise puts a number on "patient enough": the one-period gain from defecting is 1, the per-period loss from punishment is $9 - 7 = 2$, so $\delta^* = 1/(1+2) = \frac{1}{3}$. The FIR quadrilateral also contains asymmetric outcomes, e.g. near $(\frac{37}{4}, 7)$, which favour player 1 heavily. The folk theorem says these are sustainable too, which is why it explains how cooperation is possible rather than predicting which outcome occurs.

(e) Verification. `check_C_L05.py` confirms that (D, R) is the unique stage equilibrium and that $u_1(D, q) > u_1(U, q)$ for every q on a grid of step $\frac{1}{1000}$, with $\min_q \max = 7$. With sympy it computes the convex hull of the four payoff vectors (the triangle $(1, 10)$, $(10, 1)$, $(9, 9)$, with $(7, 7)$ strictly inside), the corners $(7, \frac{37}{4})$ and $(\frac{37}{4}, 7)$ and the area $\frac{9}{2}$. It solves $\frac{9}{1-\delta} = 10 + \frac{7\delta}{1-\delta}$ to get $\delta = \frac{1}{3}$ and tests the inequality on the grid $k/1000$ and at $\frac{1}{3} \pm \frac{1}{1000}$. Output: "Munoz-Garcia 6.9: stage NE (D,R) (7,7) = minimax; hull = triangle (1,10),(10,1),(9,9); FIR corners (7,7),(37/4,7),(9,9),(7,37/4); delta* = 1/3 (book: 1/3)". *Comparison with Muñoz-Garcia's answer* (pp. 305–309): the same stage equilibrium, minimax $(7, 7)$, FIR set and threshold $\delta \geq \frac{1}{3}$, and the book also shows the strategies are subgame perfect. Three slips in the book: (i) it calls the feasible set "not convex" (p. 307). The convex hull is convex by definition; what is not convex is the set of the four pure payoff vectors, which is what the public randomization device fills in. (ii) It writes $u_2(L; p) = 9p + 1(1-p) = 7 + 3p$ (p. 306); the sum is $1 + 8p$, and $7 + 3p$ is $u_2(R; p)$. (iii) It refers to "Fig. 6.7" for the figure it has just drawn (Fig. 6.11). The book takes the public randomization device as the normalization of part (b); the average-payoff scaling above is the normalization usually meant, and both are needed to draw V in stage-payoff units.

(f) Common mistakes.

- Drawing the four points, or the quadrilateral with $(7, 7)$ as a vertex, as the feasible set. $(7, 7)$ is inside the triangle, so it is not a vertex of V; it is, however, a corner of the FIR set.
- Using ≥ 7 instead of > 7 for individual rationality, or using pure-strategy minimax without checking mixed strategies. Here both give 7, but the definition uses mixed strategies.
- Mixing present values and averages in one inequality, e.g. $9 \geq 10 + \frac{7\delta}{1-\delta}$. Both sides must use the same scaling.

Answer. (a) D and R are strictly dominant: the only stage equilibrium is (D, R), payoffs $(7, 7)$. (b) With average payoffs $(1-\delta)\sum \delta^{t-1} u_i$ and public randomization, V is the convex hull of the four payoff vectors, the triangle $(1, 10)$, $(10, 1)$, $(9, 9)$; $(7, 7)$ is inside it. (c) Minimax $= (7, 7)$; FIR set $= \{\bar{v} \in V : \bar{v}_1 > 7, \bar{v}_2 > 7\}$, the quadrilateral $(7, 7)$, $(\frac{37}{4}, 7)$, $(9, 9)$, $(7, \frac{37}{4})$. (d) Grim trigger (play (U, L) until any deviation, then (D, R) forever) is an SPE, hence a Nash equilibrium, iff $\frac{9}{1-\delta} \geq 10 + \frac{7\delta}{1-\delta}$, i.e. $\delta \geq \frac{1}{3}$ (book: $\frac{1}{3}$). The same bound is necessary for any Nash equilibrium.

Item 5 (item 45 of the combined list)**: Tadelis Exercise 10.3** Exam level · difficulty 3/5

Source: Tadelis Exercise 10.3, page 215–216. The statement is reproduced verbatim in the list (item 45); it is not repeated here.

(a) What is asked. In Tadelis's repeated Prisoner's Dilemma, find the critical discount factor for (M, m) every period under a length-T punishment, (a) for $T = 1$ and (b) for $T = 2$, and (c) compare the two and explain the difference.

(b) Method. Length-T punishments checked with the one-stage deviation principle (Prop. 10.3, p. 198), against the grim-trigger benchmark (10.3), p. 200 ($\delta \geq \frac{1}{4}$ in this game) and (10.5), p. 202. The cooperation-phase test compares the two streams only over the periods in which they differ. Lecture 5 has no slides yet; the repeated-game set-up is Lecture 4, slides 21–22, and the SPE definition is slide 17. Notation as in `Notas/L05/L05-04-grim-trigger-spe.md` and `L05-05-critical-discount-factor.md`.

(c) Solution. Notation. Stage payoffs: $(M, m) = (4, 4)$, $(M, f) = (-1, 5)$, $(F, m) = (5, -1)$, $(F, f) = (1, 1)$. F and f are strictly dominant ($5 > 4$, $1 > -1$), so (F, f) with $(1, 1)$ is the unique stage equilibrium. Strategies with a length-T punishment:

- *cooperation phase*: play M (player 1) and m (player 2);
- *punishment phase*: after any deviation from the prescribed actions, play (F, f) for T periods, then return to the cooperation phase (a deviation during the punishment restarts it).

Present values are measured from the period of the deviation (period 0).

Step 1: the general cooperation-phase condition. Cooperating gives 4 in periods $0, 1, \ldots, T$; deviating gives 5 in period 0 and 1 in periods $1, \ldots, T$. From period $T + 1$ on both streams give 4 forever. So no deviation pays iff

$$4 + 4\sum_{j=1}^{T} \delta^j \;\geq\; 5 + \sum_{j=1}^{T} \delta^j \iff 3\sum_{j=1}^{T} \delta^j \;\geq\; 1.$$

The one-period gain is $5 - 4 = 1$ and each punishment period costs $4 - 1 = 3$.

Step 2: the punishment phase. The prescribed action F is a stage best reply to f. Deviating to M pays $-1 < 1$ today and does not shorten the punishment (with the restart rule it lengthens it, and punishment periods pay $1 < 4$). So no one-stage deviation pays in the punishment phase, for any δ and any T. Only Step 1 binds.

(a) $T = 1$**.** Using present values, as the book's hint suggests:

$$\frac{4}{1-\delta} \;\geq\; 5 + \delta \cdot 1 + \frac{4\delta^2}{1-\delta}.$$

Multiplying by $1 - \delta > 0$: $4 \geq 5(1-\delta) + \delta(1-\delta) + 4\delta^2 = 5 - 4\delta + 3\delta^2$, i.e. $3\delta^2 - 4\delta + 1 \leq 0$, i.e. $(3\delta - 1)(\delta - 1) \leq 0$. With $\delta < 1$ this is $\delta \geq \frac{1}{3}$. Step 1 gives the same: $3\delta \geq 1$. So

$$\delta_1 = \frac{1}{3}.$$

(b) $T = 2$**.** Present values:

$$\frac{4}{1-\delta} \;\geq\; 5 + \delta + \delta^2 + \frac{4\delta^3}{1-\delta}.$$

Multiplying by $1-\delta$: $4 \geq 5(1-\delta)+(\delta+\delta^2)(1-\delta)+4\delta^3 = 5-4\delta+3\delta^3$, i.e.

$$3\delta^3 - 4\delta + 1 \;\leq\; 0.$$

This is the hint's cubic $a\delta^3-(a+1)\delta+1$ with $a=3$. Since $\delta=1$ is a root,

$$3\delta^3 - 4\delta + 1 = (\delta-1)(3\delta^2+3\delta-1).$$

For $\delta<1$ the first factor is negative, so the inequality holds iff $3\delta^2+3\delta-1\geq 0$ (the same as Step 1: $3(\delta+\delta^2)\geq 1$). The positive root is

$$\delta_2 = \frac{-3+\sqrt{9+12}}{6} = \frac{\sqrt{21}-3}{6} \approx 0.2638,$$

and (M,m) in every period is sustained iff $\delta\geq\delta_2$. (The other root, $\frac{-3-\sqrt{21}}{6}<0$, is irrelevant.)

(c) Comparison.

$$\delta_1 = \tfrac{1}{3} \approx 0.333 \;>\; \delta_2 \approx 0.264 \;>\; \delta_\infty = \tfrac{1}{4},$$

where δ_∞ is the grim-trigger threshold ((10.3): $\frac{5-4}{5-1}=\frac{1}{4}$). A longer punishment costs the deviator more: with $T=1$ the loss is 3δ, with $T=2$ it is $3(\delta+\delta^2)$, with grim trigger $3\delta/(1-\delta)$. The gain is 1 in every case. A bigger loss covers the same gain at a lower δ, so the critical value falls as T grows, and it falls by less and less because later punishment periods are discounted more (here $\delta_1-\delta_2\approx 0.070$ but $\delta_2-\delta_\infty\approx 0.014$).

(d) Interpretation. Punishments need not be eternal. In this game the stage gain from cheating is small relative to the loss of a punished period (1 against 3), so even a one-period punishment works for moderately patient players, and two periods already get close to what grim trigger achieves. Short punishments are less costly if a "deviation" was in fact a mistake or a misread signal, since cooperation resumes after T periods. The price is a higher patience requirement.

(e) Verification. `check_C_L05.py` derives the condition $3\sum_{j=1}^{T}\delta^j\geq 1$ symbolically for $T=1,2,5$. It checks $\delta_1=\frac{1}{3}$ at the threshold and at $\pm\frac{1}{1000}$. It rebuilds the book's cubic from the present values ($3\delta^3-4\delta+1$), checks the factorization and the root $\frac{\sqrt{21}-3}{6}=0.263763$, brackets that root to 10^{-6} in exact fractions, tests the grid $\delta=j/1000$, and checks the ordering $\frac{1}{3}>\delta_2>\frac{1}{4}$. Output: "Tadelis 10.3: delta_1 = 1/3; delta_2 = (sqrt(21) - 3)/6 = 0.263763; grim 1/4". Tadelis prints no answer.

(f) Common mistakes.

- Forgetting that cooperation *resumes*: writing the deviation stream as $5+\delta+\delta^2$ and stopping there, or as $5+\frac{\delta}{1-\delta}$ (which is grim trigger).
- Keeping the root $\delta=1$ of the cubic as the answer. It is the trivial root that the hint tells you to factor out; the threshold is the root of the quadratic factor.
- Saying that a shorter punishment is "more lenient, so cooperation is easier". It is the opposite: a weaker threat needs more patience.

Answer. With a length-T reversion to (F,f), (M,m) forever is an SPE iff $3\sum_{j=1}^{T}\delta^j\geq 1$ (the punishment phase never binds). (a) $T=1$: $\delta_1=\frac{1}{3}$. (b) $T=2$: $3\delta^3-4\delta+1=(\delta-1)(3\delta^2+3\delta-1)\leq 0$, so $\delta_2=\frac{\sqrt{21}-3}{6}\approx 0.264$. (c) $\delta_1>\delta_2>\frac{1}{4}$ (grim trigger): a longer punishment makes deviating costlier for the same one-period gain, so less patience is needed, with diminishing effect as T grows.

Item 6 (item 46 of the combined list)**: Tadelis Exercise 9.1** Exam level · difficulty 4/5

Source: Tadelis Exercise 9.1, page 186–187. The statement is reproduced verbatim in the list (item 46); it is not repeated here.

(a) What is asked. For the 3×3 game played twice with the first-period outcome observed: (a) all pure-strategy SPE with $\delta = 1$, with history-contingent strategies; (b) for each, the smallest δ that supports it.

(b) Method. Multistage games (Tadelis §9.1–9.4, pp. 175–184) and the finitely repeated game $G(2, \delta)$ (§10.1, pp. 190–192). Period-2 play must be a stage Nash equilibrium after every history (Prop. 9.2, p. 181). A first-period profile that is not a stage equilibrium can be supported only by a carrot (a good stage equilibrium on the path) and a stick (a bad one after a deviation), and only when $\delta \geq \delta^* = g/(c - s)$ (`Notas/L04/L04-02`, `Notas/L05/L05-02`; Tadelis's own two-stage example, p. 191). Deviations are checked one stage at a time (one-stage deviation principle, Thm. 9.1, p. 185). Slides: Lecture 4, slide 22 ($G(T, \delta)$, payoff $\sum_t \delta^{t-1} u_i^t$, each period starts a subgame) and slide 17 (SPE).

(c) Solution. Notation. $v_i = u_i(a^1) + \delta\, u_i(a^2)$, where a^t is the period-t action profile. A pure strategy for player i is a period-1 action and a map from the nine possible period-1 outcomes to a period-2 action. $g_i(a)$ is player i's best one-shot gain from deviating from a:

$$g_1(a) = \max_r u_1(r, a_2) - u_1(a), \qquad g_2(a) = \max_c u_2(a_1, c) - u_2(a).$$

Best replies are underlined:

	L	C	R
T	10, 10	2, 12	0, $\underline{13}$
M	12, 2	$\underline{5}$, $\underline{5}$	0, 0
B	$\underline{13}$, 0	0, 0	$\underline{1}$, $\underline{1}$

Step 1: stage equilibria. Player 1's best replies: to L, B (13); to C, M (5); to R, B (1). Player 2's: to T, R (13); to M, C (5); to B, R (1). The mutual best replies are

$$(M, C) \text{ with } (5, 5) \quad \text{(the carrot)}, \qquad (B, R) \text{ with } (1, 1) \quad \text{(the stick)}.$$

Each player gets 5 in the carrot and 1 in the stick, so the largest punishment one period can deliver is $c - s = 5 - 1 = 4$.

Step 2: period 2 (the last phase). After any first-period outcome the subgame is the stage game (plus a sunk constant), so it must be (M, C) or (B, R). No one-stage deviation in period 2 pays, because both are stage equilibria and nothing follows.

Step 3: period 1 (which profiles can be supported). A first-period profile a is supportable if and only if each player's gain is at most what the threat can take away:

$$g_i(a) \;\leq\; \delta\,(5 - 1) = 4\delta, \qquad i = 1, 2, \tag{13}$$

using (M, C) after a and (B, R) after any deviation. No continuation can do better, since 5 is the most and 1 the least a player can get in period 2. For a stage equilibrium a, $g_i(a) = 0$ and any continuation works. The gains are:

a	$u(a)$	$g_1(a)$	$g_2(a)$	supportable ($\delta=1$)?	$\delta^*=\max_i g_i/4$
(T,L)	$(10,10)$	$13-10=3$	$13-10=3$	yes	$3/4$
(T,C)	$(2,12)$	$5-2=3$	$13-12=1$	yes	$3/4$
(T,R)	$(0,13)$	$1-0=1$	0	yes	$1/4$
(M,L)	$(12,2)$	$13-12=1$	$5-2=3$	yes	$3/4$
(M,C)	$(5,5)$	0	0	yes	0
(M,R)	$(0,0)$	$1-0=1$	$5-0=5$	no	–
(B,L)	$(13,0)$	0	$1-0=1$	yes	$1/4$
(B,C)	$(0,0)$	$5-0=5$	$1-0=1$	no	–
(B,R)	$(1,1)$	0	0	yes	0

(a) The pure SPE with $\delta=1$. There are nine SPE *paths*:

- the four sequences of stage equilibria, played regardless of history (Prop. 9.1, p. 180): (M,C) then (M,C); (M,C) then (B,R); (B,R) then (M,C); (B,R) then (B,R);
- the five carrot-and-stick paths: a then (M,C), for $a\in\{(T,L),(T,C),(T,R),(M,L),(B,L)\}$.

For a carrot-and-stick path the strategies are, for $\hat a=(\hat a_1,\hat a_2)$ one of those five:

$$s_1:\ \text{play } \hat a_1;\ \text{in period 2 play } M \text{ if } \hat a \text{ was played, } B \text{ otherwise;}$$
$$s_2:\ \text{play } \hat a_2;\ \text{in period 2 play } C \text{ if } \hat a \text{ was played, } R \text{ otherwise.}$$

One-stage deviation checks for each path (with $\delta=1$; path payoff against best deviation):

$\hat a$	P1: path	P1: best deviation	P2: path	P2: best deviation
(T,L)	$10+5=15$	B: $13+1=14$	$10+5=15$	R: $13+1=14$
(T,C)	$2+5=7$	M: $5+1=6$	$12+5=17$	R: $13+1=14$
(T,R)	$0+5=5$	B: $1+1=2$	$13+5=18$	none (best reply)
(M,L)	$12+5=17$	B: $13+1=14$	$2+5=7$	C: $5+1=6$
(B,L)	$13+5=18$	none (best reply)	$0+5=5$	R: $1+1=2$

In period 2 both players play a stage equilibrium after every history, so no deviation pays there. (M,R) and (B,C) cannot be supported: a gain of 5 exceeds the largest punishment, 4. For instance, at (M,R) player 2 gets $0+5=5$ on the path but $5+1=6$ by deviating to C.

These strategies are not the only ones. Any assignment of (M,C) or (B,R) to the nine period-1 outcomes that satisfies (13) for the deviations that would be profitable also works (e.g. for (T,R) only player 1's deviation to B must be punished). Counting every such map, `check_C_L05.py` finds 992 pure SPE strategy profiles at $\delta=1$, all with one of the nine paths above.

(b) The smallest discount factor. From (13), $\delta^*(a)=\max_i g_i(a)/4$ (last column of the first table). Solving each binding inequality:

- (T,L): $10+5\delta\geq 13+\delta \iff 4\delta\geq 3 \iff \delta\geq\frac{3}{4}$ (both players);
- (T,C): player 1 binds, $2+5\delta\geq 5+\delta \iff \delta\geq\frac{3}{4}$; player 2 needs $12+5\delta\geq 13+\delta \iff \delta\geq\frac{1}{4}$;
- (M,L): the mirror image of (T,C), $\delta\geq\frac{3}{4}$;
- (T,R): only player 1, $0+5\delta\geq 1+\delta \iff \delta\geq\frac{1}{4}$;
- (B,L): only player 2, $0+5\delta\geq 1+\delta \iff \delta\geq\frac{1}{4}$;
- the four sequences of stage equilibria: any $\delta\geq 0$.

Below δ^* no continuation map works, because (M, C) after $\hat{a}$ and (B, R) after the deviation is already the largest reward gap available.

A caveat: renegotiation. The stick (B, R) is Pareto-dominated by (M, C). After a deviation both players would like to "forgive" and play (M, C) in period 2, and if they can renegotiate the threat fails (Gibbons §2.3.A, pp. 87–88; `Notas/L05/L05-02`). The SPE above take the threat as given.

(d) Interpretation. Two stage equilibria with different payoffs give the second period a carrot and a stick. That lets the players reach (T, L), which pays $(10, 10)$, twice the $(5, 5)$ of the good stage equilibrium, in the first period. Profiles with a gain above 4 cannot be enforced by a single future period, however patient the players are. This is the finite-horizon version of the logic of Lecture 5: what can be sustained today is limited by how much the future can reward and punish. With a unique stage equilibrium (item Cabral 7.13) that gap is zero and nothing can be sustained.

(e) Verification. `check_C_L05.py` finds the two stage equilibria and enumerates all 9×2^9 combinations of first-period profile and continuation map at $\delta = 1$ (992 SPE, 9 paths; (M, R) and (B, C) excluded). For every threshold it checks that each path is an SPE at δ^* and at $\delta^* + \frac{1}{1000}$, that no continuation map at all supports it at $\delta^* - \frac{1}{1000}$, and that on the grid $\delta = k/1000$ the carrot-and-stick map works exactly when $\delta \geq \delta^*$. Output: "Tadelis 9.1: stage NE (M,C),(B,R); 992 pure SPE at delta=1 on 9 paths; delta* = 3/4 (TL, TC, ML), 1/4 (TR, BL); grid and +-1/1000 pass". Tadelis prints no answer.

(f) Common mistakes.

- Listing only the outcome "(T, L) then (M, C)" without the off-path play. The question asks for history-contingent strategies, and the threat to play (B, R) after a deviation is the whole point.
- Using the smallest gain instead of the largest: the threshold is $\max_i g_i/4$, so in (T, C) player 1's gain of 3 binds, not player 2's gain of 1.
- Claiming (M, R) or (B, C) can be sustained with enough patience. With one period left the punishment is capped at $5 - 1 = 4$, which is below the gain of 5 even at $\delta = 1$.

Answer. Stage equilibria: (M, C) $(5, 5)$ and (B, R) $(1, 1)$. (a) Pure SPE paths with $\delta = 1$: the four sequences of stage equilibria, and a then (M, C) for $a \in \{(T, L), (T, C), (T, R), (M, L), (B, L)\}$, supported by "play (M, C) after a, (B, R) after any other outcome". (M, R) and (B, C) cannot be supported. (b) $\delta^* = \frac{3}{4}$ for (T, L), (T, C), (M, L); $\frac{1}{4}$ for (T, R), (B, L); 0 for the sequences of stage equilibria.

Item 7 (item 47 of the combined list)**: Gibbons Problem 2.11** Exam level · difficulty 4/5

Source: Gibbons Problem 2.11, page 135. The statement is reproduced verbatim in the list (item 47); it is not repeated here.

(a) What is asked. For the 3×3 game played twice without discounting, decide whether $(4, 4)$ can be obtained in the first stage of a pure-strategy SPE, and give the strategies or a proof.

(b) Method. Two-stage repeated game with several stage equilibria (Gibbons §2.3.A, pp. 82–88, especially the three-strategy example of pp. 84–87; Tadelis §9.4, pp. 180–184, and §10.1, p. 191). In the second stage a stage Nash equilibrium must be played after every history (Tadelis Prop. 9.2, p. 181). A first-stage profile that is not a stage equilibrium is supported by a *reward* equilibrium after it and a *punishment* equilibrium after deviations. Deviations are checked one stage at a time (Tadelis Thm. 9.1, p. 185). Slides: Lecture 4, slide 22 ($G(T, \delta)$, each period starts a subgame) and slide 17 (SPE). Notation as in `Notas/L05/L05-02-two-stage-carrot-stick.md`.

(c) Solution. Notation. $v_i = u_i(a^1) + u_i(a^2)$ (no discounting). Best replies are underlined:

	L	C	R
T	$\underline{3},\ \underline{1}$	$0,\ 0$	$\underline{5},\ 0$
M	$2,\ 1$	$\underline{1},\ \underline{2}$	$3,\ 1$
B	$1,\ 2$	$0,\ 1$	$4,\ \underline{4}$

Step 1: stage equilibria. Player 1's best replies: to L, T (3); to C, M (1); to R, T (5). Player 2's: to T, L (1); to M, C (2); to B, R (4). The mutual best replies are

$$(T, L) \text{ with } (3, 1), \qquad (M, C) \text{ with } (1, 2).$$

(B, R) is *not* a stage equilibrium: against R, player 1 prefers T ($5 > 4$). Since the question asks for a pure-strategy SPE, second-stage play must be one of these two pure equilibria after every history.

Step 2: who must be deterred. At (B, R), player 2 is already best-responding (4 is the most she can get against B), so only player 1 has a profitable one-stage deviation: to T, gaining $5 - 4 = 1$ (to M he would get $3 < 4$). Player 1 gets 3 in (T, L) and 1 in (M, C), so the second stage can reward him with 3 and punish him with 1: a swing of $2 > 1$.

Step 3: the strategies.

- Player 1: play B in stage 1; in stage 2 play T if the stage-1 outcome was (B, R), and M otherwise.
- Player 2: play R in stage 1; in stage 2 play L if the stage-1 outcome was (B, R), and C otherwise.

Step 4: one-stage deviation checks. *Stage 2 (every history).* The prescribed profile is (T, L) or (M, C), a stage equilibrium, and nothing follows, so no deviation pays.

Stage 1, player 1. On the path he gets $4 + 3 = 7$. Deviating to T gives 5 now and then (M, C), so $5 + 1 = 6 < 7$; deviating to M gives $3 + 1 = 4 < 7$.

Stage 1, player 2. On the path she gets $4 + 1 = 5$. Deviating to L gives 2 now and then (M, C), so $2 + 2 = 4 < 5$; deviating to C gives $1 + 2 = 3 < 5$. (The "punishment" (M, C) is actually better for player 2 than (T, L), but she has nothing to gain in stage 1, so it does not matter.)

All one-stage deviations lose, so the profile is a pure-strategy SPE and the answer is **yes**: $(4, 4)$ is reached in the first stage, with total payoffs $(7, 5)$.

How much room there is. With a discount factor δ, player 1's constraint is $4 + 3\delta \geq 5 + \delta \iff \delta \geq \frac{1}{2}$, so the construction survives some impatience; player 2's constraint, $4+\delta \geq 2+2\delta \iff \delta \leq 2$, holds for every $\delta \in [0, 1]$. Among all $2^9 = 512$ assignments of (T, L) or (M, C) to the nine stage-1 outcomes, exactly $128 = 2^7$ support (B, R) first: those with (T, L) after (B, R) and (M, C) after player 1's deviation to T. The other seven outcomes can be assigned either way.

(d) Interpretation. The second stage offers two equilibria that player 1 ranks differently, and that difference is a credible incentive. The players can coordinate on the efficient (B, R) today because both expect player 1 to be "paid" with his preferred equilibrium (T, L) tomorrow if he behaves, and pushed to (M, C) if he grabs the extra 1. Gibbons (pp. 87–88) warns that such threats may be renegotiated when the punishment equilibrium is Pareto-dominated by the reward equilibrium. That is not the case here: player 2 prefers (M, C) to (T, L), so after player 1's deviation she would not agree to forgive him, and the threat survives that critique.

(e) Verification. `check_C_L05.py` finds the two pure stage equilibria, confirms that player 1's best reply to R is T, and enumerates all 512 continuation maps with (B, R) first at $\delta = 1$. It finds 128 SPE, all with (T, L) after (B, R) and (M, C) after (T, R). It checks the payoffs 7 against 6, and on the grid $\delta = k/1000$ that the constructed profile works exactly for $\delta \geq \frac{1}{2}$. Output: "Gibbons 2.11: stage NE (T,L),(M,C); (B,R) first is a pure SPE: 128 of 512 maps work, all with (T,L) after (B,R) and (M,C) after (T,R)". Gibbons prints no answer. *Comparison with Muñoz-Garcia Ch. 6 Ex. 7* (pp. 299–303, the same game): the same answer (yes, (B, R) then (T, L), with $7 > 6$ for player 1). Two differences. (i) The book punishes with the stage game's mixed equilibrium (each player mixes $\frac{1}{2}$–$\frac{1}{2}$ over $\{T, M\}$ and $\{L, C\}$, payoffs $(\frac{3}{2}, 1)$), giving $5 + \frac{3}{2} = 6.5 < 7$. That works as an SPE, but it is not a *pure-strategy* SPE as the question asks; the pure equilibrium (M, C) is both admissible and a harsher punishment. (ii) The book lists $BR_1(R) = B$; in fact $BR_1(R) = T$ $(5 > 4)$, which is exactly why (B, R) needs support. Its list of pure equilibria, (T, L) and (M, C), is right.

(f) Common mistakes.

- Answering "no, because (B, R) is not a Nash equilibrium of the stage game". In a repeated game with several stage equilibria, non-equilibrium play *can* be supported in early stages.
- Using a mixed equilibrium as the punishment when the question restricts to pure strategies, or a non-equilibrium profile (e.g. (B, C)) in the last stage.
- Checking only player 2 (whose payoff 4 is high) and missing that player 1 is the one tempted.

Answer. Yes. Stage equilibria: (T, L) $(3, 1)$ and (M, C) $(1, 2)$. Strategies: play (B, R) in stage 1; in stage 2 play (T, L) if stage 1 was (B, R) and (M, C) otherwise. Player 1: $4 + 3 = 7 > 5 + 1 = 6$ (deviation to T) and $> 3 + 1$ (to M); player 2: $4 + 1 = 5 > 2 + 2$ (to L) and $> 1 + 2$ (to C); stage 2 is always a stage equilibrium. This is a pure-strategy SPE with $(4, 4)$ in the first stage (it would survive discounting down to $\delta = \frac{1}{2}$).

Item 8 (item 48 of the combined list)**: Tadelis Exercise 11.5** Exam level · difficulty 4/5

Source: Tadelis Exercise 11.5, page 236–237. The statement is reproduced verbatim in the list (item 48); it is not repeated here.

(a) What is asked. In three-period alternating-offer bargaining over a pie of 1, with player 1 proposing first and discount factors $\delta_1 \neq \delta_2$: (a) the SPE outcome; (b) player 1's advantage when $\delta_1 = \delta_2$; (c) when player 2 has the advantage, and why.

(b) Method. Backward induction in finite alternating-offer bargaining (Tadelis §11.2, pp. 224–227; Gibbons §2.1.D, pp. 68–70). In the last period the proposer takes everything. In each earlier period the proposer offers the responder exactly the discounted value of what the responder would get next period, and the responder accepts at indifference (the argument of Prop. 11.2, p. 223; agreement in round one, Prop. 11.3, p. 226). Slides: Lecture 4, slides 9–10 (backward induction) and slide 17 (SPE). Lecture 5's bargaining part has no slides yet. Notation as in `Notas/L04/L04-06-finite-alternating-offers.md`.

(c) Solution. Notation. x_t = player 1's share of the pie when agreement is reached in period $t \in \{1, 2, 3\}$, so player 2 gets $1 - x_t$. Player i's payoff from agreement $(x, 1-x)$ in period t is δ_i^{t-1} times his share. Players 1, 2, 1 propose in periods 1, 2, 3, and no agreement in period 3 gives $(0, 0)$. "Advantage" means a share above $\frac{1}{2}$.

Step 1: period 3 (player 1 proposes, last period). Player 2 compares accepting $1 - x$ with rejecting, which gives 0. She accepts every $x < 1$ strictly. At $x = 1$ she is indifferent, and in an SPE she must accept: if she rejected $x = 1$, player 1 would want the largest $x < 1$, which does not exist, so he would have no best offer. Player 1 offers $x_3 = 1$. Period-3 values: player 1 gets 1, player 2 gets 0; in period-1 terms, $(\delta_1^2,\ 0)$.

Step 2: period 2 (player 2 proposes). Player 1 accepts x iff x is at least what rejecting gives him, namely 1 one period later, worth $\delta_1 \cdot 1$ today:

$$x \ \geq \ \delta_1 \, x_3 = \delta_1. \tag{14}$$

Player 2 keeps the most she can by offering exactly $x_2 = \delta_1$ (by the same argument as in Step 1, player 1 accepts at equality). She then gets $1 - \delta_1$. Making an offer player 1 rejects gives her $\delta_2 \cdot 0 = 0 < 1 - \delta_1$, so she makes the acceptable offer.

Step 3: period 1 (player 1 proposes). Player 2 accepts x iff her share now is at least the value of rejecting, i.e. $1 - x_2$ one period later, discounted by her own δ_2:

$$1 - x \ \geq \ \delta_2 \, (1 - x_2) = \delta_2 \, (1 - \delta_1).$$

Player 1 keeps as much as possible:

$$x_1 = 1 - \delta_2(1 - \delta_1) = 1 - \delta_2 + \delta_1\delta_2. \tag{15}$$

He prefers this to a rejected offer, which gives him $\delta_1 x_2 = \delta_1^2$: indeed $x_1 - \delta_1^2 = (1-\delta_1)(1+\delta_1-\delta_2) > 0$.

(a) The SPE. Strategies at every node:

- player 1: offer $x_1 = 1 - \delta_2(1 - \delta_1)$ in period 1; in period 2 accept iff offered $x \geq \delta_1$; in period 3 offer $x_3 = 1$;
- player 2: in period 1 accept iff $1 - x \geq \delta_2(1 - \delta_1)$; in period 2 offer $x_2 = \delta_1$; in period 3 accept every offer.

Each acceptance rule holds after every possible offer, including offers never made on the path, so every subgame is played optimally. The outcome is immediate agreement:

$$(v_1, v_2) = \big(1 - \delta_2(1-\delta_1),\ \ \delta_2(1-\delta_1)\big).$$

The tie-breaking argument of Step 1 applies at every round, so this SPE is unique.

(b) Equal discount factors. Setting $\delta_1 = \delta_2 = \delta$ in (15):

$$v_1 = 1 - \delta + \delta^2, \qquad v_2 = \delta - \delta^2,$$

$$v_1 - v_2 = 1 - 2\delta + 2\delta^2 = (1-\delta)^2 + \delta^2 \ > \ 0.$$

Player 1 gets strictly more than half for every δ. He proposes both first and last. The edge $(1-\delta)^2 + \delta^2$ is smallest at $\delta = \frac{1}{2}$ (derivative $-2(1-\delta) + 2\delta = 0$), where it equals $\frac{1}{2}$ and $v_1 = \frac{3}{4}$.

(c) When player 2 is ahead. Player 2 gets more than half iff

$$\delta_2\,(1-\delta_1) > \tfrac{1}{2} \iff \delta_1 < 1 - \frac{1}{2\delta_2}.$$

This needs $\delta_2 > \frac{1}{2}$ (otherwise $\delta_2(1-\delta_1) \le \frac{1}{2}$ for every δ_1) and δ_1 low enough. Example: $\delta_2 = 0.9$ gives the cut-off $\delta_1 < 1 - \frac{1}{1.8} = \frac{4}{9}$; with $\delta_1 = 0.2$, $v_2 = 0.9 \times 0.8 = 0.72$. *Why.* Player 1's power comes from the last period. An impatient player 1 values it at only δ_1 from period 2's viewpoint, so player 2 keeps $1 - \delta_1$ when she proposes. A patient player 2 values that period-2 position almost fully (δ_2), so in period 1 player 1 must concede $\delta_2(1-\delta_1)$. Patience is bargaining power:

$$\frac{\partial v_1}{\partial \delta_1} = \delta_2 > 0, \qquad \frac{\partial v_1}{\partial \delta_2} = -(1-\delta_1) < 0.$$

The infinite-horizon formula $v_1 = \frac{1-\delta_2}{1-\delta_1\delta_2}$ (Tadelis §11.3; `Notas/L04/L04-07`) has the same signs.

(d) Interpretation. First-mover and last-mover advantages both favour player 1, so only a large patience gap in player 2's favour can overturn them. As $\delta \to 0$ (equal patience) the first-mover advantage takes over and $v_1 \to 1$. As $\delta \to 1$ the last-mover advantage takes over and again $v_1 \to 1$, because waiting for period 3 costs player 1 nothing. Patience matters because each player's threat to reject is worth what the next round is worth *to him*: the more patient player loses less by waiting, so the other side must pay him more to settle now.

(e) Verification. `check_C_L05.py` redoes the recursion symbolically ($x_1 = 1 - \delta_2 + \delta_1\delta_2$; $v_1 - v_2 = (1-\delta)^2 + \delta^2$ with minimum $\frac{1}{2}$ at $\delta = \frac{1}{2}$; limits 1 at both ends; the two derivatives). It tests the condition in (c) at the boundary $\delta_1 = \frac{4}{9}$ for $\delta_2 = \frac{9}{10}$ and at $\frac{4}{9} \pm \frac{1}{1000}$, confirms that player 2 can never be ahead when $\delta_2 \le \frac{1}{2}$ (grid of step $\frac{1}{100}$), and repeats the backward induction numerically with offers on a grid of step $\frac{1}{1000}$ for three pairs; e.g. $(\delta_1, \delta_2) = (0.9, 0.7)$ gives $x_1 = 0.93$. Output: "Tadelis 11.5: v1 = 1 - d2(1 - d1), v2 = d2(1 - d1); equal delta edge (1-d)^2 + d^2 >= 1/2; P2 ahead iff d2(1-d1) > 1/2 (boundary 4/9 at d2 = 9/10 checked +-1/1000)". Tadelis prints no answer.

(f) Common mistakes.

- Discounting player 2's continuation with δ_1 (or with a common δ). Each player values the future with *his own* factor: player 2's period-1 threat is $\delta_2(1-\delta_1)$.
- Letting player 2 reject at indifference in period 3 (or player 1 in period 2). Then the proposer has no best offer and there is no SPE; acceptance at indifference is forced.

- Answering (b) with "player 1 moves first" only. With three periods he also moves last, which is why his edge does not vanish as $\delta \to 1$.

Answer. (a) Immediate agreement: player 1 gets $1-\delta_2(1-\delta_1)$, player 2 gets $\delta_2(1-\delta_1)$ (offers: $x_3 = 1$, $x_2 = \delta_1$, $x_1 = 1-\delta_2+\delta_1\delta_2$). (b) With $\delta_1 = \delta_2 = \delta$: $v_1 - v_2 = (1-\delta)^2 + \delta^2 > 0$ (at least $\frac{1}{2}$). (c) Player 2 is ahead iff $\delta_2(1-\delta_1) > \frac{1}{2}$, which needs player 2 patient ($\delta_2 > \frac{1}{2}$) and player 1 impatient ($\delta_1 < 1 - \frac{1}{2\delta_2}$). Player 1's last-period power is worth little to an impatient player 1, while player 2's period-2 power is worth a lot to a patient player 2.

Item 9 (item 49 of the combined list)**: Tadelis Exercise 10.7** Advanced · difficulty 4/5

Source: Tadelis Exercise 10.7, page 217–218. The statement is reproduced verbatim in the list (item 49); it is not repeated here.

(a) What is asked. In the bartender–customer game: (a) all stage Nash equilibria; (b) all SPE when it is played on 10 evenings without discounting; (c) with infinite repetition, the prices p at which an SPE has $x = 1$ and a purchase every day; (d) the values of δ for which that price range is non-empty.

(b) Method. (a) Strict dominance (Tadelis §4.1). (b) Finite repetition of a game with a unique stage equilibrium: Prop. 10.1, p. 192 (§10.1). (c)–(d) Grim trigger and the one-stage deviation principle (Prop. 10.3, p. 198; (10.3) p. 200; (10.5) p. 202), applied to each player separately because the game is asymmetric, plus the minimax argument showing the punishment is the harshest available (`Notas/L05/L05-05-critical-discount-factor.md`, §1 and §5). Slides: Lecture 4, slides 21–22 (repeated games, $G(T, \delta)$) and slide 17 (SPE). Lecture 5 has no slides yet.

(c) Solution. Notation. The bartender (B) chooses $x \in [0, 1]$; the customer (K) buys with probability $\beta \in [0, 1]$, without seeing x, so within a day the moves are effectively simultaneous. Stage payoffs:

$$u_B(x, \beta) = \beta p - cx, \qquad u_K(x, \beta) = \beta(vx - p),$$

with $v > c > 0$. The price p is fixed by the market; take $p > 0$ (the case $p = 0$ is noted where it matters). The customer learns x only if he buys. $\delta \in (0, 1)$ is the discount factor (the book's "discount rate").

(a) Stage Nash equilibria. For the bartender, $\partial u_B/\partial x = -c < 0$ whatever β is, so $x = 0$ is strictly dominant: a diluted drink is cheaper, and the customer cannot see it before paying. Against $x = 0$, buying gives the customer $v \cdot 0 - p = -p < 0$ and not buying gives 0, so he does not buy ($\beta = 0$). The unique Nash equilibrium is

$$(x, \beta) = (0, 0), \qquad \text{payoffs } (u_B, u_K) = (0, 0).$$

(If $p = 0$ the customer is indifferent, and every $\beta \in [0, 1]$ with $x = 0$ is an equilibrium, all with payoffs $(0, 0)$.) The efficient outcome, $x = 1$ with a purchase, creates surplus $v - c > 0$ but is not an equilibrium: this is a one-sided Prisoner's Dilemma (a trust game).

(b) Ten evenings. The stage game has a unique Nash equilibrium, so by backward induction (Prop. 10.1) the unique SPE is $x = 0$ and no purchase on every evening after every history. Step by step: on evening 10 nothing follows, $x = 0$ is strictly dominant and so the customer does not buy, whatever he knows about earlier evenings. On evening $k < 10$, by induction play from $k + 1$ on is (0, no purchase) whatever happens on evening k, so evening k is the stage game plus a constant, and the same argument applies. The customer's imperfect information about x on days he does not buy does not matter, because $x = 0$ is strictly dominant at every information set. (With $p = 0$ the bartender's payoff on every evening is $-cx$ whatever the customer does, so he still chooses $x = 0$ every evening, and the customer may buy or not.) Payoffs: $(0, 0)$.

(c) Infinite repetition: the price range. Strategies (grim trigger):

- bartender: $x = 1$ on day 1 and as long as every past day had $x = 1$ and a purchase; otherwise $x = 0$ forever;
- customer: buy on day 1 and as long as every past day had $x = 1$ and a purchase; otherwise never buy again.

On the path the customer buys every day, so he observes every x and any dilution is detected; a refusal to buy is seen by the bartender. In the punishment phase the customer no longer observes x, but nothing there depends on it.

Bartender, cooperation phase. Staying gives $p - c$ every day: $\frac{p-c}{1-\delta}$. The customer buys today without seeing x, so the best deviation is $x = 0$ (any $x \in (0, 1)$ earns $p - cx < p$). It pays p today and 0 forever after:

$$\frac{p-c}{1-\delta} \;\geq\; p \iff p - c \geq (1-\delta)p \iff \delta p \geq c \iff p \;\geq\; \frac{c}{\delta}.$$

Customer, cooperation phase. Staying gives $v - p$ every day: $\frac{v-p}{1-\delta}$. The deviation, not buying, gives 0 today and 0 forever:

$$\frac{v-p}{1-\delta} \;\geq\; 0 \iff p \;\leq\; v.$$

Punishment phase. The prescribed play is the stage equilibrium (0, no purchase) and the future does not depend on today's action, so a deviation only changes today's payoff: the bartender would pay $cx > 0$ for nothing, and the customer would pay p for a drink with $x = 0$. Neither gains.

So the grim-trigger profile is an SPE with $x = 1$ and a purchase every day iff

$$\frac{c}{\delta} \;\leq\; p \;\leq\; v.$$

No other SPE does better. The bartender can guarantee himself 0 every day by choosing $x = 0$ (his minimax payoff, reached when the customer stops buying), and the customer can guarantee 0 by not buying. Grim trigger already pushes each deviator to 0, so no other punishment can widen the range: $[c/\delta, v]$ is the set of prices for which *any* SPE with this path exists.

(d) When the range is non-empty.

$$\frac{c}{\delta} \leq v \iff \delta \geq \frac{c}{v}.$$

Because $v > c$, $\frac{c}{v} < 1$, so the condition holds for all $\delta \in [\frac{c}{v}, 1)$. At $\delta = \frac{c}{v}$ the only price is $p = v$, which leaves the customer no surplus. For example, with $c = 1$, $v = 3$, $\delta = \frac{1}{2}$: $p \in [2, 3]$.

(d) Interpretation. Honest service is sustained by the value of the relationship, and that value has to come from the price. The bartender must earn a premium, $p - c \geq \frac{1-\delta}{\delta} c$ above cost, large enough that losing a regular customer outweighs the one-time saving c from watering one drink. A tourist, on a known finite visit, is cheated on every drink and so never buys; a local, valued as a long-term customer, is served full drinks at a price above cost. This is the logic of quality-assuring prices and relational contracts (Cabral §7.3, p. 174): with p too low there is no rent to lose, and with p above v the customer would rather not buy. The more patient the bartender, the smaller the premium needed.

(e) Verification. `check_C_L05.py` checks with sympy that u_B falls in x for every β (so $x = 0$ is strictly dominant) and that the bartender's condition reduces to $\delta p - c \geq 0$. For $(v, c) = (3, 1)$ and $(10, 9)$ and $\delta \in \{\frac{1}{2}, \frac{9}{10}, \frac{c}{v}, \frac{c}{v} + \frac{1}{1000}\}$ it confirms in exact fractions that both incentive constraints hold at $p = c/\delta$ and $p = v$ and fail at $c/\delta - \frac{1}{1000}$ and $v + \frac{1}{1000}$, or that no price works when $c/\delta > v$. On the grid $\delta = k/1000$ it checks that the range is non-empty exactly when $\delta \geq c/v$. Output: "Tadelis 10.7: stage NE x = 0, no purchase (p > 0); SPE price range c/delta <= p <= v; non-empty iff delta >= c/v (checked at c/v and +-1/1000)". Tadelis prints no answer.

(f) Common mistakes.

- Letting the customer react to x on the same day. He pays before he tastes, so the bartender's deviation earns the full price p that day; the payoff from deviating is p, not 0.
- Giving only the bartender's constraint ($p \geq c/\delta$) and forgetting the customer's ($p \leq v$), or the reverse; the game is asymmetric and each player has his own condition.
- In (b), arguing that ten evenings is "long enough" for reputation to work. With a known last evening and a unique stage equilibrium the game unravels completely.

Answer. (a) $x = 0$ is strictly dominant, so for $p > 0$ the unique NE is $x = 0$ and no purchase, payoffs $(0, 0)$ (for $p = 0$, $x = 0$ with any purchase probability). (b) Unique SPE: $x = 0$ and no purchase on all 10 evenings after every history. (c) With grim trigger, the bartender needs $\frac{p-c}{1-\delta} \geq p$, i.e. $p \geq c/\delta$, and the customer needs $p \leq v$; since these punishments are each player's minimax, the range is exactly $\frac{c}{\delta} \leq p \leq v$. (d) It is non-empty iff $\delta \geq c/v$ (which is below 1 because $v > c$).

Item 10 (item 50 of the combined list)**: Gibbons Problem 2.20** Advanced · difficulty 5/5

Source: Gibbons Problem 2.20, page 137–138. The statement is reproduced verbatim in the list (item 50); it is not repeated here.

(a) What is asked. Show that the stationary strategies given in the problem (player 1 offers $(s^*, 1-s^*)$ and accepts iff $s \geq \delta s^*$; player 2 offers $(1-s^*, s^*)$ and accepts iff $1-s \geq \delta s^*$, with $s^* = 1/(1+\delta)$) are first a Nash equilibrium and then a subgame-perfect equilibrium of the infinite-horizon Rubinstein game.

(b) Method. Nash equilibrium: bound each player's payoff over every possible agreement date against the opponent's fixed strategy. Subgame perfection: the one-stage deviation principle (Tadelis Thm. 9.1, p. 185; its infinite-horizon form, Prop. 10.3, p. 198). It applies because payoffs are discounted and bounded, so a deviation's effect far in the future is negligible. Because the strategies ignore history, there are only four kinds of subgame to check. Model: Gibbons §2.1.D, pp. 68–71 (the infinite-horizon value $1/(1+\delta)$ and the Shaked–Sutton argument, pp. 70–71); Tadelis §11.3, pp. 228–229 (eqs. (11.4)–(11.5)). Slides: Lecture 4, slide 17 (SPE: Nash equilibrium in *every* subgame) and slides 9–10 (backward induction). Notation as in `Notas/L04/L04-07-rubinstein-infinite-horizon.md`.

(c) Solution. Notation and two identities. s is always player 1's share. In period t an agreement $(s, 1-s)$ is worth $\delta^{t-1}s$ to player 1 and $\delta^{t-1}(1-s)$ to player 2; perpetual disagreement gives $(0,0)$. Player 1 proposes in odd periods, player 2 in even ones. With $s^* = \frac{1}{1+\delta}$,

$$1 - s^* = \frac{\delta}{1+\delta} = \delta s^*, \qquad 1 - \delta s^* = \frac{1}{1+\delta} = s^*. \tag{16}$$

So player 2's offer $(1-s^*, s^*)$ gives player 1 exactly δs^*, his acceptance threshold, and player 1's offer gives player 2 exactly $1-s^* = \delta s^*$, her threshold. On the path, player 1 offers $(s^*, 1-s^*)$ in period 1 and player 2 accepts: payoffs $(s^*, 1-s^*) = \left(\frac{1}{1+\delta}, \frac{\delta}{1+\delta}\right)$.

Part 1: Nash equilibrium. *Player 1's best response.* Fix player 2's strategy. Whatever player 1 does, any agreement happens either in an odd period t, on an offer of player 1 that player 2 accepts, so $1-s \geq \delta s^*$, i.e. $s \leq 1-\delta s^* = s^*$ by (16); or in an even period t, on player 2's only offer, $s = 1-s^* = \delta s^*$. Player 1's payoff is therefore at most

$$\max\Big\{ \max_{t \text{ odd}} \delta^{t-1} s^*, \quad \max_{t \text{ even}} \delta^{t-1}\, \delta s^*, \quad 0 \Big\} \;=\; \max\{s^*,\ \delta^2 s^*,\ 0\} = s^*,$$

reached at $t=1$, and his strategy gets exactly s^*. It is a best response.

Player 2's best response. Fix player 1's strategy. Agreement in an odd period t is on player 1's only offer, giving player 2 $\delta^{t-1}(1-s^*)$. Agreement in an even period t needs an offer player 1 accepts, $s \geq \delta s^*$, so player 2 gets at most $\delta^{t-1}(1-\delta s^*) = \delta^{t-1}s^*$. The best she can do is

$$\max\{\, 1-s^* \ (t=1),\ \ \delta s^* \ (t=2),\ \text{smaller values later}\,\} = 1-s^* = \delta s^*,$$

the two candidates being equal by (16). Her strategy (accept in period 1) gets $1-s^*$, so it is a best response. She is indifferent between accepting now and rejecting to propose next period, which is the property that pins down s^*. So the strategies form a Nash equilibrium.

Part 2: subgame perfection. The strategies do not depend on history, so a subgame is characterized only by whose move it is and, for a responder, by the offer on the table. Up to the common factor δ^{t-1} there are four types:

(i) *player 1 about to propose* (odd period);

(ii) *player 2 responding to an arbitrary offer* $(s, 1-s)$ (odd period);

(iii) *player 2 about to propose* (even period);

(iv) *player 1 responding to an arbitrary offer* $(s, 1-s)$ (even period).

Continuation values. Starting a type-(i) subgame, the strategies give agreement at once on $(s^*, 1-s^*)$. Starting a type-(iii) subgame, they give agreement at once on $(1-s^*, s^*)$: player 2 gets s^* and player 1 gets $1-s^* = \delta s^*$. By the one-stage deviation principle it is enough to check single deviations at each type.

(ii) Player 2 responding to $(s, 1-s)$. Accepting gives $1-s$. Rejecting leads to a type-(iii) subgame next period, where she gets s^*, worth δs^* now. Accepting is optimal iff $1-s \geq \delta s^*$, which is exactly her rule (at equality she is indifferent and accepting is a best reply). No one-stage deviation pays, for *every* offer s, on or off the path.

(iv) Player 1 responding to $(s, 1-s)$. Accepting gives s. Rejecting leads to a type-(i) subgame next period, worth δs^*. Accepting is optimal iff $s \geq \delta s^*$: his rule.

(i) Player 1 proposing. If he offers $s \leq s^*$, player 2 accepts $(1-s \geq 1-s^* = \delta s^*)$ and he gets $s \leq s^*$. If he offers $s > s^*$, player 2 rejects; next period player 2 offers $(\delta s^*, \cdot)$, he accepts, and he gets $\delta \cdot \delta s^* = \delta^2 s^* < s^*$. The best offer is $s = s^*$, as prescribed.

(iii) Player 2 proposing. Symmetrically, offering player 1 $s \geq \delta s^*$ is accepted and leaves her $1-s \leq 1-\delta s^* = s^*$; offering less is rejected; player 1 then proposes $(s^*, 1-s^*)$, she accepts, and she gets $\delta(1-s^*) = \delta^2 s^* < s^*$. Her best offer is $s = 1-s^* = \delta s^*$, as prescribed.

No player gains from a single deviation in any subgame, so by the one-stage deviation principle the strategy profile is a Nash equilibrium in every subgame: it is subgame perfect.

Why s^* and not another number. The two indifference conditions are $1-s_1 = \delta(1-s_2)$ (player 2 indifferent in odd periods) and $s_2 = \delta s_1$ (player 1 indifferent in even periods). Substituting, $s_1 = 1-\delta(1-\delta s_1)$, whose only solution is $s_1 = \frac{1}{1+\delta} = s^*$. The same fixed point appears in Gibbons's backward-induction argument, pp. 70–71. The problem does not ask for uniqueness; Rubinstein (1982), via the Shaked–Sutton argument in Gibbons, shows this is the *only* SPE.

(d) Interpretation. With an infinite horizon there is no last period to anchor backward induction, but the game looks the same every time a given player proposes. Each proposer offers the responder exactly what the responder could get by waiting one period and proposing himself, δs^*, and keeps the rest. The first proposer's advantage is $s^* - (1-s^*) = \frac{1-\delta}{1+\delta}$, the cost of one period of delay, and it vanishes as $\delta \to 1$, where the split tends to $(\frac{1}{2}, \frac{1}{2})$. Off the path the strategies tell each responder to accept anything at least as good as the value of waiting: these threats are credible, which is exactly what subgame perfection requires beyond Nash.

(e) Verification. `check_C_L05.py` verifies the identities (16) and the unique solution $s^* = 1/(1+\delta)$ of $s = 1-\delta(1-\delta s)$ with sympy. Then, for $\delta \in \{\frac{1}{2}, \frac{3}{4}, \frac{9}{10}, \frac{99}{100}\}$ in exact fractions, it checks: each responder accepts at the threshold and switches exactly at threshold $\pm\frac{1}{1000}$; the proposers' best one-stage offers, on a grid of step $\frac{1}{1000}$ plus $s^* \pm \frac{1}{1000}$, are the prescribed ones, and a rejected offer yields $\delta^2 s^*$; and the best agreement date against the opponent's strategy is $t = 1$ for player 1 and $t = 1$ (tied with $t = 2$) for player 2, over 59 periods. Output: "Gibbons 2.20: s* = 1/(1+delta), 1 - s* = delta s*; one-stage deviations by proposers and responders unprofitable for delta in {1/2, 3/4, 9/10, 99/100}; cutoffs +-1/1000 pass". Gibbons prints no answer.

(f) Common mistakes.

- Proving only the Nash part, or checking only on-path behaviour. Subgame perfection needs the responder's rule to be optimal after *every* offer, including offers never made.
- Valuing a rejection by the responder's share of the *current* offer, or forgetting to discount: rejecting today is worth δs^* (the proposer's share next period, one period later), not s^*.
- Applying the one-stage deviation principle without saying why it holds in an infinite game: it needs discounting with bounded payoffs (continuity at infinity).

Answer. With $s^* = \frac{1}{1+\delta}$ we have $1 - s^* = \delta s^*$ and $1 - \delta s^* = s^*$. Nash: against player 2's strategy, player 1 can get at most s^* (odd periods) or $\delta^2 s^*$ (even periods), and his strategy gets s^*; against player 1's strategy, player 2 can get at most $1 - s^* = \delta s^*$, and accepting in period 1 gets it. SPE: in each of the four subgame types (proposer 1, responder 2, proposer 2, responder 1), no one-stage deviation pays, because each responder accepts exactly when the offer is worth at least waiting (δs^*) and each proposer's best offer is the prescribed one (a rejected offer gives $\delta^2 s^* < s^*$). By the one-stage deviation principle the profile is subgame perfect, with immediate agreement on $(\frac{1}{1+\delta}, \frac{\delta}{1+\delta})$.